

Assignment 3

Problem 2Exp

System Parameters

Plant 3				
R	κ	τ	K_m	b
10.9	451	0.195	0.0539	-0.000149

Table 1: Plant 3 identified parameters and derived device metrics.

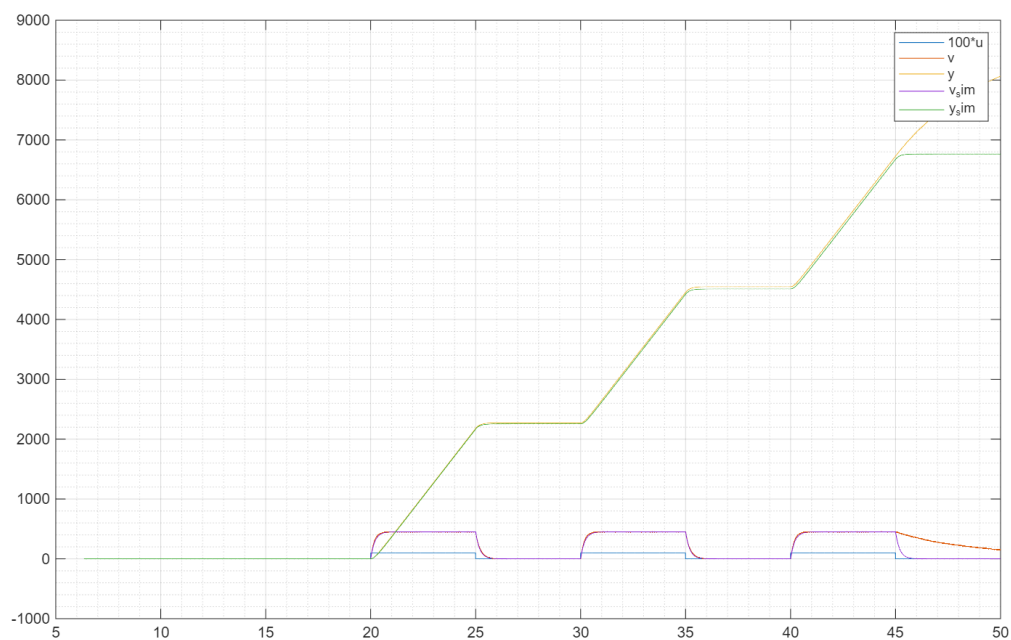


Figure 1: Experimental open loop velocity and position data with simulated plant velocity and position response to same input.

Velocity and Control comparison

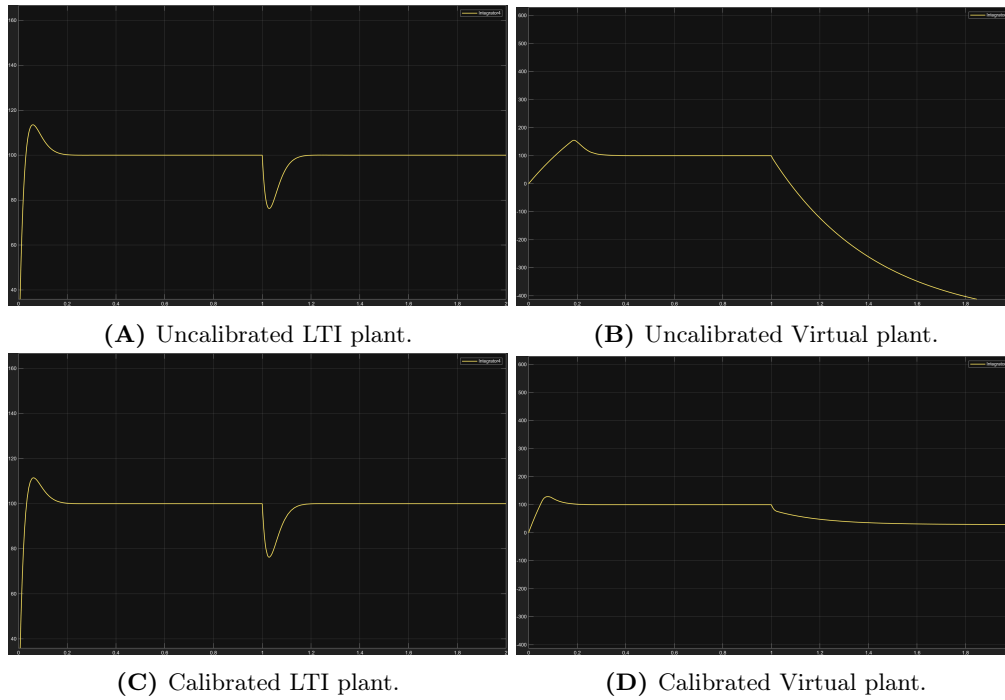


Figure 2: Simulated velocity output scopes.

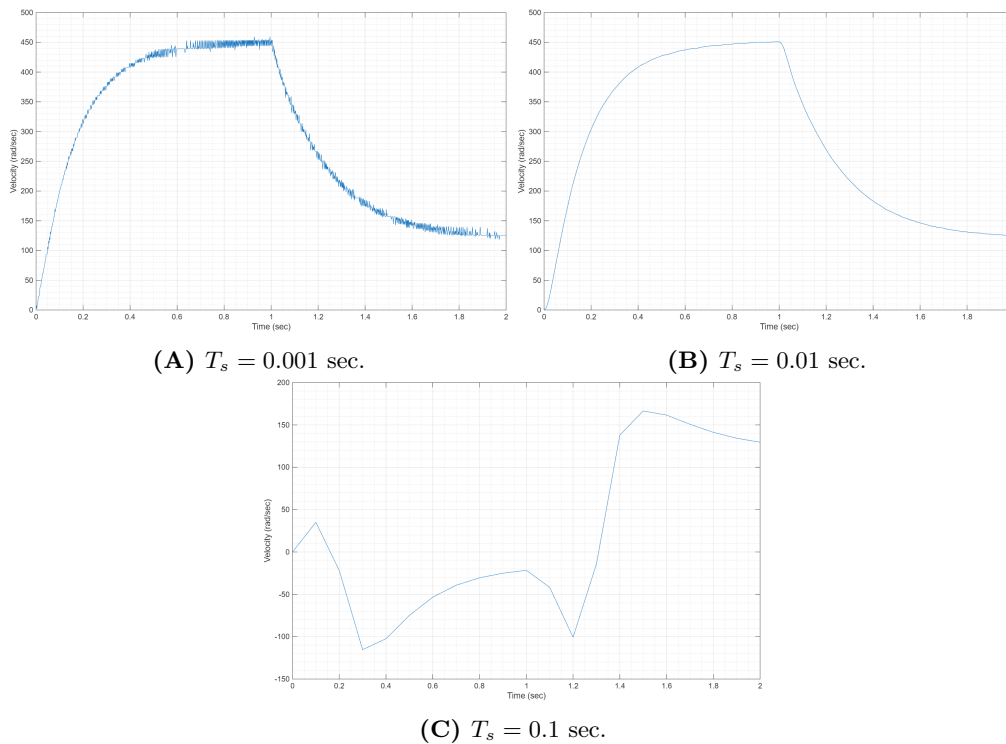


Figure 3: Experimental velocity output plots.

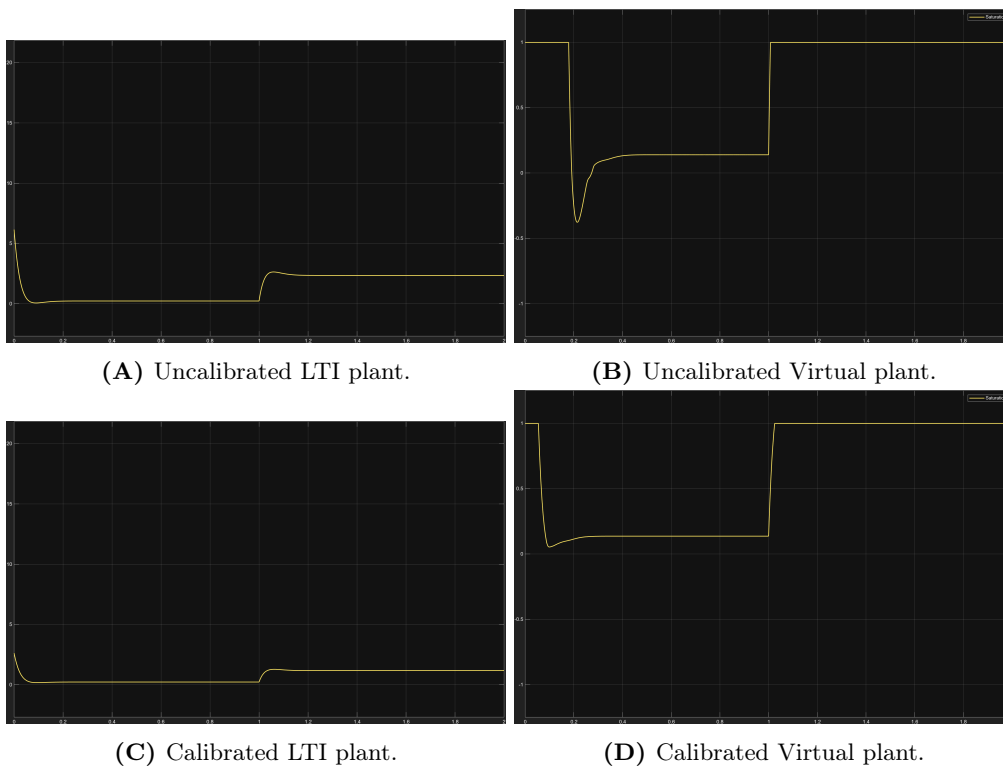


Figure 4: Simulated control scopes.

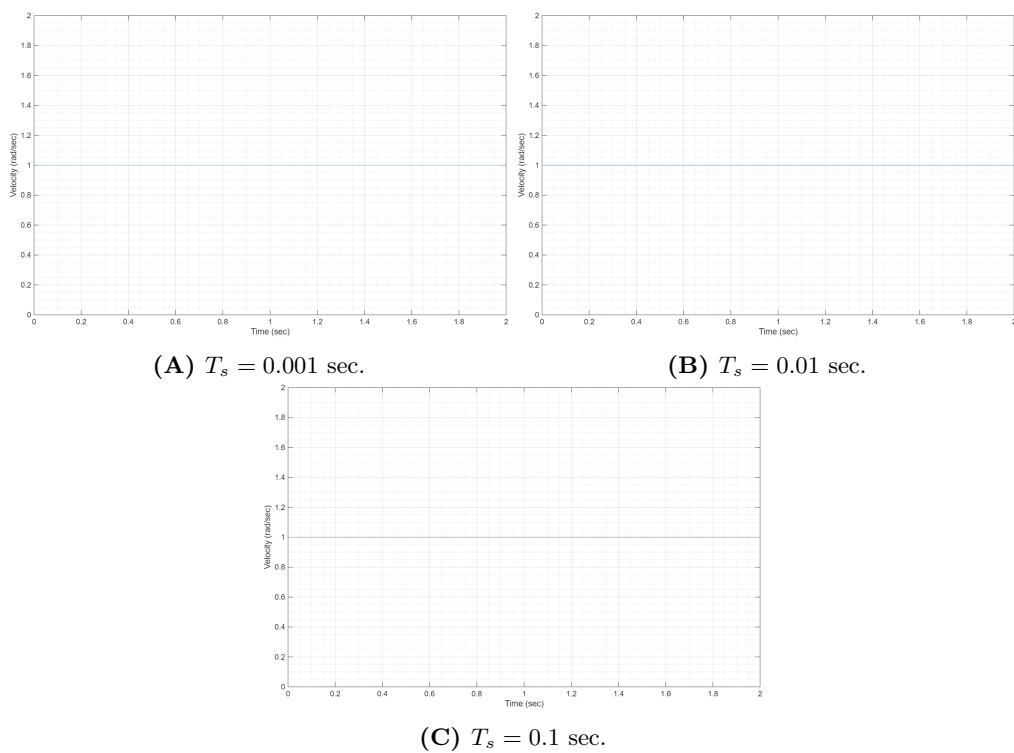


Figure 5: Experimental control plots.

Discussion

These results are very interesting, if not confusing. We were able to identify and match the system parameters through the open loop behavior of the physical setup. However, moving to the closed loop behavior demonstrates some strange behavior. The velocity behavior of all physical trials does not quite match the behavior of either the linear calibrated nor uncalibrated simulations – the simulations predict some overshoot in response to a step reference input. There is some resemblance with the virtual plants, with an exponential decay occurring in response to a step disturbance input and a non-zero error in response to the same. What is also interesting is the control signal for physical trials seem to show that they are experiencing control saturation through the whole experiment. The Virtual plants do show some saturation, but not to the extent that the experiments are showing. Perhaps this is due to a variance in how the duty cycle is mapped to the control system – we are aware that this relationship is not one-to-one, however it may be necessary to gather more data to have better representation in the virtual plant simulations.

Assignment 4

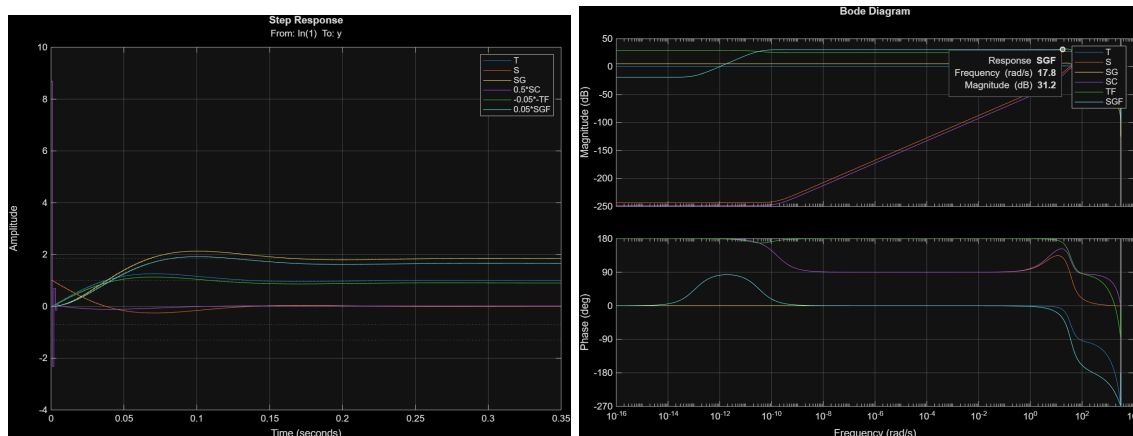
Problem 4: Indirect Digital Control

Specifications and Performance Metrics

T_s (sec)	ω_n $\frac{\text{rad}}{\text{sec}}$	ζ	T_r (sec)	PO (%)	P_m ($^\circ$)	G_m
0.001	36	0.5169	0.028	25.78	51.97	62.59
0.01	36	0.5169	0.02	34.61	41.26	6.30
0.1	unstable	unstable	unstable	unstable	unstable	unstable

Table 1: System parameters and performance metrics.

Steady State Performance



(A) Step response plot

(B) Frequency response plot

Figure 1: Step and frequency behavior for $T_s = 0.001\text{sec}$.

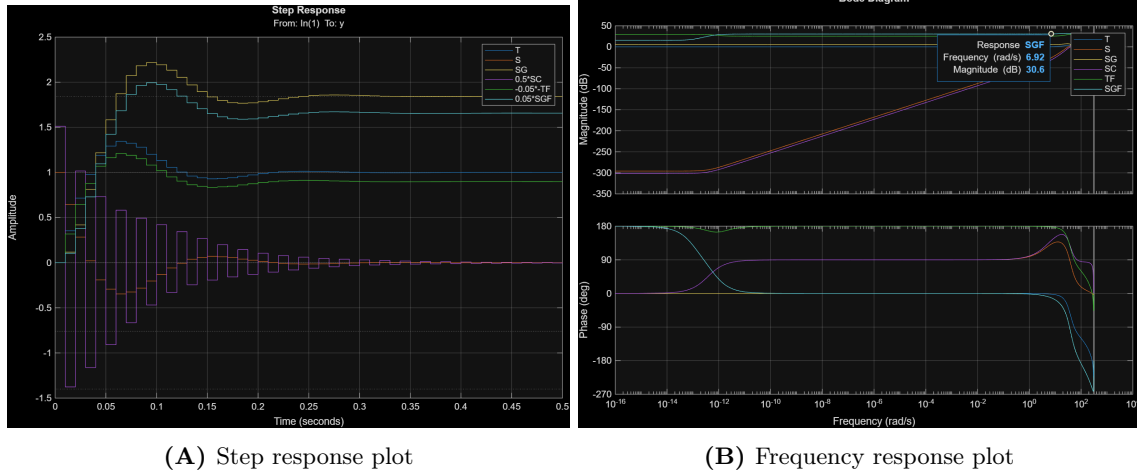


Figure 2: Step and frequency behavior for $T_s = 0.01\text{sec}$.

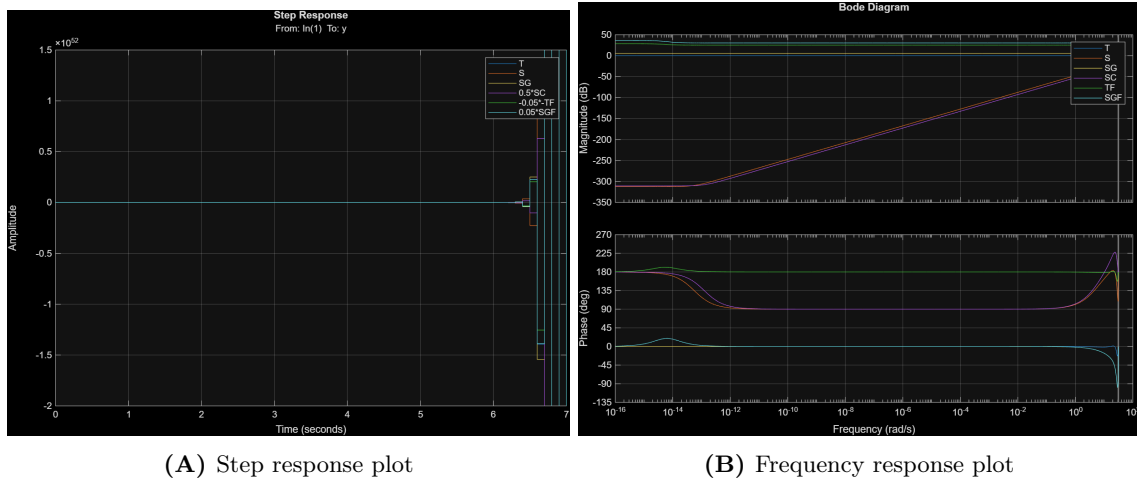


Figure 3: Step and frequency behavior for $T_s = 0.1\text{sec}$.

Step Input Steady State Values and Control Saturation

T_s (sec)	$\frac{u(k)}{r(k)}$	$\frac{y(k)}{r(k)}$	$\frac{e(k)}{r(k)}$	$\frac{u(k)}{d(k)}$	$\frac{y(k)}{d(k)}$	$\frac{e(k)}{d(k)}$	r_{ss} saturation?	$T_{d,ss}$ saturation?
	SC	T	S	-T	SG	-SG	SC	-TF
0.001	0.03	1 rad	0	-1	1.84 rad	-1.84 rad	yes	barely yes
0.01	0	1 rad	0	-1	1.84 rad	-1.84 rad	yes	yes
0.1	unstable	unstable	unstable	unstable	unstable	unstable	unstable	unstable

Table 2: Indirect digital control steady state values and control saturation taken from step response plot.

Sinusoidal Steady State Response

T_s (sec)	$\frac{y(k)}{r(k)}$	$\frac{w(k)}{r(k)}$	$\frac{e(k)}{r(k)}$	$\frac{u(k)}{r(k)}$	$\frac{y(k)}{T_d(k)}$	$\frac{w(k)}{T_d(k)}$	$\frac{e(k)}{T_d(k)}$	$\frac{u(k)}{T_d(k)}$
	T	SG	S	SC	SGF	SF	SGF	TF
0.001	0/0	5dB/0°	-90dB/90°	-95dB/90°	30dB/0°	-55dB/270°	30dB/0°	25dB/180°
0.01	0/0	5dB/0°	-70dB/90°	-75dB/90°	30dB/0°	-45dB/270°	30dB/0°	25dB/180°
0.1	unstable	unstable	unstable	unstable	unstable	unstable	unstable	unstable

Table 3: Sinusoidal steady state gain and phase values taken from frequency response plot.

At the provided frequencies, a number of the gains become very small, approximating zero. The steady state sinusoids for $T = 0.001\text{sec}$ are the following:

$$\begin{aligned}
 y(k) &= \sin(10Tk) + 0.316 \sin(20Tk) \\
 w(k) &\approx 1.77 \sin(10Tk) + 0 \\
 e(k) &\approx 0 + 0.316 \sin(20kT) \\
 u(k) &\approx 0 + 0.178 \sin(20kT + 180^\circ)
 \end{aligned}$$

The steady state sinusoids for $T = 0.01\text{sec}$ are the following:

$$\begin{aligned}
 y(k) &= \sin(10Tk) + 0.316 \sin(20Tk) \\
 w(k) &= 1.77 \sin(10Tk) + 0.00562 \sin(20kT + 270^\circ) \\
 e(k) &\approx 0 + 0.316 \sin(20kT) \\
 u(k) &\approx 0 + 0.178 \sin(20kT + 180^\circ)
 \end{aligned}$$

The system is unstable at $T = 0.1\text{sec}$, therefore no steady state response can be generated.

Analysis and Simulation

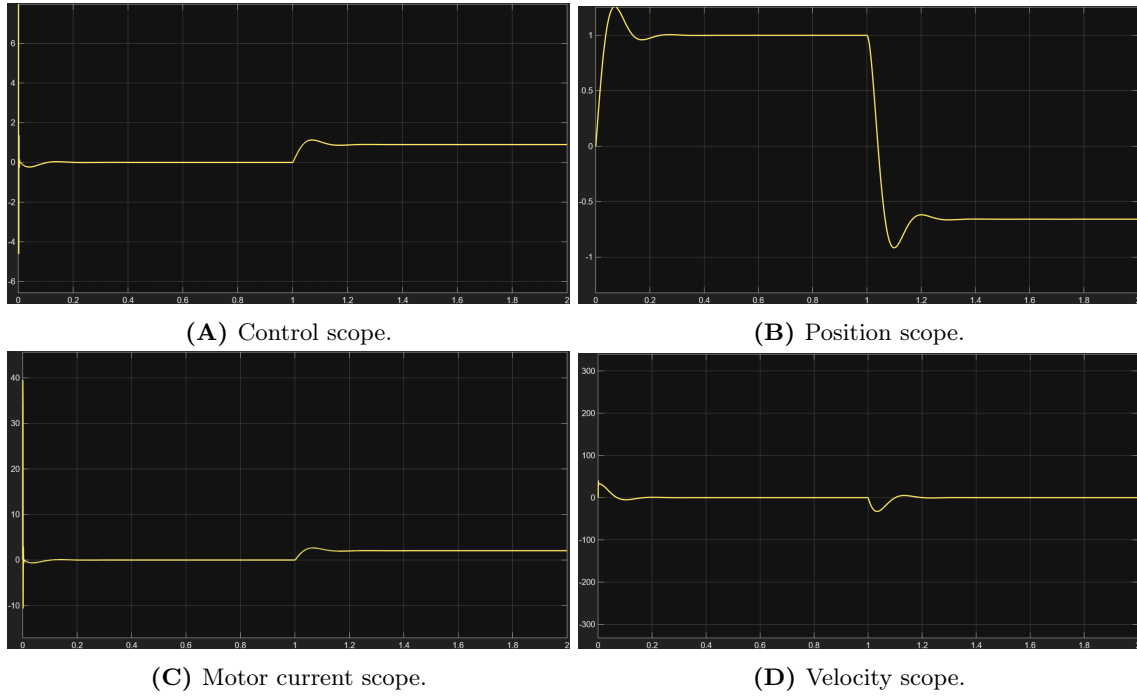


Figure 4: Linear plant simulation for $T_s = 0.001\text{sec}$.

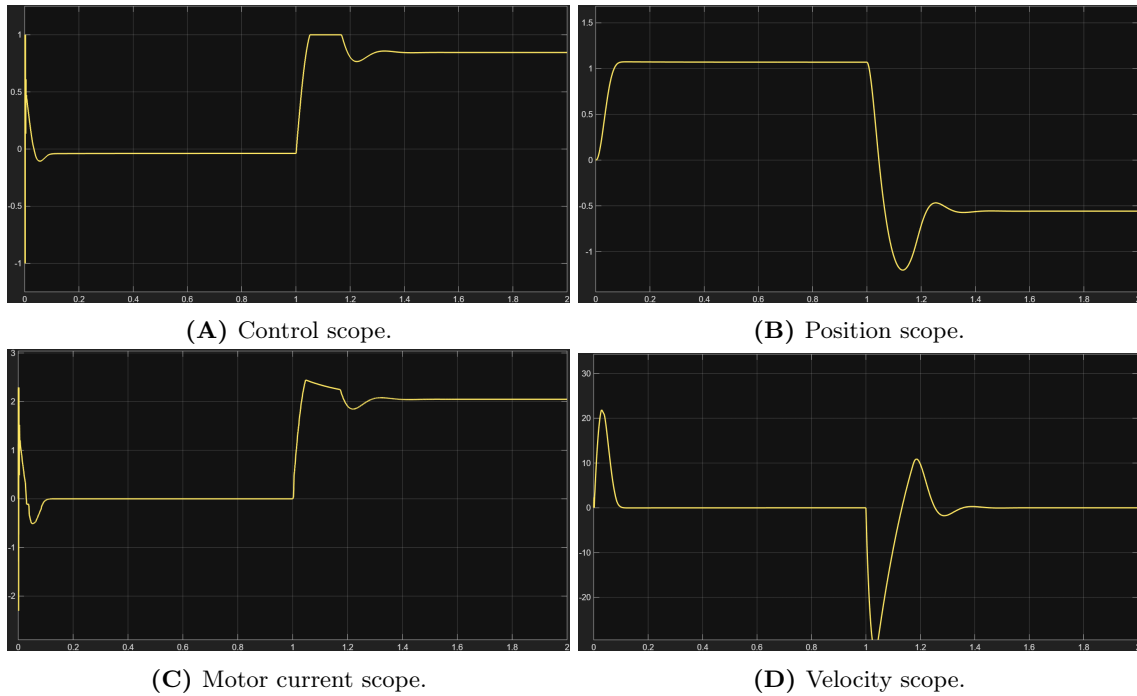


Figure 5: Virtual plant simulation for $T_s = 0.001\text{sec}$.

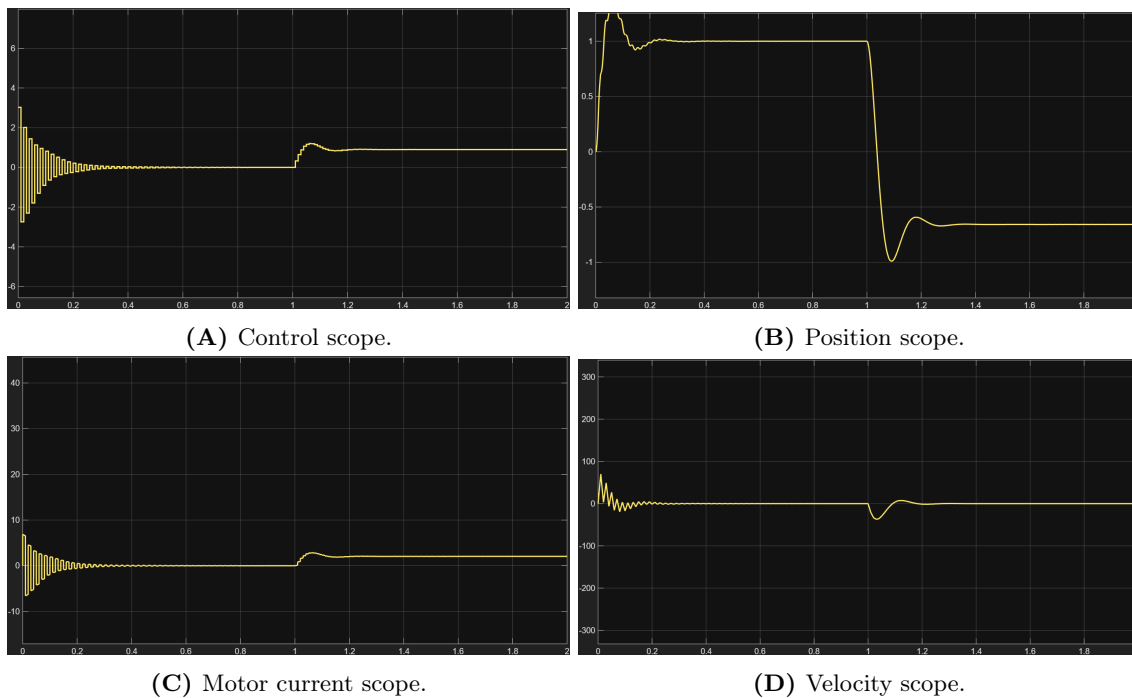


Figure 6: Linear plant simulation for $T_s = 0.01\text{sec}$.

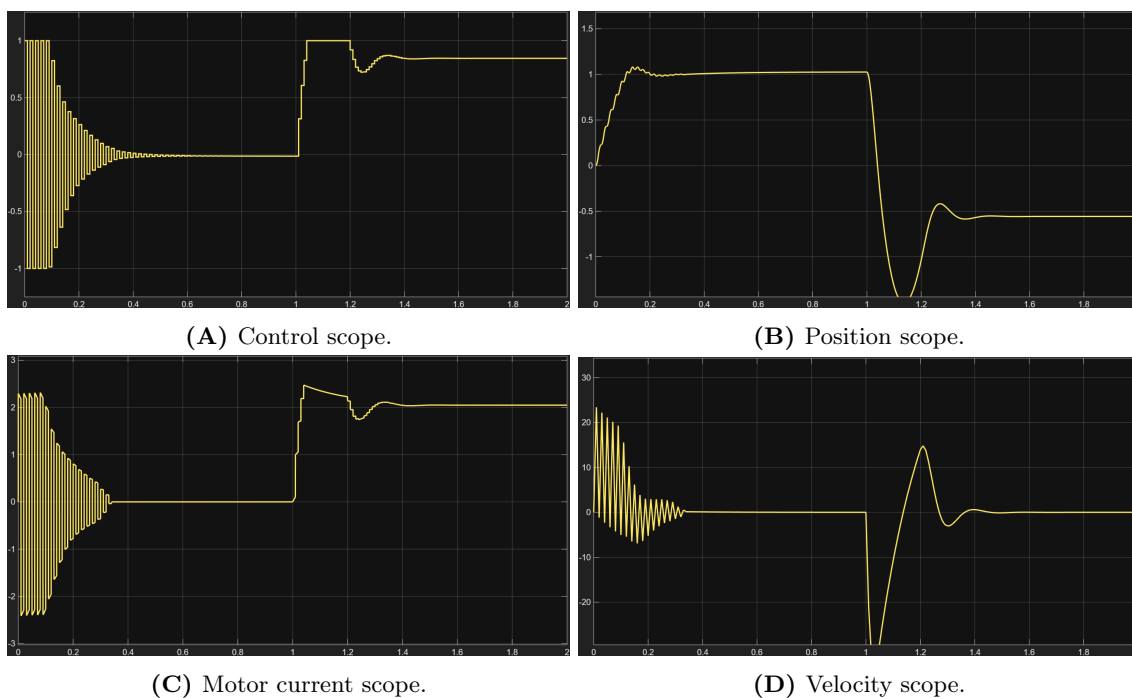


Figure 7: Virtual plant simulation for $T_s = 0.01\text{sec}$.

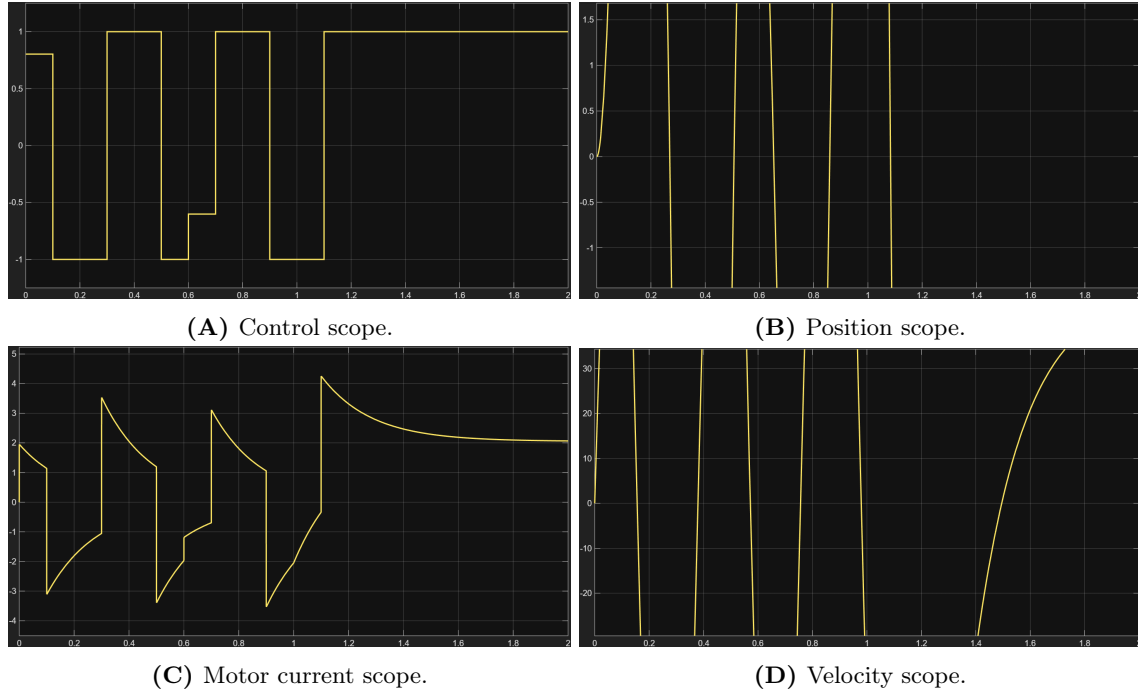


Figure 8: Virtual plant simulation for $T_s = 0.1\text{sec}$. The system is unstable.

Discussion

The analytical results partially match with the expectations of the plant, as there is zero steady state error for the step input, but is non-zero for the an input disturbance/torque. This is due to the integrator being involved with the forward control gain for the disturbance and torque transfer functions.

For the smaller sampling period, we can observe that the linear and virtual plants both produce fairly similar results, which can be attributed to the little saturation that occurs. The same cannot be said for the $T_s = 0.01\text{sec}$ case, where there is heavy saturation for the first 0.1sec of the simulation. However, the results do eventually converge after the transient response to the reference decays.

Problem 5: Direct Digital Control

PD Controller: $T_r = 0.05\text{sec}$, $PO_{max} = 15\%$

Specifications and Performance Metrics

T_s	0.001 sec	0.01 sec	0.1 sec
PD controller	$\frac{21.06z-20.64}{z}$	$\frac{1.6z-1.392}{z}$	$\frac{0.05z-0.035}{z}$

Table 4: Tuned controllers for each sampling time.

T_s (sec)	$\omega_n \frac{\text{rad}}{\text{sec}}$	ζ	T_r (sec)	PO (%)	P_m (°)	G_m
0.001	31.3	0.86	0.027	12.07	72.09	39.9
0.01	41.6	1	0.03	14.8	56.89	5.398
0.1	13.8	0.336	0.1	13.91	52.84	2.456

Table 5: Relevant performance metrics.

Steady State Performance

Step Input Steady State Values

T_s (sec)	$\frac{u(k)}{r(k)}$	$\frac{y(k)}{r(k)}$	$\frac{e(k)}{r(k)}$	$\frac{u(k)}{d(k)}$	$\frac{y(k)}{d(k)}$	$\frac{e(k)}{d(k)}$	r_{ss} saturation?	$T_{d,ss}$ saturation?
	SC	T	S	-T	SG	-SG	SC	-TF
0.001	0	1 rad	0	-1	2.38 rad	-2.38 rad	yes	barely yes
0.01	0	1 rad	0	-1	4.81 rad	-4.81 rad	yes	barely yes
0.1	0	1 rad	0	-1	66.7 rad	-66.7 rad	no	barely yes

Table 6: Direct digital PD control steady state values to step inputs.

Sinusoidal Steady State Response

T_s (sec)	$\frac{y(k)}{r(k)}$	$\frac{w(k)}{r(k)}$	$\frac{e(k)}{r(k)}$	$\frac{u(k)}{r(k)}$	$\frac{y(k)}{T_d(k)}$	$\frac{w(k)}{T_d(k)}$	$\frac{e(k)}{T_d(k)}$	$\frac{u(k)}{T_d(k)}$
	T	SG	S	SC	SGF	SF	SGF	TF
0.001	0/0	7.5dB/0°	-85dB/90°	-92dB/90°	32dB/0°	-50dB/270°	32dB/0°	25dB/180°
0.01	0/0	13.5dB/0°	-60dB/90°	-70dB/90°	39dB/0°	-27dB/270°	39dB/0°	25dB/180°
0.1	0/-10°	36dB/-20°	-16dB/80°	-50dB/95°	60dB/-45°	12dB/270°	60dB/-45°	25dB/160°

Table 7: Sinusoidal steady state gain and phase values taken from frequency response plot.

$T_s = 0.001\text{sec}$:

$$y(k) = \sin(10Tk) + 0.398 \sin(20Tk)$$

$$w(k) \approx 2.37 \sin(10Tk) + 0$$

$$e(k) \approx 0 + 0.398 \sin(20kT)$$

$$u(k) \approx 0 + 0.178 \sin(20kT + 180^\circ)$$

$T_s = 0.01\text{sec}$:

$$y(k) = \sin(10Tk) - 0.891 \sin(20Tk)$$

$$w(k) \approx 4.73 \sin(10Tk) + 0$$

$$e(k) \approx 0 + 0.891 \sin(20kT)$$

$$u(k) \approx 0 + 0.178 \sin(20kT + 180^\circ)$$

$T_s = 0.1\text{sec}$:

$$\begin{aligned}
y(k) &= \sin(10Tk - 10^\circ) + 10 \sin(20Tk - 45^\circ) \\
w(k) &= 63.1 \sin(10Tk - 20^\circ) + 0.03981 \sin(20Tk + 270^\circ) \\
e(k) &= 0.158 \sin(10Tk + 80^\circ) + 10 \sin(20kT - 45^\circ) \\
u(k) &\approx 0 + 0.178 \sin(20kT + 160^\circ)
\end{aligned}$$

Analysis and Simulation

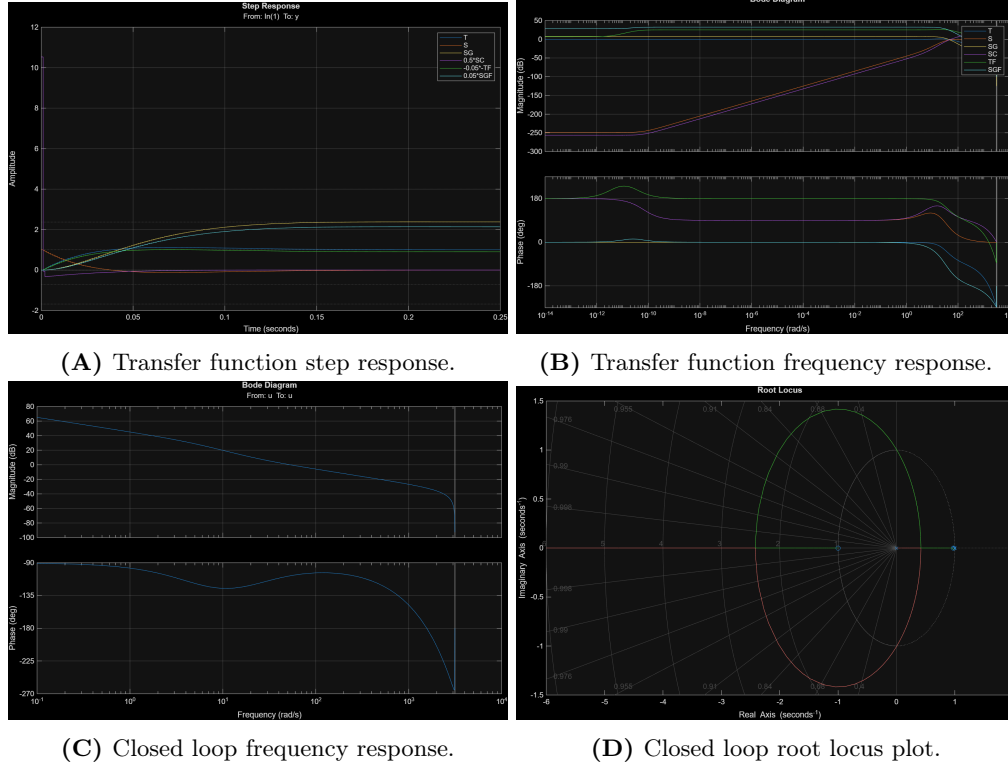


Figure 9: Reduced order system analysis for $T_s = 0.001\text{sec}$.

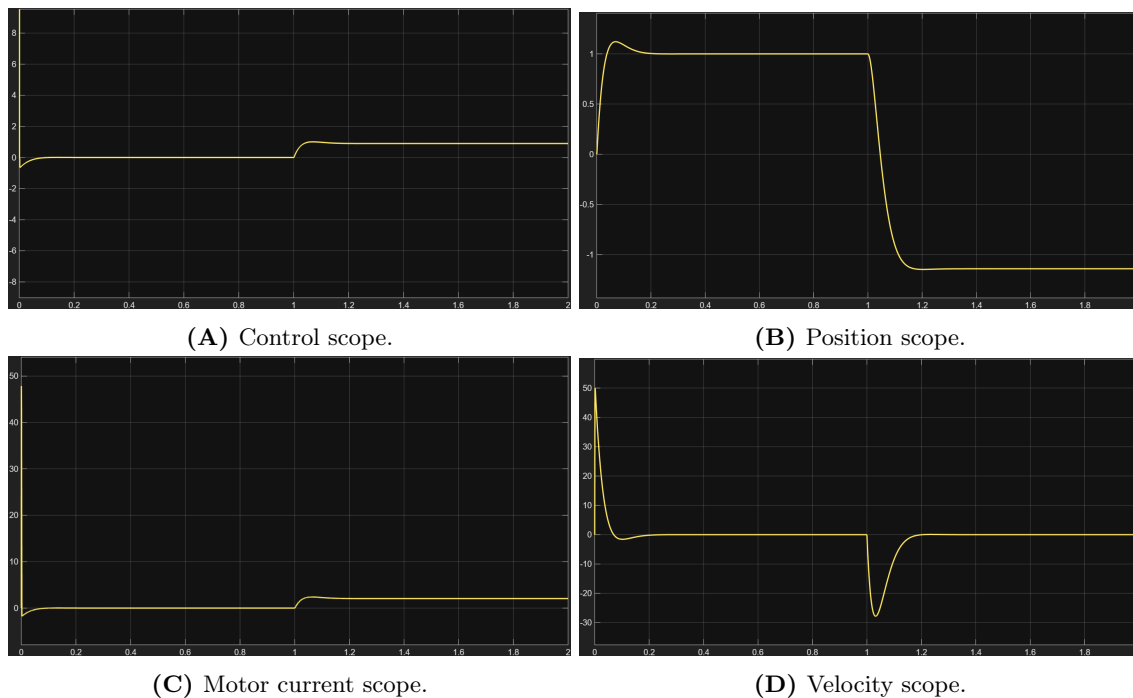


Figure 10: Linear plant simulation for $T_s = 0.001\text{sec}$.

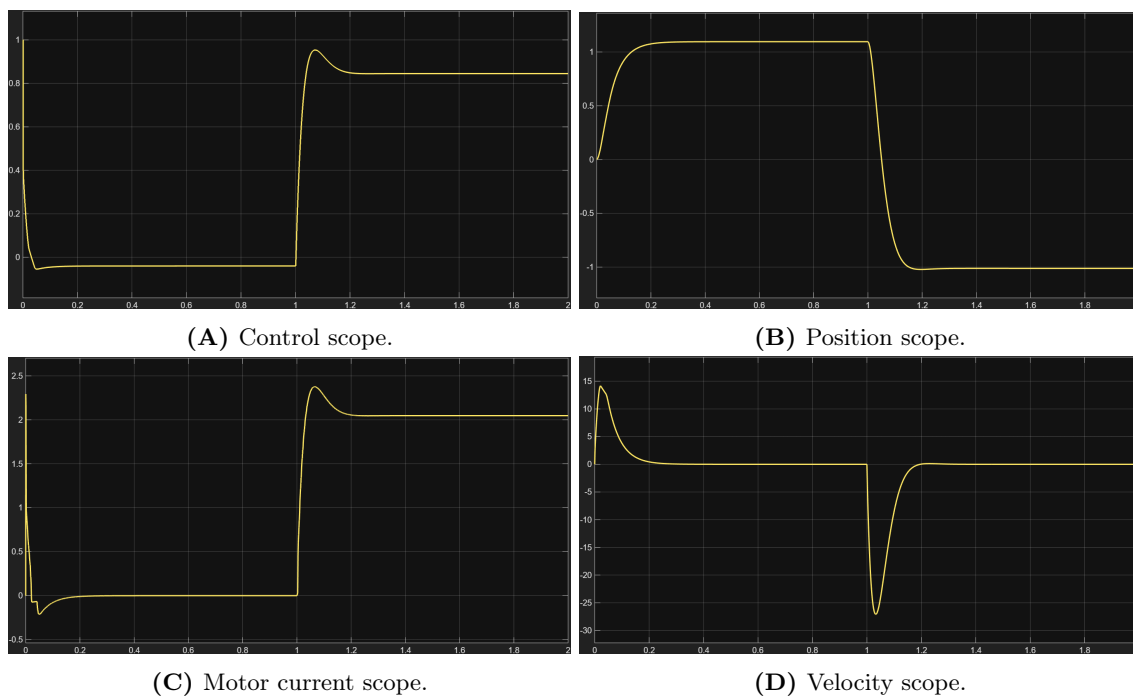


Figure 11: Virtual plant simulation for $T_s = 0.001\text{sec}$.

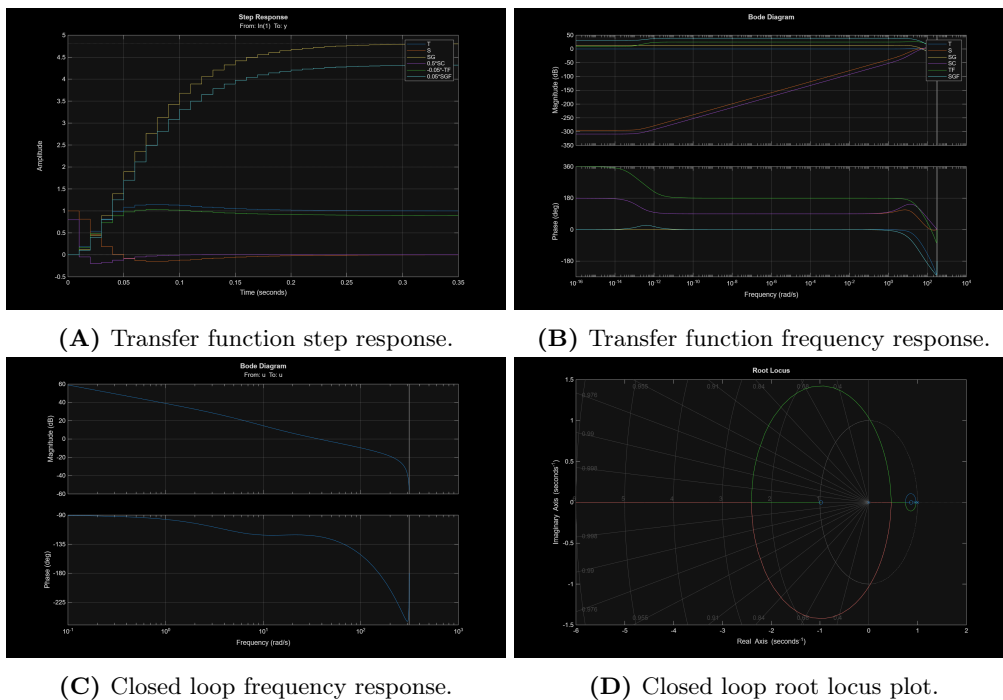


Figure 12: Reduced order system analysis for $T_s = 0.01\text{sec}$.

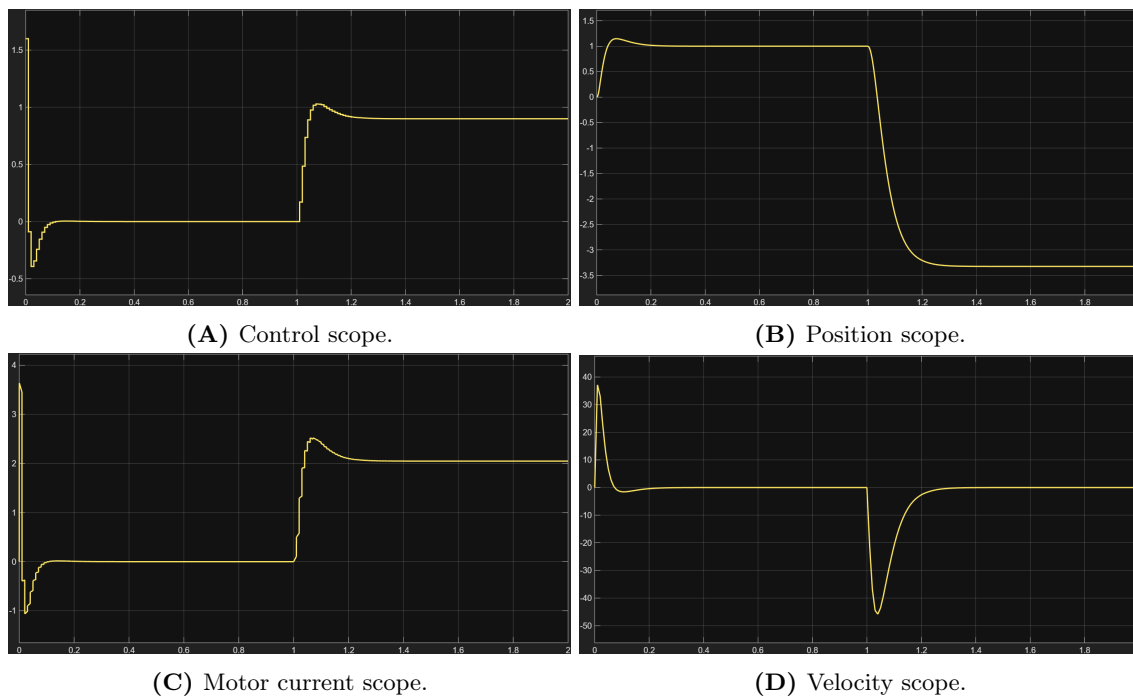


Figure 13: Linear plant simulation for $T_s = 0.01\text{sec}$.

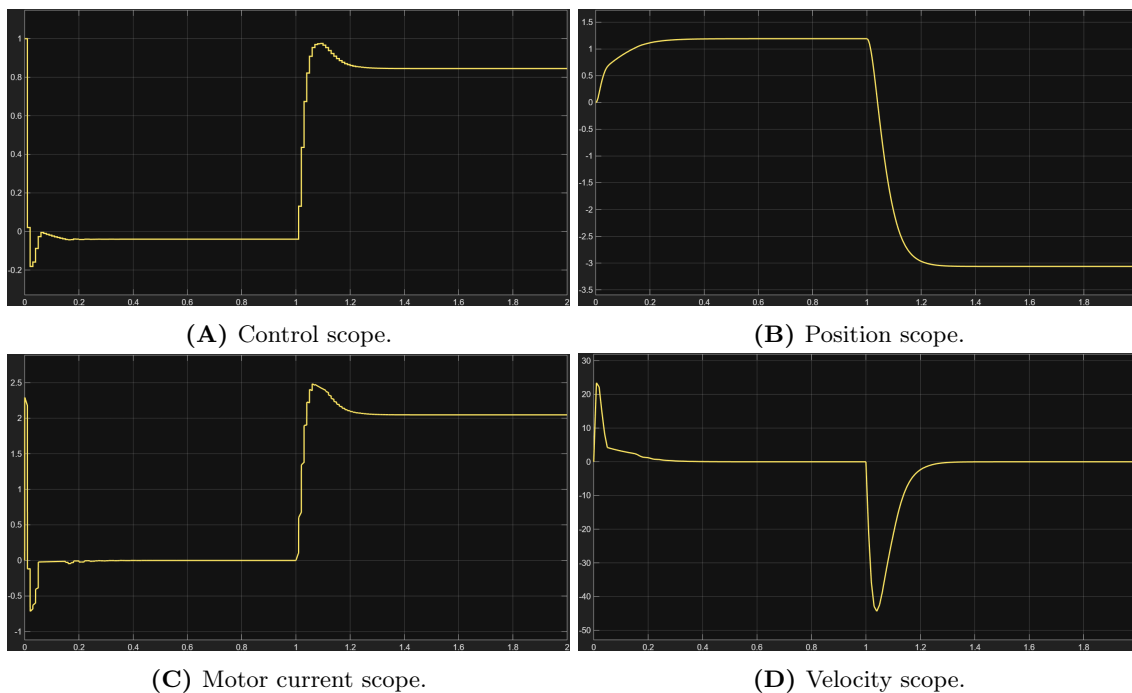


Figure 14: Virtual plant simulation for $T_s = 0.01\text{sec}$.

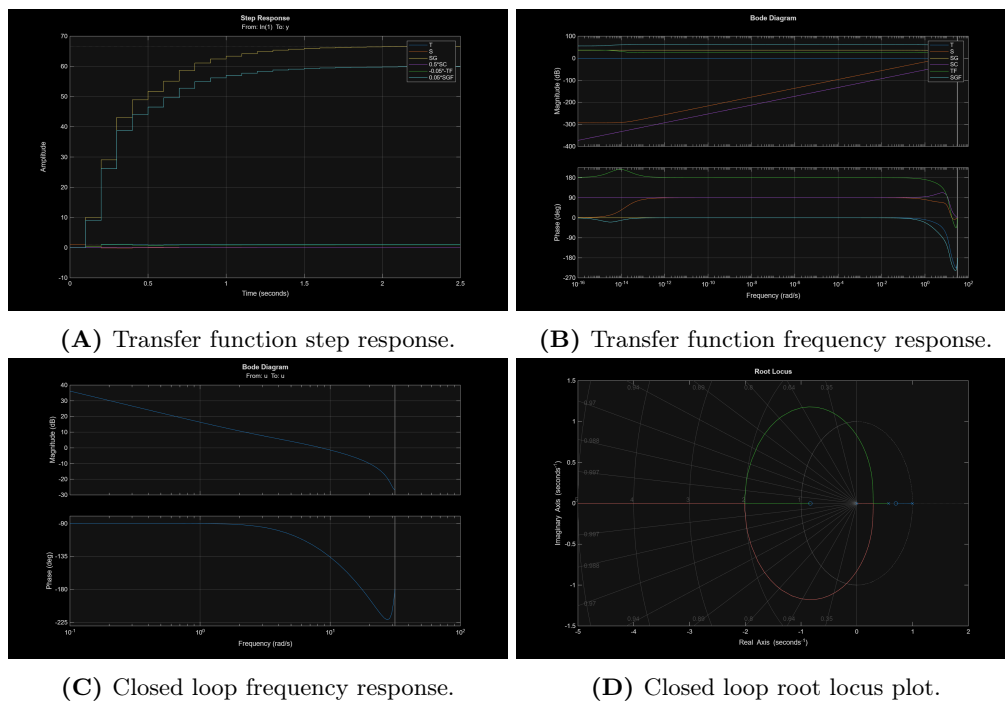


Figure 15: Reduced order system analysis for $T_s = 0.1\text{sec}$.

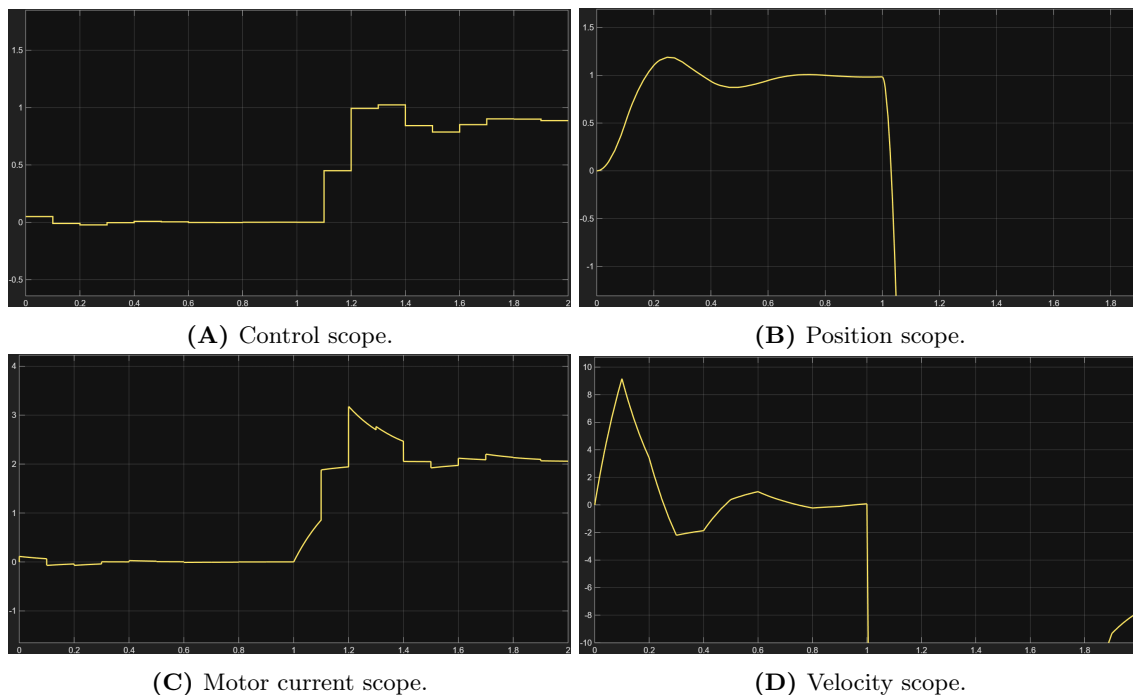


Figure 16: Linear plant simulation for $T_s = 0.1\text{sec}$.

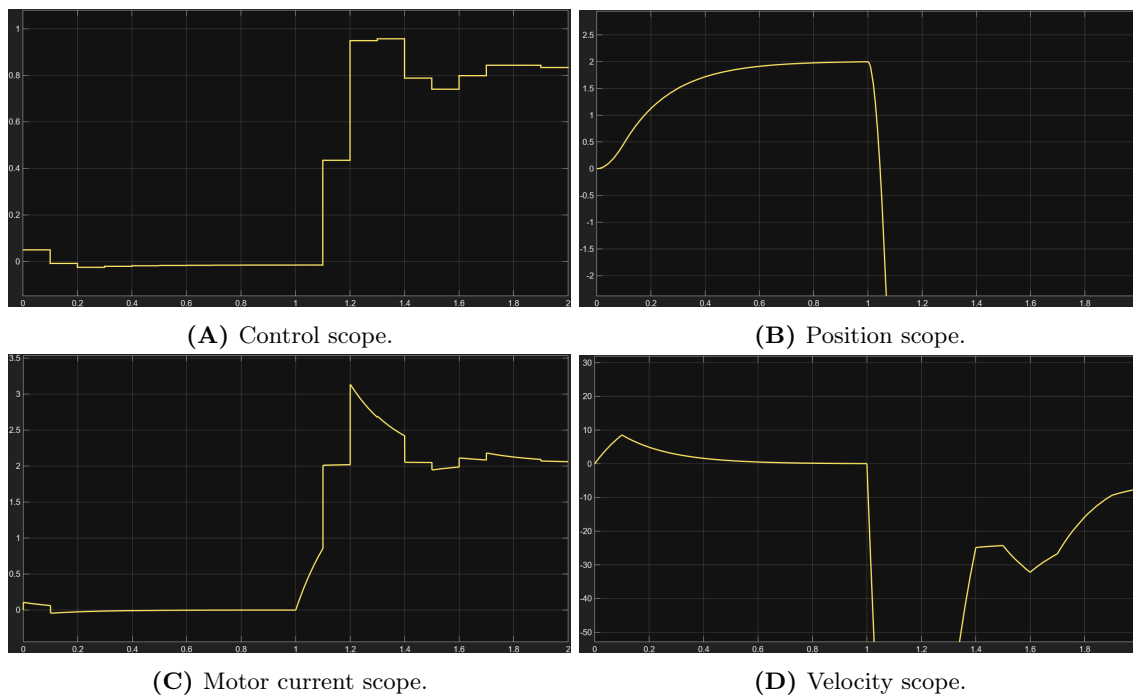


Figure 17: Virtual plant simulation for $T_s = 0.1\text{sec}$.

Discussion

After tuning the gains for the controllers, each one is able to out-perform their respective indirect digital control counterparts. With the exception of the largest sampling time, each system was able to meet all

performance specifications (because the sampling time is larger than the target rise time, the lowest analytical rise time possible is the sampling time itself). Important to note is that while the error response to the reference is zero, achieving the performance specs comes at a cost of an incredibly large gain for the error in response to an input disturbance.

PID Controller: $T_r = 0.05\text{sec}$, $PO_{max} = 15\%$

Specifications and Performance Metrics

T_s	0.001 sec	0.01 sec	0.1 sec
PD controller	$\frac{27z^2-53.43z+26.44}{z^2-z}$	$\frac{1.8z^2-3.411z+1.611}{z^2-z}$	$\frac{0.045z^2-0.0756z+0.03087}{z^2-z}$

Table 8: Tuned controllers for each sampling time.

T_s (sec)	$\omega_n \frac{\text{rad}}{\text{sec}}$	ζ	T_r (sec)	PO (%)	P_m (°)	G_m
0.001	42.9	1	0.021	14.06	72.86	31.12
0.01	1.66	1	0.03	11.352	59.42	4.8155
0.1	0.224	1	0.1	12.84	57.41	2.733

Table 9: Relevant performance metrics.

Steady State Performance

Step Input Steady State Values

T_s (sec)	$\frac{u(k)}{r(k)}$	$\frac{y(k)}{r(k)}$	$\frac{e(k)}{r(k)}$	$\frac{u(k)}{d(k)}$	$\frac{y(k)}{d(k)}$	$\frac{e(k)}{d(k)}$	r_{ss} saturation?	$T_{d,ss}$ saturation?
	SC	T	S	-T	SG	-SG	SC	-TF
0.001	0	1 rad	0	-1	0	0	yes	barely yes
0.01	0	1 rad	0	-1	0	0	yes	barely no
0.1	0	1 rad	0	-1	0	0	no	barely yes

Table 10: Direct digital PID control steady state values to step input.

Sinusoidal Steady State Response

T_s (sec)	$\frac{y(k)}{r(k)}$	$\frac{w(k)}{r(k)}$	$\frac{e(k)}{r(k)}$	$\frac{u(k)}{r(k)}$	$\frac{y(k)}{T_d(k)}$	$\frac{w(k)}{T_d(k)}$	$\frac{e(k)}{T_d(k)}$	$\frac{u(k)}{T_d(k)}$
	T	SG	S	SC	SGF	SF	SGF	TF
0.001	0/0	-48dB/90°	-130dB/180°	-92dB/90°	-43dB/90°	-115dB/360°	-43dB/90°	25dB/180°
0.01	0/0	-8dB/90°	-80dB/180°	-70dB/90°	25dB/80°	-50dB/360°	25dB/80°	25dB/180°
0.1	0/0	37dB/90°	-15dB/90°	-52dB/90°	60dB/-40°	15dB/240°	60dB/-40°	25dB/160°

Table 11: Sinusoidal steady state gain and phase values taken from frequency response plot.

$T_s = 0.001\text{sec}$:

$$\begin{aligned}
y(k) &\approx \sin(10Tk) + 0 \\
w(k) &\approx 2.37 \sin(10Tk) + 0 \\
e(k) &\approx 0 + 0 \\
u(k) &\approx 0 + 0.178 \sin(20kT + 180^\circ)
\end{aligned}$$

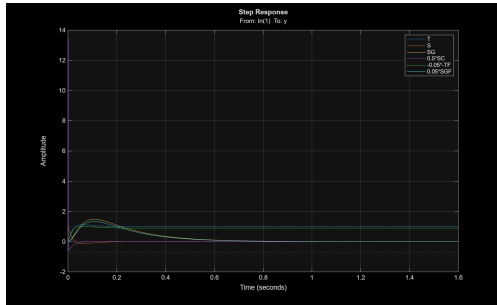
$T_s = 0.01\text{sec}$:

$$\begin{aligned}
y(k) &= \sin(10Tk) + 0.178 \sin(20Tk) \\
w(k) &\approx 0.398 \sin(10Tk) + 0 \\
e(k) &\approx 0 + 0.178 \sin(20kT + 80^\circ) \\
u(k) &\approx 0 + 0.178 \sin(20kT + 180^\circ)
\end{aligned}$$

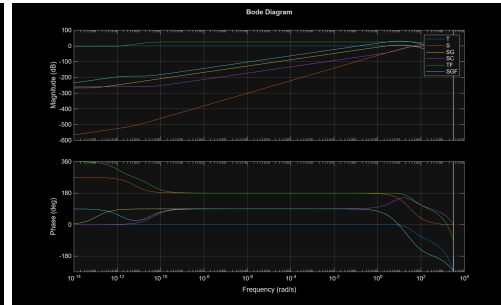
$T_s = 0.1\text{sec}$:

$$\begin{aligned}
y(k) &= \sin(10Tk - 10^\circ) + 10 \sin(20Tk - 40^\circ) \\
w(k) &= 70.79 \sin(10Tk + 90^\circ) + 0.0562 \sin(20Tk + 240^\circ) \\
e(k) &= 0.178 \sin(10Tk + 90^\circ) + 10 \sin(20kT - 40^\circ) \\
u(k) &\approx 0 + 0.178 \sin(20kT + 160^\circ)
\end{aligned}$$

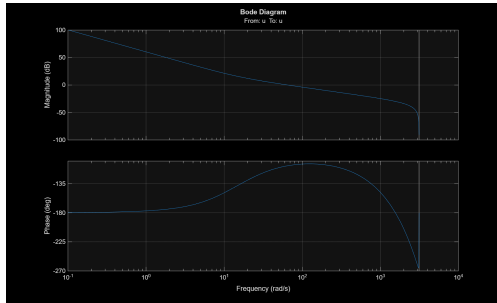
Analysis and Simulation



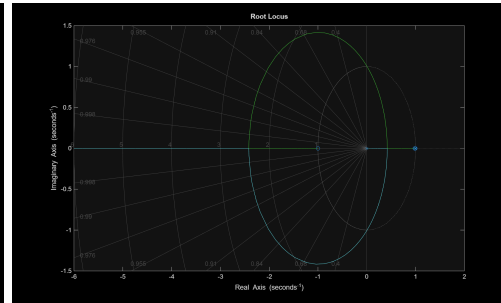
(A) Transfer function step response.



(B) Transfer function frequency response.



(C) Closed loop frequency response.



(D) Closed loop root locus plot.

Figure 18: Reduced order system analysis for $T_s = 0.001\text{sec}$.

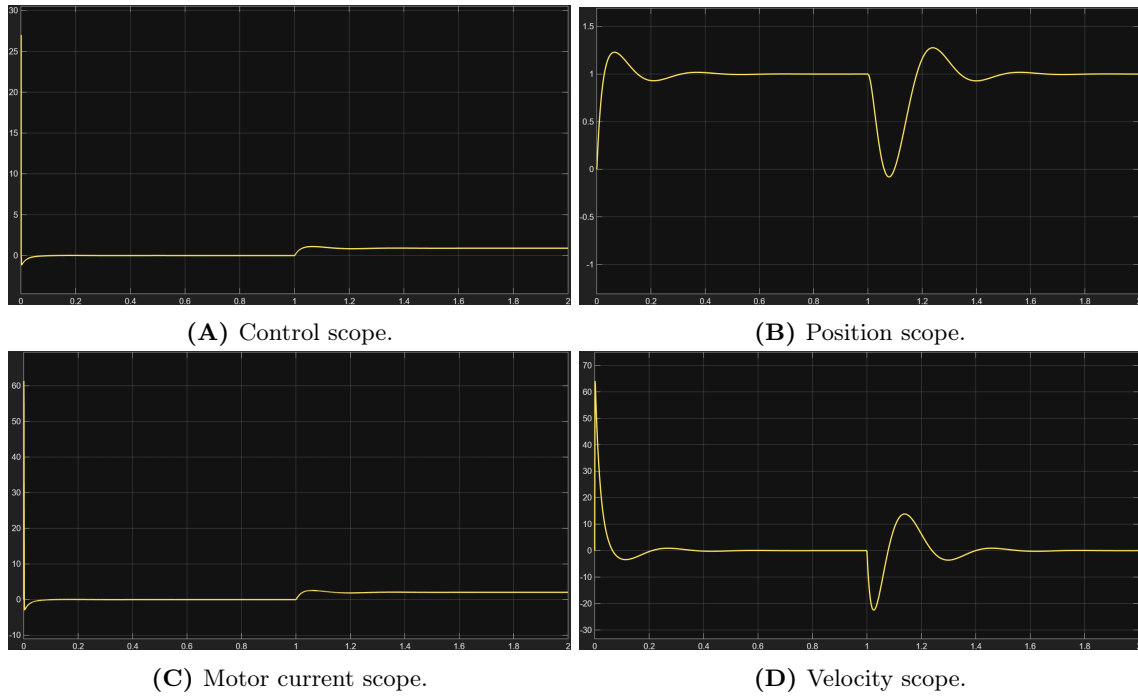


Figure 19: Linear plant simulation for $T_s = 0.001\text{sec}$.

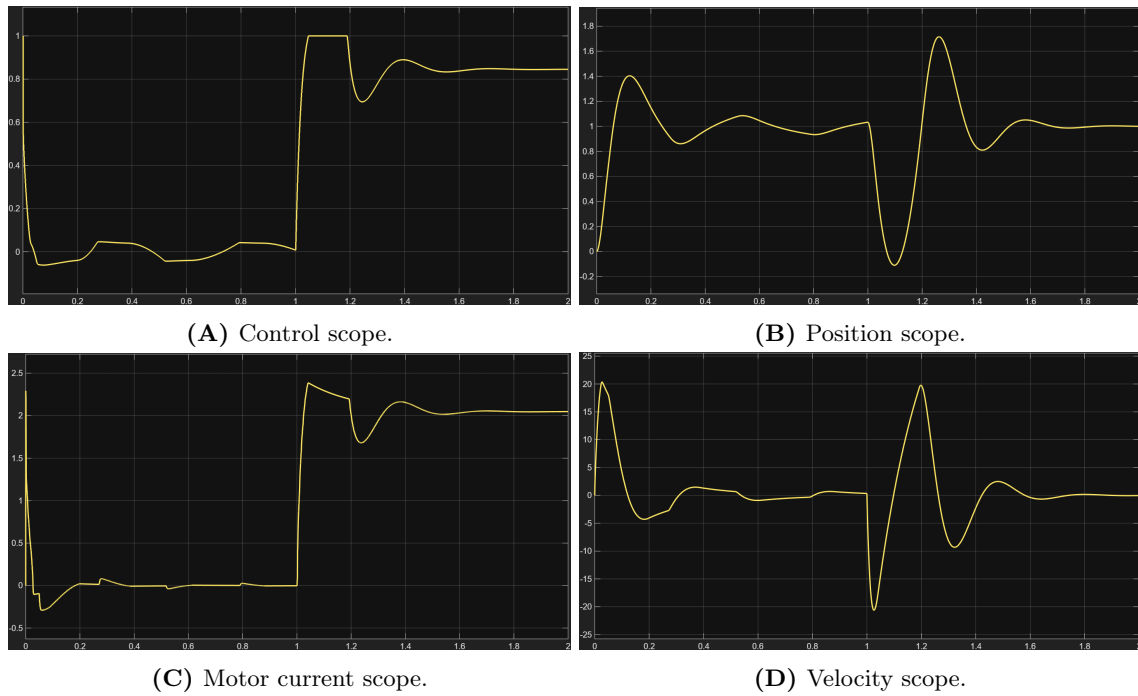


Figure 20: Virtual plant simulation for $T_s = 0.001\text{sec}$.

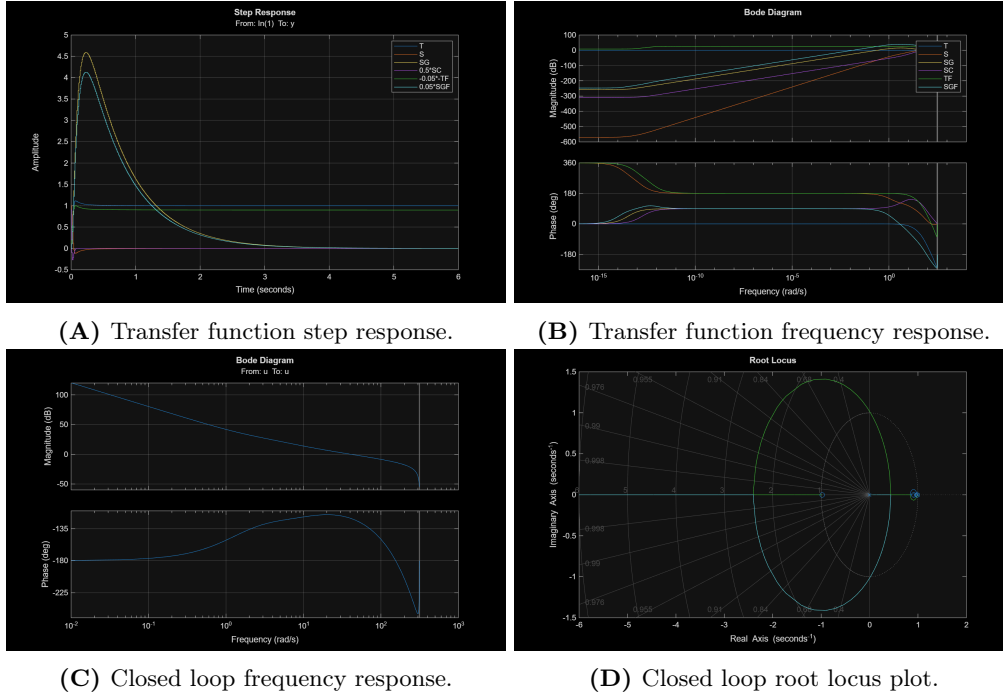


Figure 21: Reduced order system analysis for $T_s = 0.01\text{sec}$.

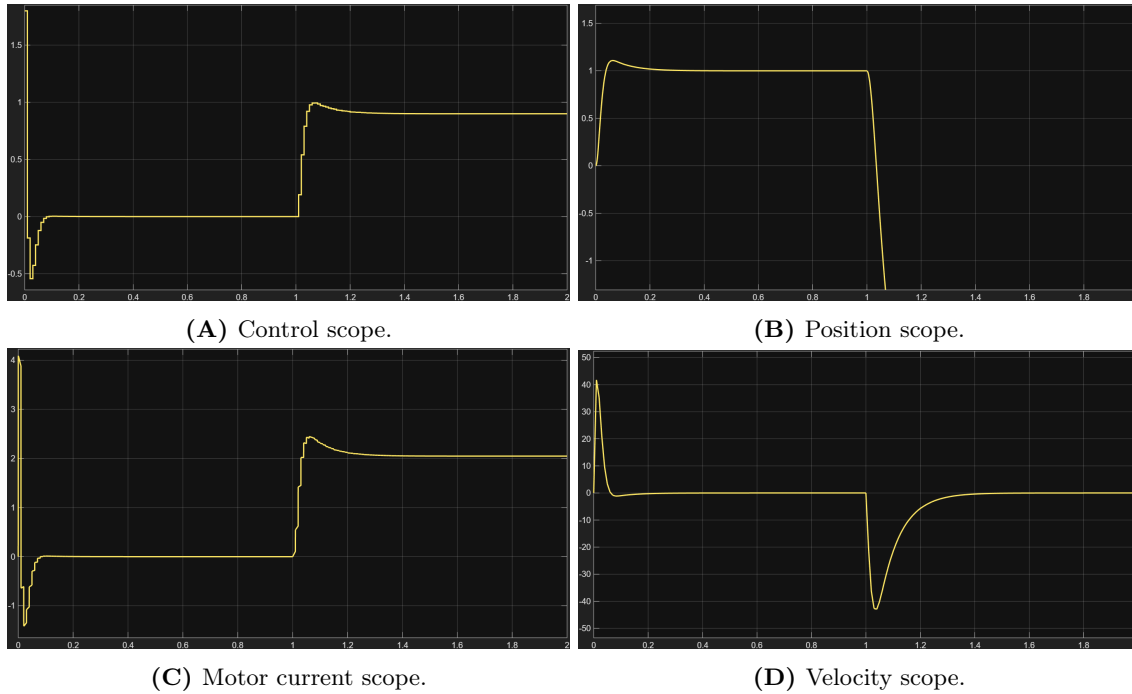


Figure 22: Linear plant simulation for $T_s = 0.01\text{sec}$.

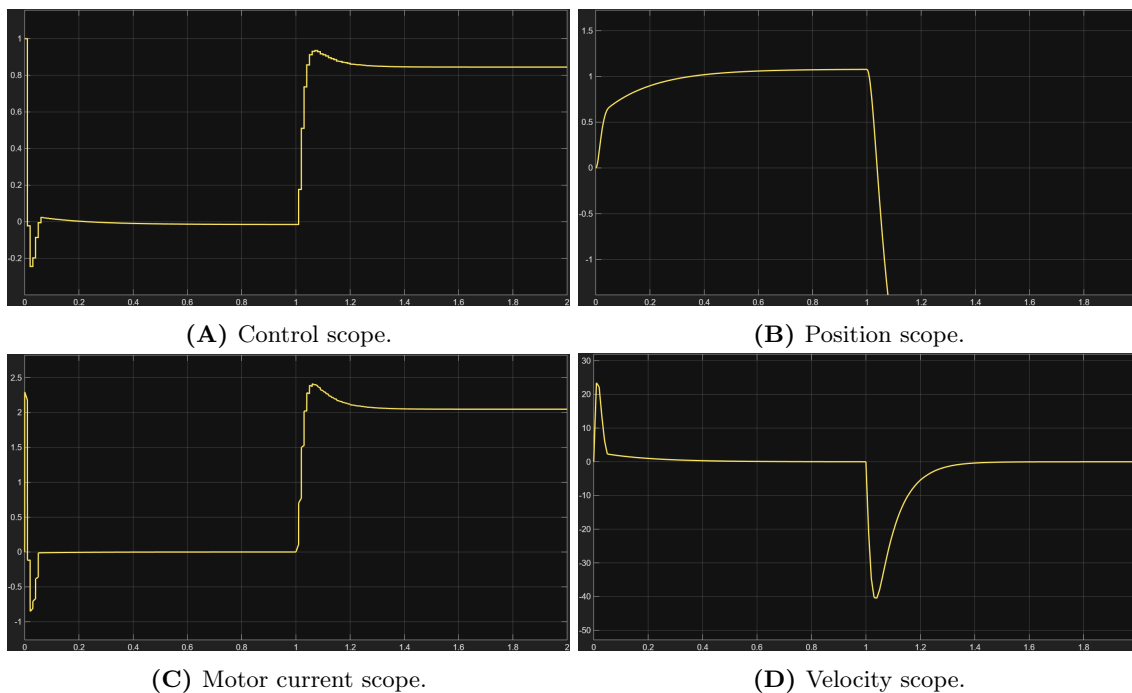


Figure 23: Virtual plant simulation for $T_s = 0.01\text{sec}$.

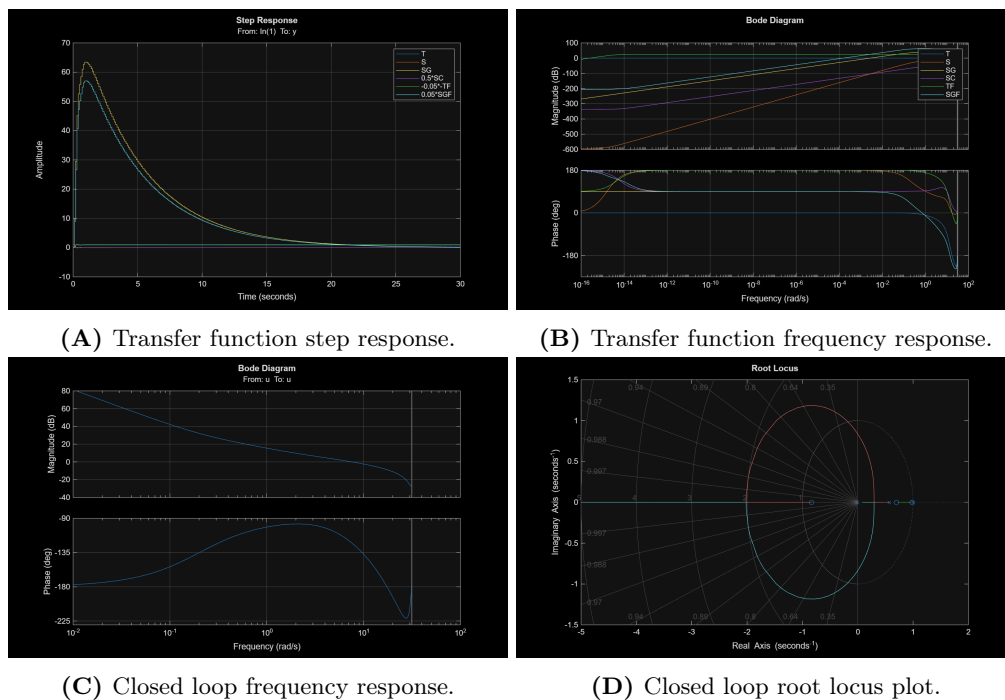


Figure 24: Reduced order system analysis for $T_s = 0.1\text{sec}$.

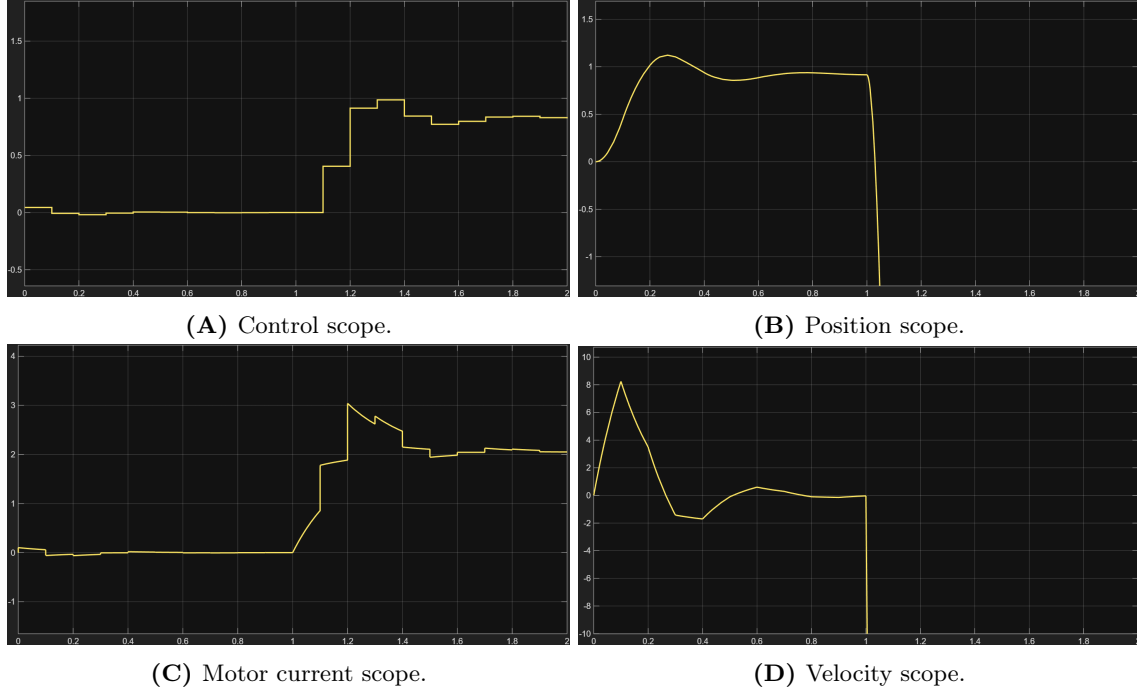


Figure 25: Linear plant simulation for $T_s = 0.1\text{sec}$.

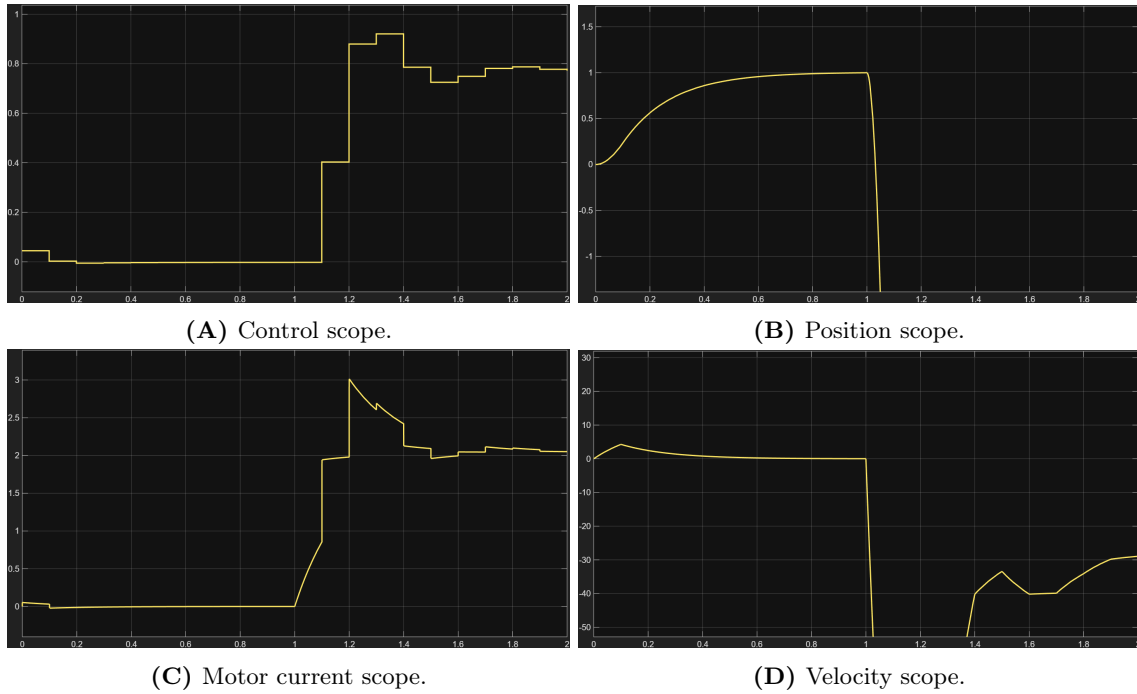


Figure 26: Virtual plant simulation for $T_s = 0.1\text{sec}$.

Discussion

The switch to a tuned PID controller is theoretically able to remove tracking error to step disturbance inputs, as indicated by Table 10. This can be attributed to the inclusion of a new pole at $z - 1$. However, in the

simulated plants we can observe that the high gain values attributed to these transfer functions result in tremendous overshooting as the sampling time decreases, similar to the PD controllers.

PD Controller: $T_p = 0.01\text{sec}$, $T_s = 0.05\text{sec}$

Specifications and Performance Metrics

T_s	0.001 sec	0.01 sec	0.1 sec
PD controller	$\frac{96z-76.8}{z}$	$\frac{2.1z-2.01}{z}$	$\frac{0.031z-0.01934}{z}$

Table 12: Tuned controllers for each sampling time.

T_s (sec)	$\omega_n \frac{\text{rad}}{\text{sec}}$	ζ	T_p (sec)	T_{set} (sec)	P_m ($^\circ$)	G_m
0.001	23.9	0.416	0.01	0.042	36.27	8.5176
0.01	4.31	1	0.05	0.04	63.96	4.1378
0.1	4	1	0.4	0.3	65.34	3.9246

Table 13: Relevant performance metrics.

Steady State Performance

Step Input Steady State Values

T_s (sec)	$\frac{u(k)}{r(k)}$	$\frac{y(k)}{r(k)}$	$\frac{e(k)}{r(k)}$	$\frac{u(k)}{d(k)}$	$\frac{y(k)}{d(k)}$	$\frac{e(k)}{d(k)}$	r_{ss} saturation?	$T_{d,ss}$ saturation?
	SC	T	S	-T	SG	-SG	SC	-TF
0.001	0	1 rad	0	-1	0.0468	-0.0468	yes	yes
0.01	0	1 rad	0	-1	11.1	-11.1	yes	no
0.1	0	1 rad	0	-1	85.7	-85.7	no	no

Table 14: Direct digital PD control steady state values to step inputs.

Sinusoidal Steady State Response

T_s (sec)	$\frac{y(k)}{r(k)}$	$\frac{w(k)}{r(k)}$	$\frac{e(k)}{r(k)}$	$\frac{u(k)}{r(k)}$	$\frac{y(k)}{T_d(k)}$	$\frac{w(k)}{T_d(k)}$	$\frac{e(k)}{T_d(k)}$	$\frac{u(k)}{T_d(k)}$
	T	SG	S	SC	SGF	SF	SGF	TF
0.001	0/0	-25dB/0	-120dB/90°	-90dB/90°	-0.6dB/0	-85dB/270°	-0.6dB/0	25dB/180°
0.01	0/0	21dB/0	-50dB/90°	-70dB/90°	45dB/0	-20dB/270°	45dB/0	25dB/180°
0.1	0/-10°	39dB/-10°	-14dB/82.4°	-50dB/90°	64dB/-20°	12dB/250°	64dB/-20°	25dB/170°

Table 15: Sinusoidal steady state gain and phase values taken from frequency response plot.

$T_s = 0.001\text{sec}$:

$$y(k) \approx \sin(10Tk) + 0$$

$$w(k) \approx 0.0562 \sin(10Tk) + 0$$

$$e(k) \approx 0 + 0$$

$$u(k) \approx 0 + 0.178 \sin(20kT + 180^\circ)$$

$$T_s = 0.01\text{sec:}$$

$$y(k) = \sin(10Tk) + 0.178 \sin(20Tk)$$

$$w(k) \approx 11.22 \sin(10Tk) + 0$$

$$e(k) \approx 0 + 0.178 \sin(20kT + 80^\circ)$$

$$u(k) \approx 0 + 0.178 \sin(20kT + 180^\circ)$$

$$T_s = 0.1\text{sec:}$$

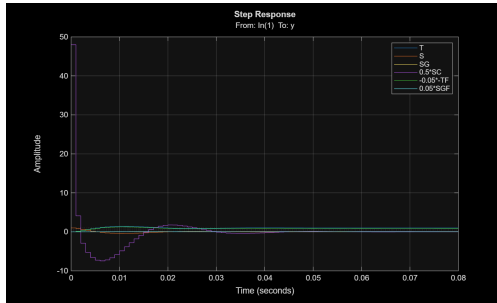
$$y(k) = \sin(10Tk - 10^\circ) + 15.85 \sin(20Tk - 20^\circ)$$

$$w(k) = 89.13 \sin(10Tk - 10^\circ) + 0.0398 \sin(20Tk + 270^\circ)$$

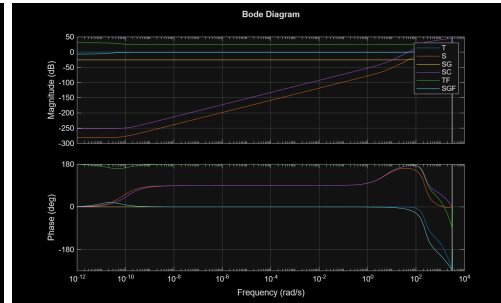
$$e(k) = 0.200 \sin(10Tk + 82.4^\circ) + 15.85 \sin(20kT - 20^\circ)$$

$$u(k) \approx 0 + 0.178 \sin(20kT + 170^\circ)$$

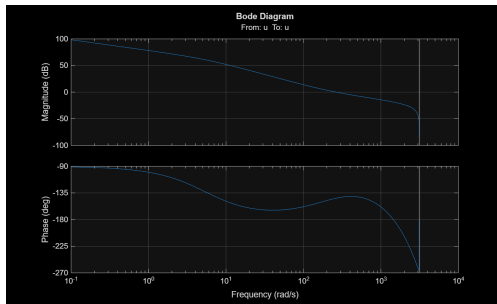
Analysis and Simulation



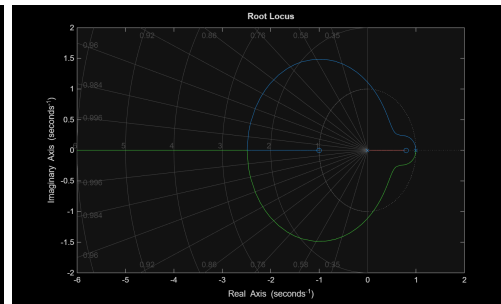
(A) Transfer function step response.



(B) Transfer function frequency response.



(C) Closed loop frequency response.



(D) Closed loop root locus plot.

Figure 27: Reduced order system analysis for $T_s = 0.001\text{sec}$.

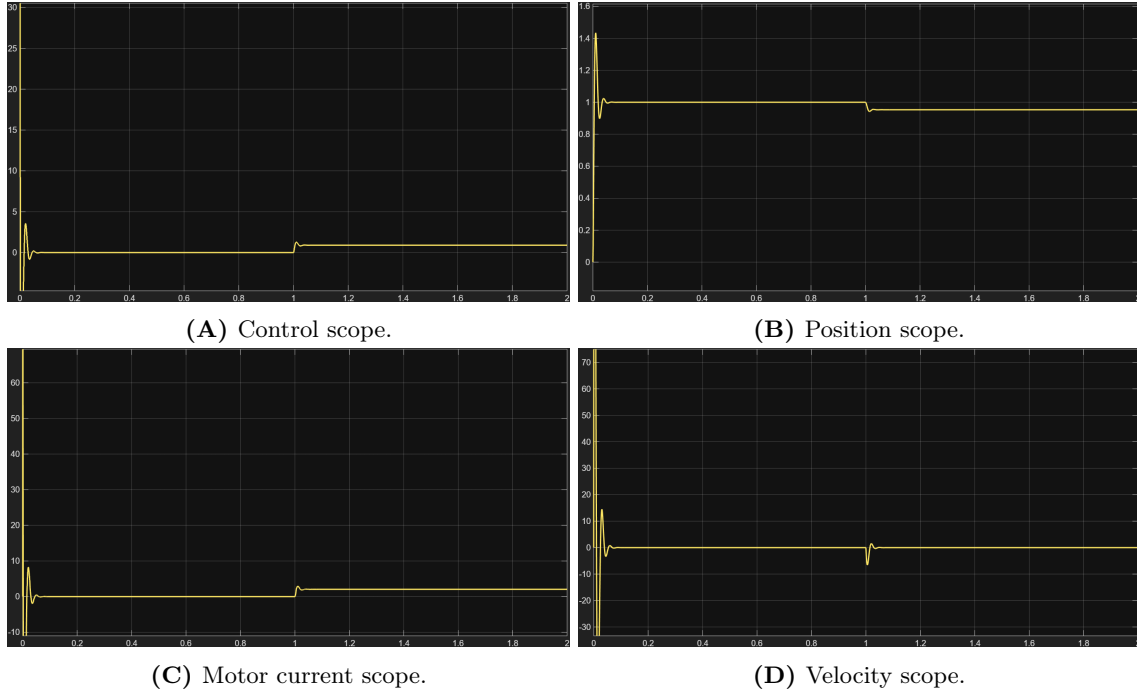
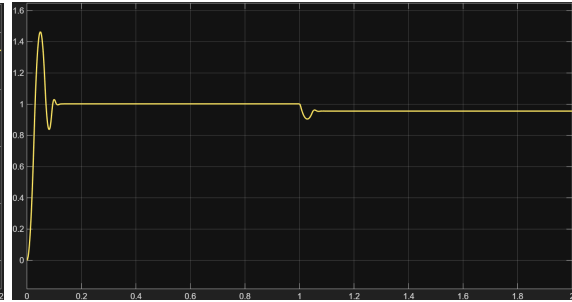


Figure 28: Linear plant simulation for $T_s = 0.001\text{sec}$.



(A) Control scope.



(B) Position scope.

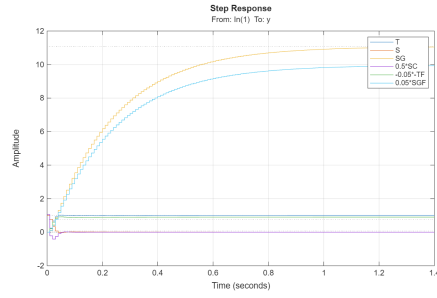


(C) Motor current scope.

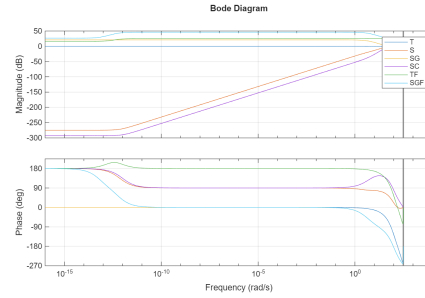


(D) Velocity scope.

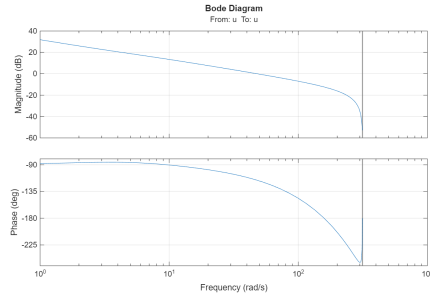
Figure 29: Virtual plant simulation for $T_s = 0.001\text{sec}$.



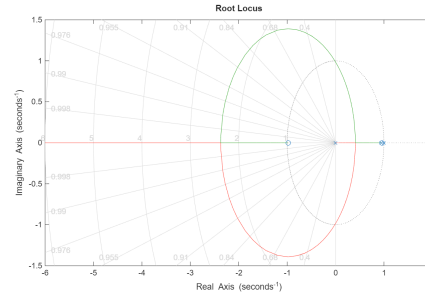
(A) Transfer function step response.



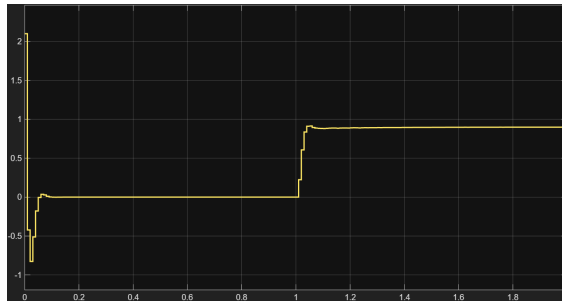
(B) Transfer function frequency response.



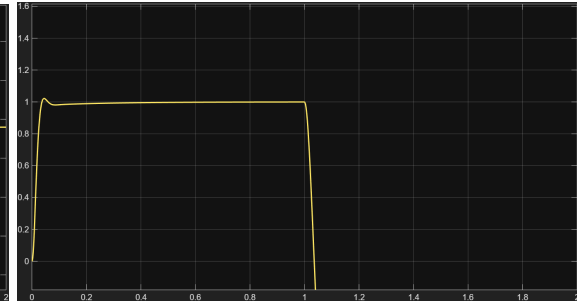
(C) Closed loop frequency response.



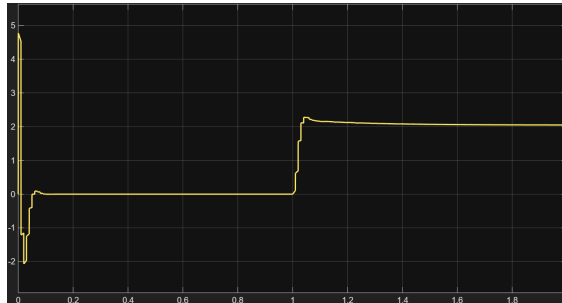
(D) Closed loop root locus plot.

Figure 30: Reduced order system analysis for $T_s = 0.01\text{sec}$.

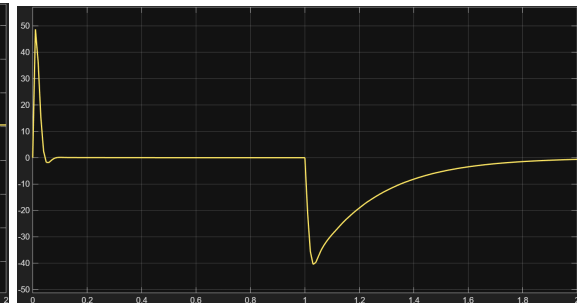
(A) Control scope.



(B) Position scope.



(C) Motor current scope.



(D) Velocity scope.

Figure 31: Linear plant simulation for $T_s = 0.01\text{sec}$.

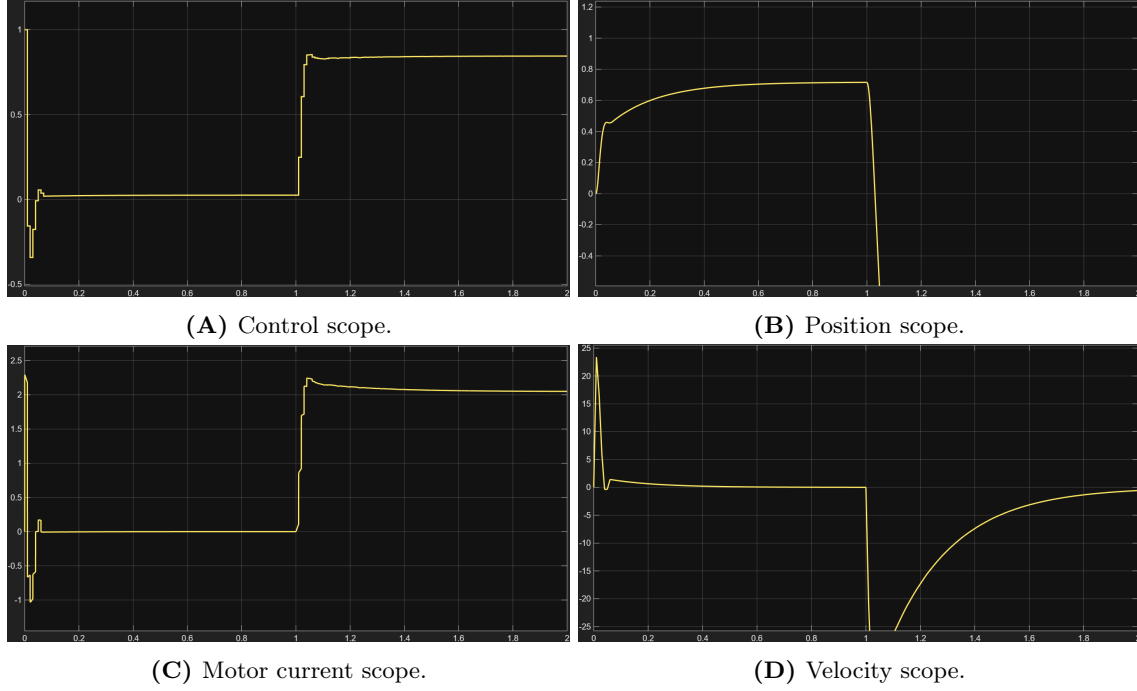


Figure 32: Virtual plant simulation for $T_s = 0.01\text{sec}$.

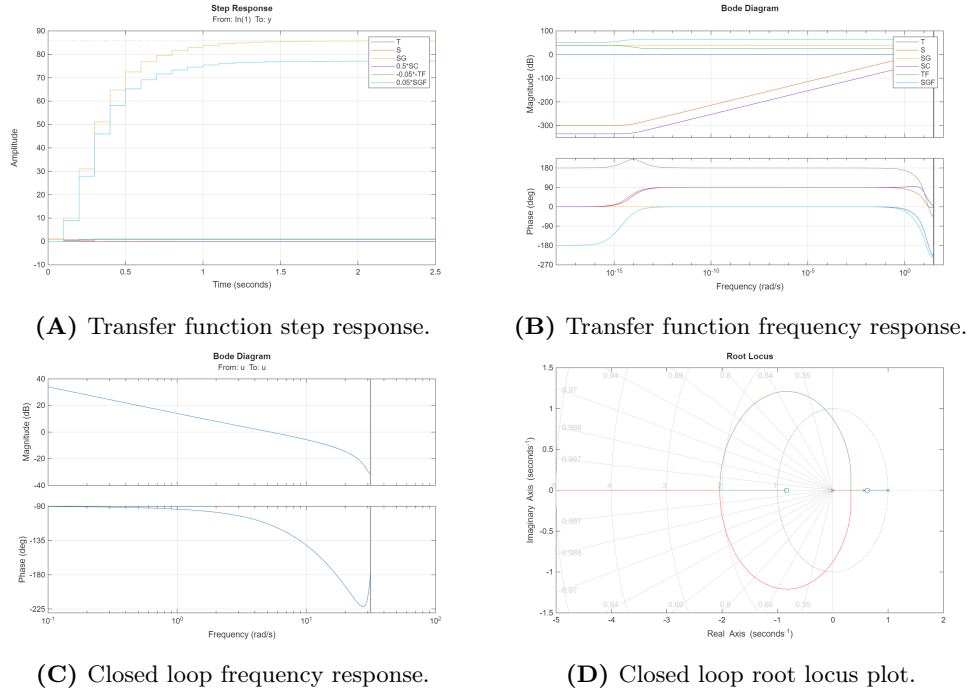


Figure 33: Reduced order system analysis for $T_s = 0.1\text{sec}$.

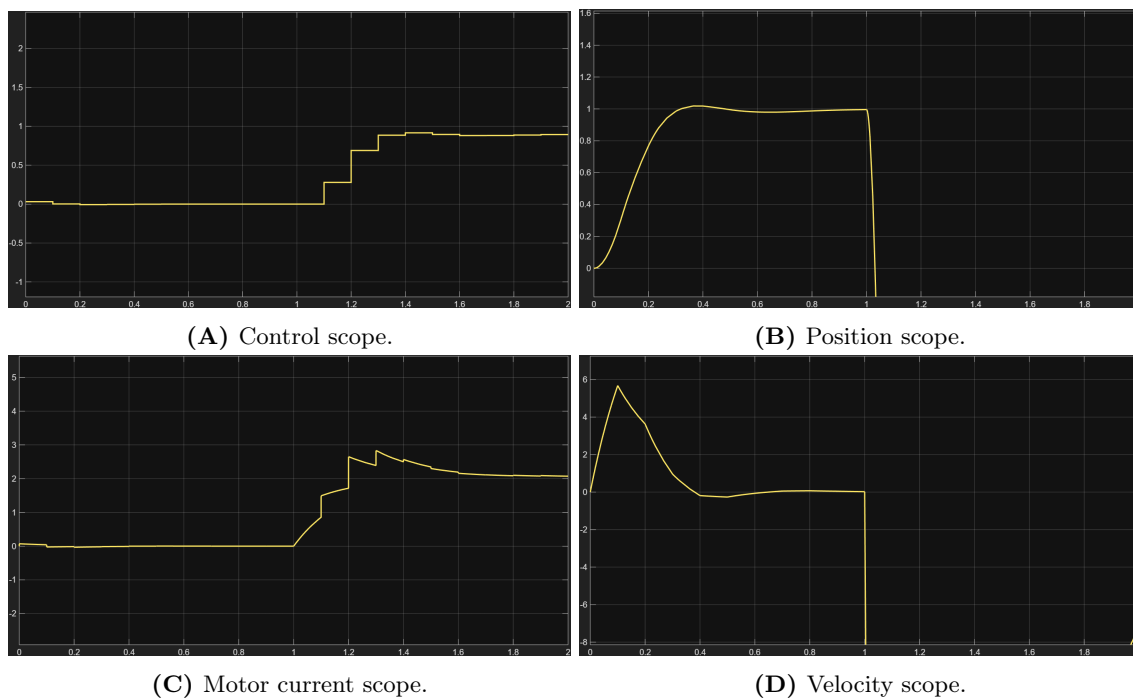


Figure 34: Linear plant simulation for $T_s = 0.1\text{sec}$.

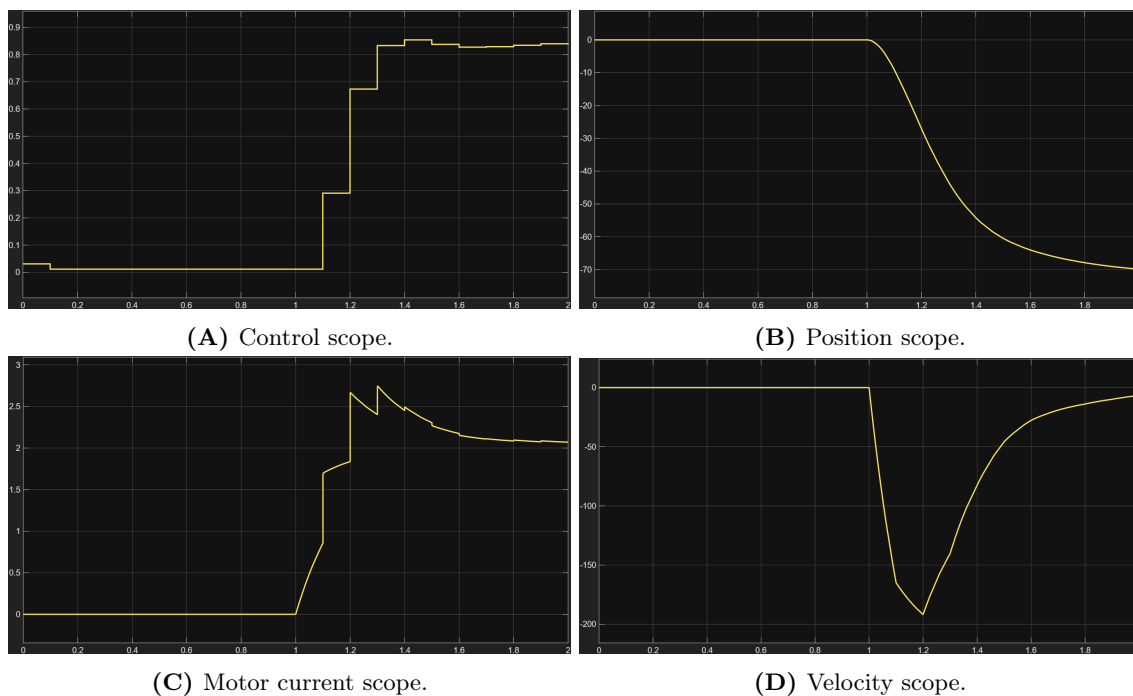


Figure 35: Virtual plant simulation for $T_s = 0.1\text{sec}$.

Discussion

We can observe that changing the design specifications may not be possible. The criteria for these controllers required a quick-acting control scheme. Though it was possible to achieve with the smallest sampling period,

$T_S = 0.001\text{sec}$, neither of the other two controllers were able to meet both design specifications. Moreover, for the largest sampling time $T_s = 0.1\text{sec}$ the virtual plant was unable to generate a proper output for the step reference input, which may be due to the dead zone within the signal mapping.

Something interesting to note is that the demanding peak time was unable to be met analytically by the larger sampling periods. This may be attributed to the fact that the sampling time was greater than or equal to the goal peak time, therefore being largely impractical (if not impossible) to meet through analysis alone.

Assignment 5

Problem 5Exp: $T_r = 0.05\text{sec}$, $PO_{max} = 15\%$

Controllers

T_s	0.001 sec	0.01 sec	0.1 sec
PD controller	$\frac{21.06z-20.64}{z}$	$\frac{1.6z-1.392}{z}$	$\frac{0.05z-0.035}{z}$

Table 1: Tuned PD controllers for each sampling time.

T_s	0.001 sec	0.01 sec	0.1 sec
PD controller	$\frac{27z^2-53.43z+26.44}{z^2-z}$	$\frac{1.8z^2-3.411z+1.611}{z^2-z}$	$\frac{0.045z^2-0.0756z+0.03087}{z^2-z}$

Table 2: Tuned PID controllers for each sampling time.

PD Virtual plant and Experimental Results

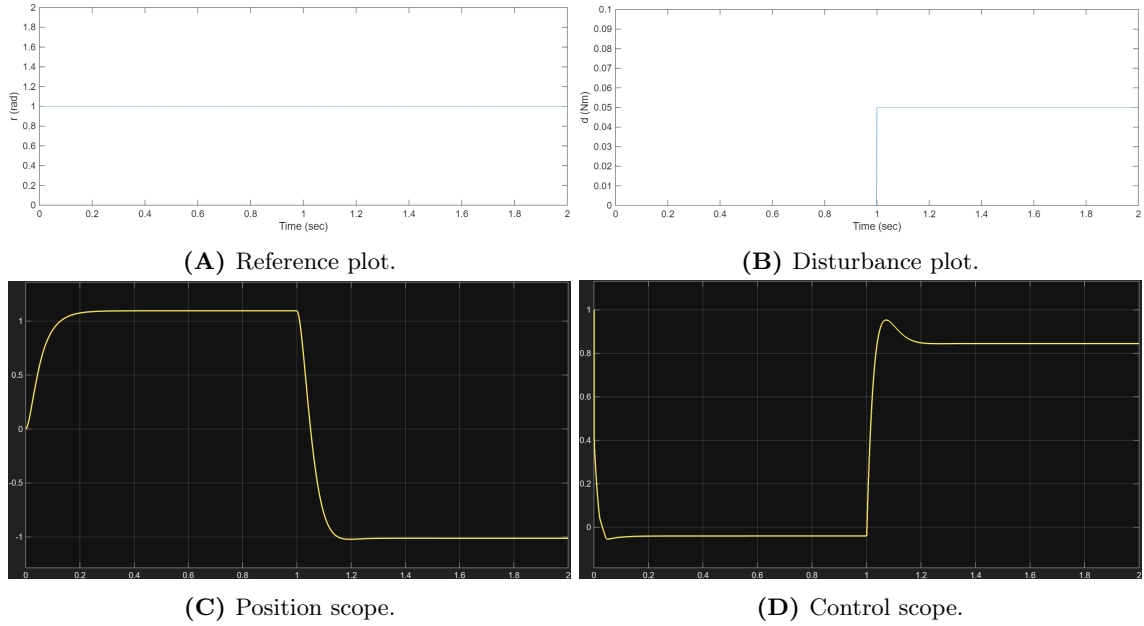


Figure 1: Virtual plant simulation for $T_s = 0.001\text{sec}$.

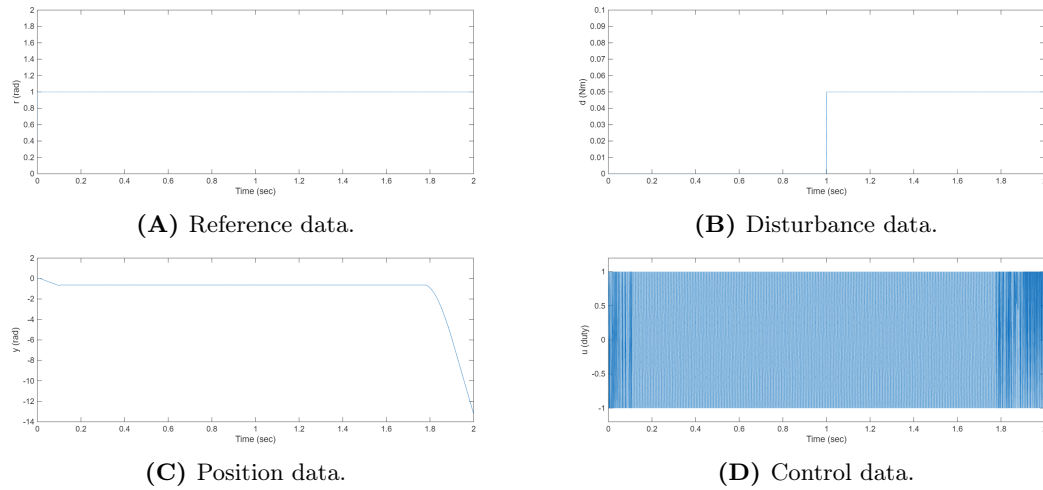


Figure 2: Experimental results for $T_s = 0.001\text{sec}$.

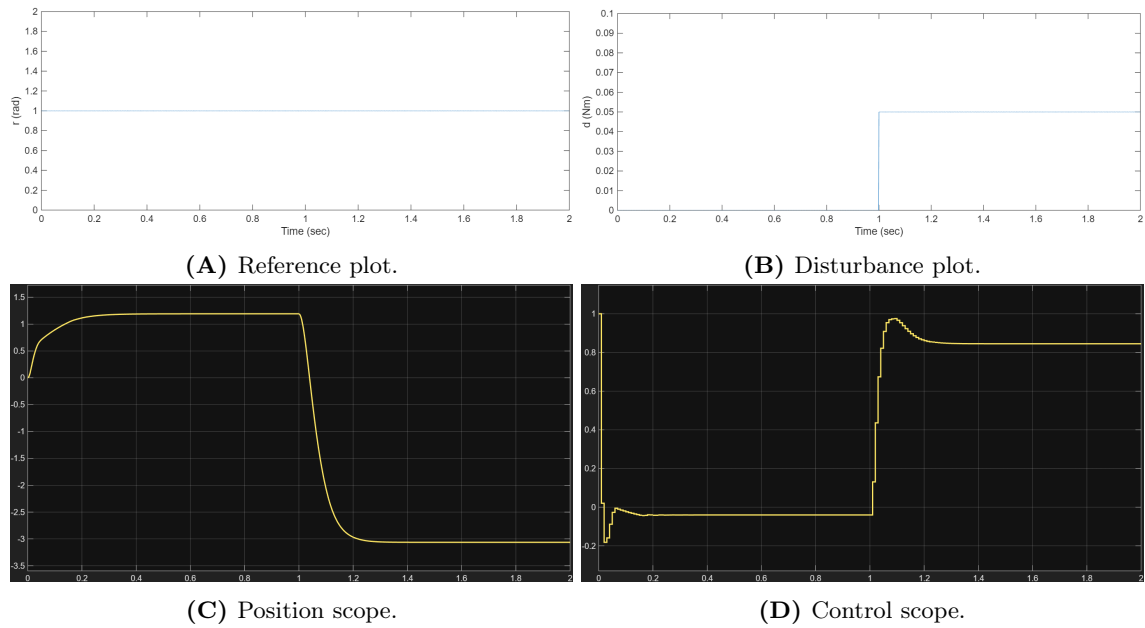


Figure 3: Virtual plant simulation for $T_s = 0.01\text{sec}$.

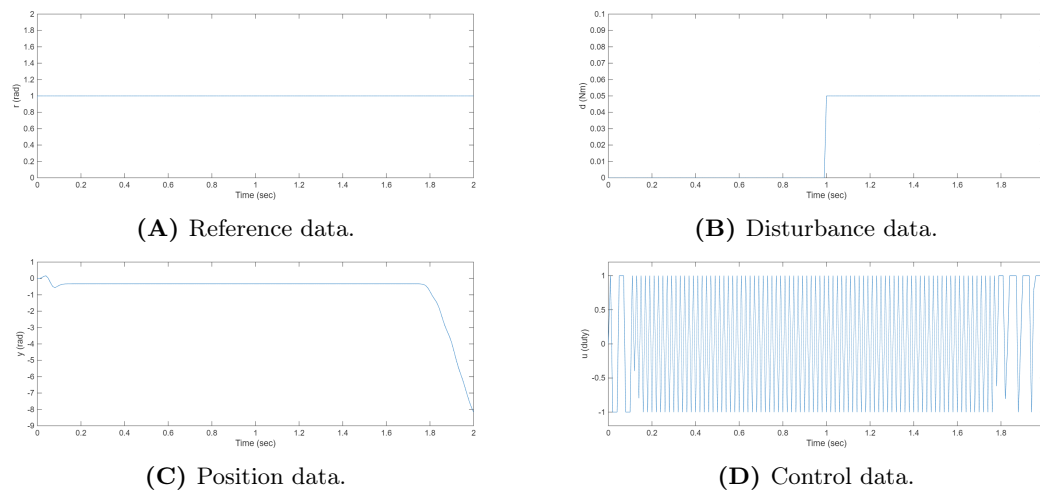


Figure 4: Experimental results for $T_s = 0.01\text{sec}$.

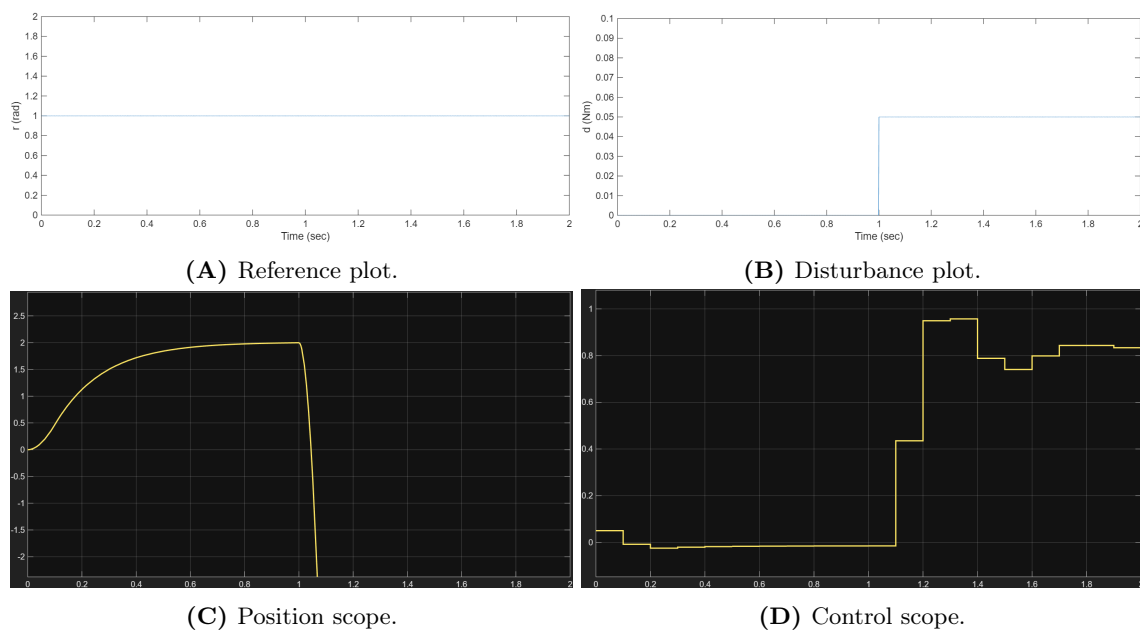


Figure 5: Virtual plant simulation for $T_s = 0.1\text{sec}$.

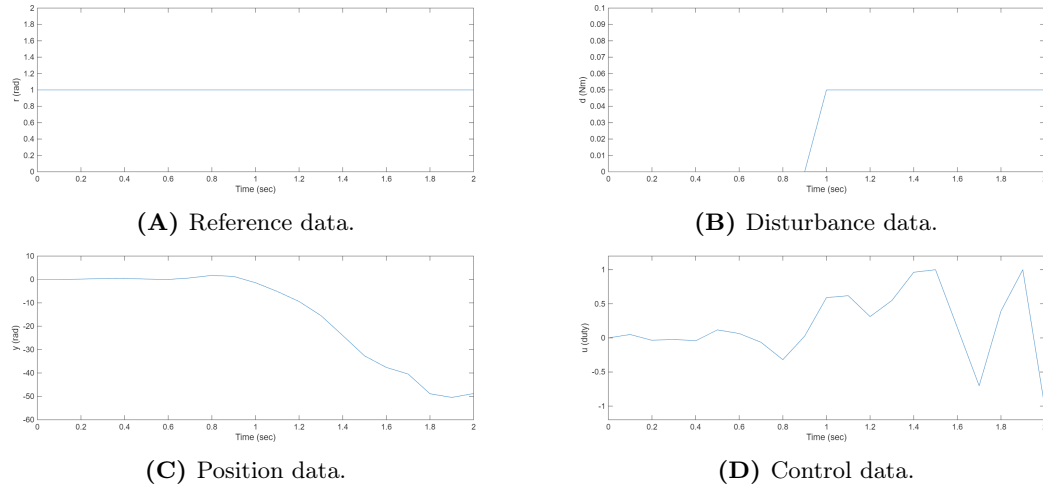


Figure 6: Experimental results for $T_s = 0.1\text{sec}$.

PID Virtual plant and Experimental Results

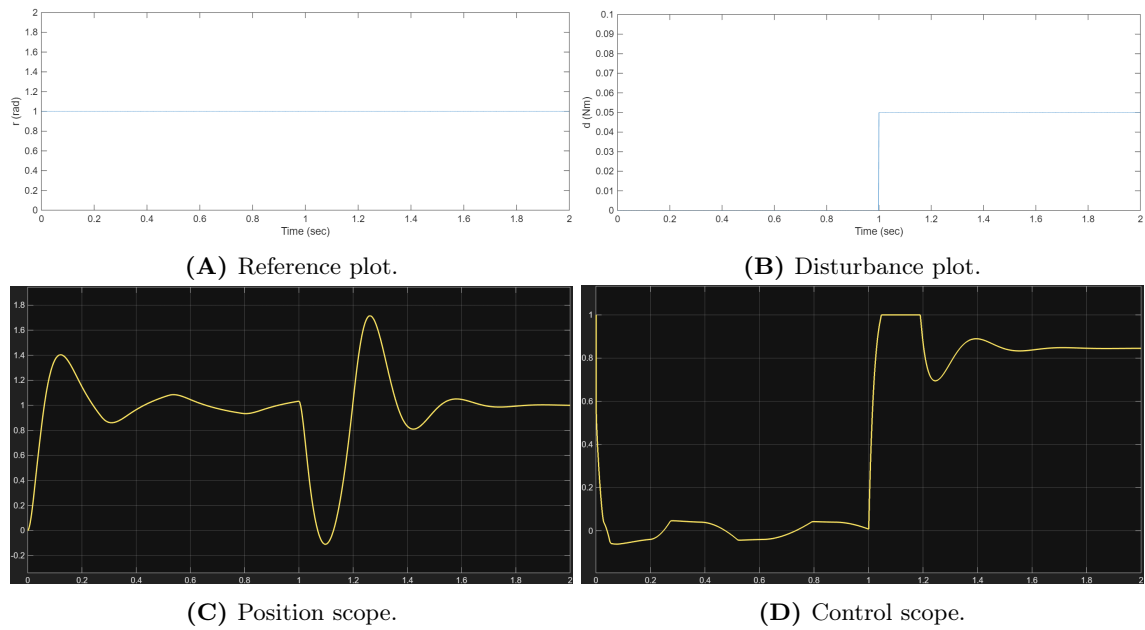
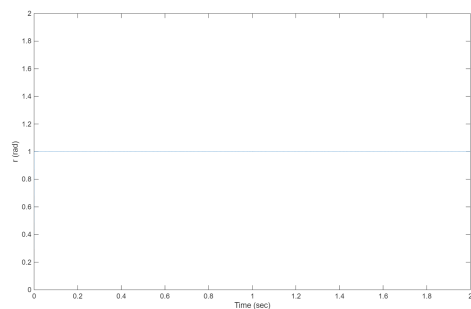
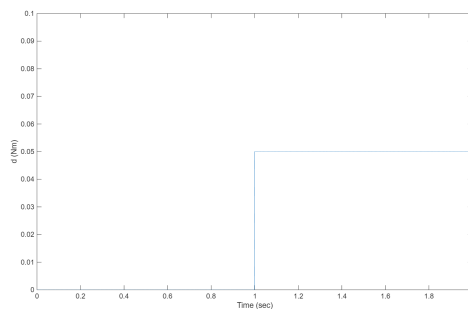


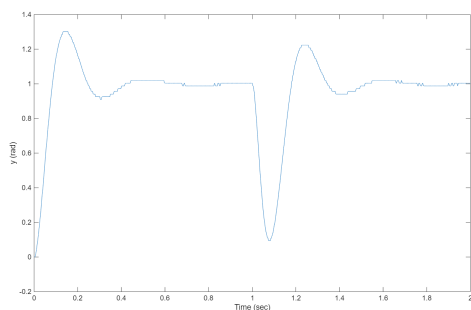
Figure 7: Virtual plant simulation for $T_s = 0.001\text{sec}$.



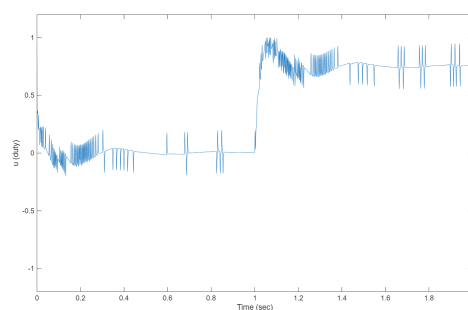
(A) Reference data.



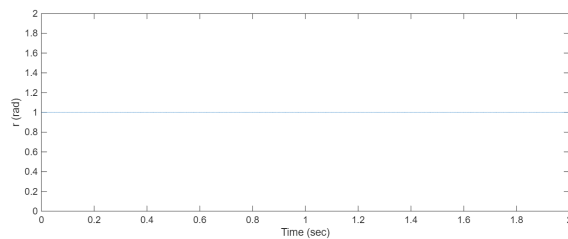
(B) Disturbance data.



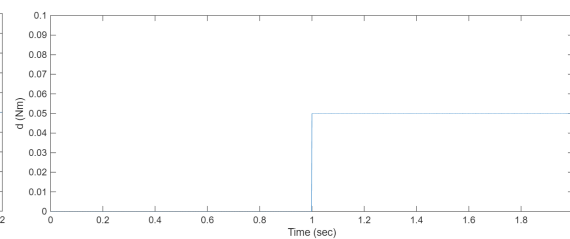
(C) Position data.



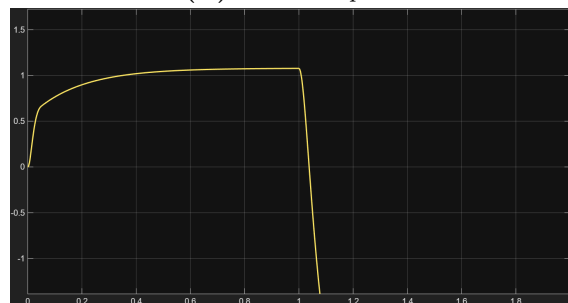
(D) Control data.

Figure 8: Experimental results for $T_s = 0.001\text{sec}$.

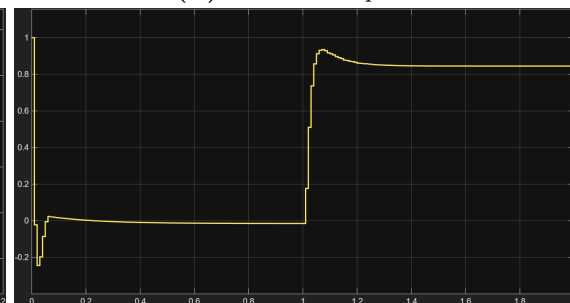
(A) Reference plot.



(B) Disturbance plot.

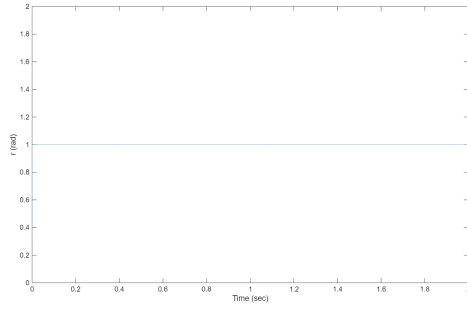


(C) Position scope.

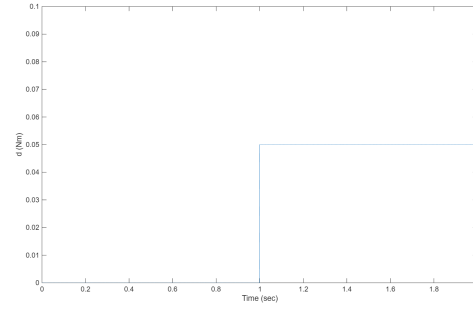


(D) Control scope.

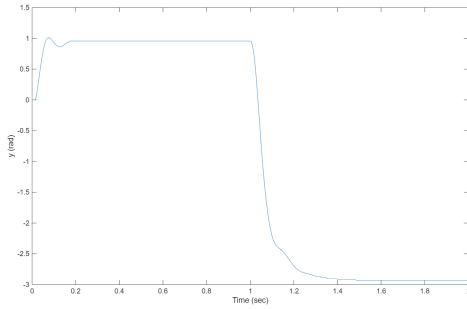
Figure 9: Virtual plant simulation for $T_s = 0.01\text{sec}$.



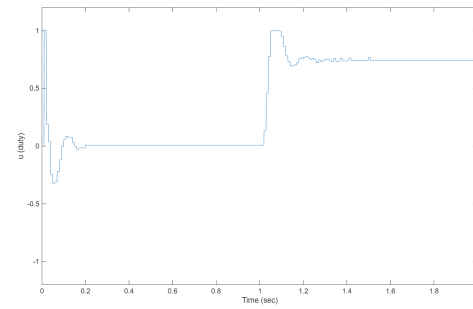
(A) Reference data.



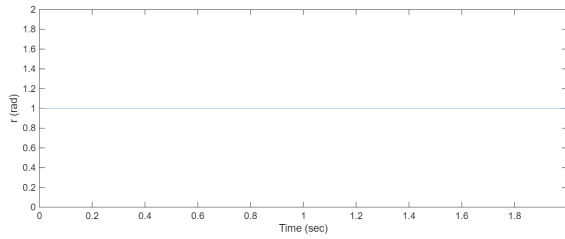
(B) Disturbance data.



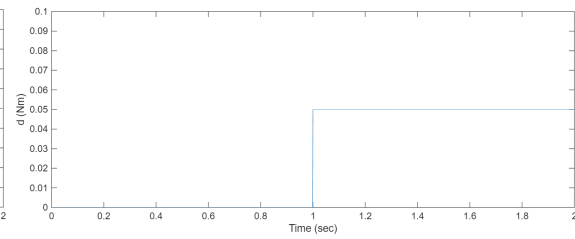
(C) Position data.



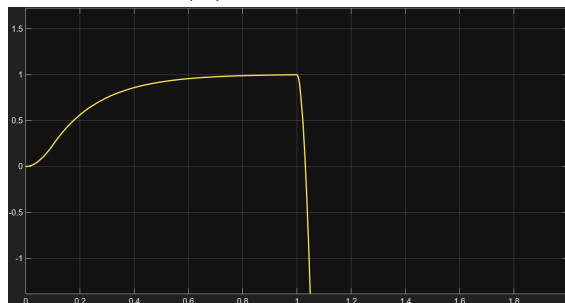
(D) Control data.

Figure 10: Experimental results for $T_s = 0.01\text{sec}$.

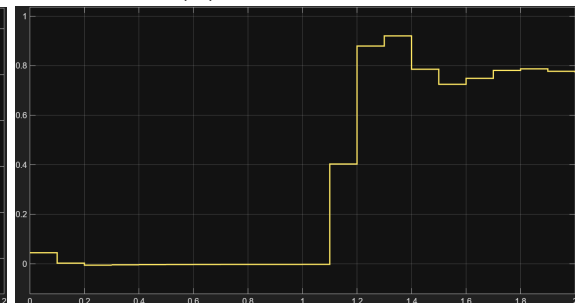
(A) Reference plot.



(B) Disturbance plot.



(C) Position scope.



(D) Control scope.

Figure 11: Virtual plant simulation for $T_s = 0.1\text{sec}$.

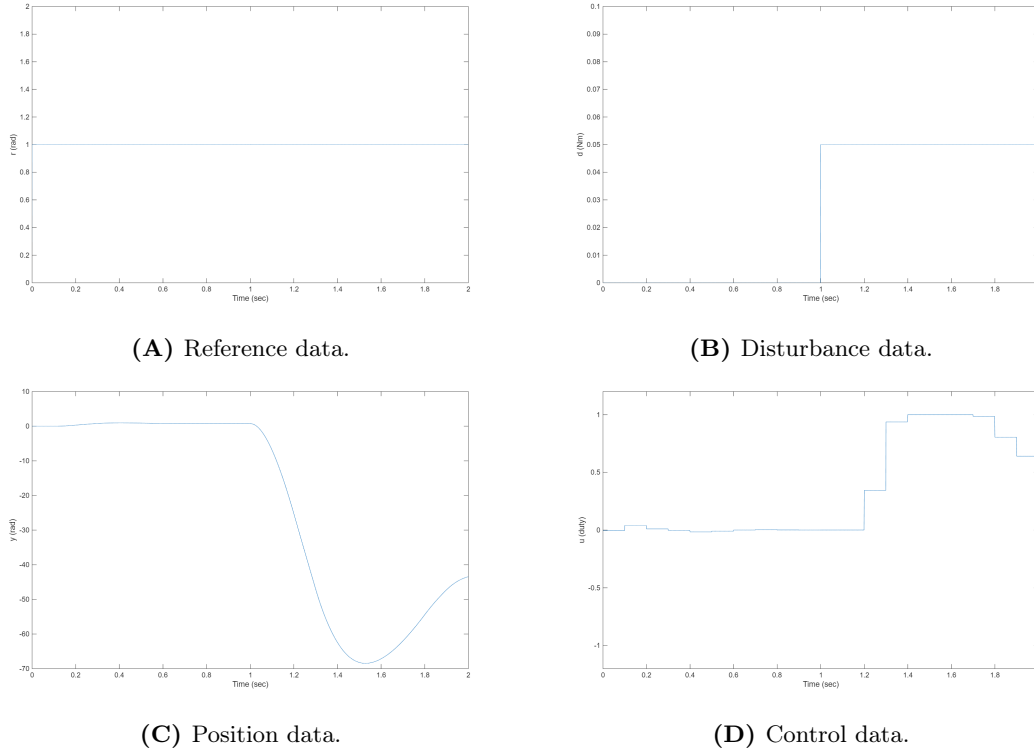


Figure 12: Experimental results for $T_s = 0.1\text{sec}$.

Discussion

I don't quite understand what happened with my experiments. The first thing we can see is the typically static behavior of the output plots. The position of the motor does not move until the disturbance input is applied, if at all. It seems that the reference input does not produce any visible output. My gut feeling is that there was some error on my behalf when implementing the controller into the setup. However, if there was something wrong with the experiment itself, it could be possible that the reference input was too low of a magnitude and only produced a control signal within the dead-zone of the physical system. If there is any similarity to the virtual plant position, it would be the eventual extreme steady state gain when the disturbance input is applied, where the position appears to react extremely to this small input.

The next observation would be with the control data retrieved from the experimental results. It appears that the control signal alternates between its two saturation points (-1 and 1) at or near the sampling frequency. To me, I can draw similarities to the behavior of a system experiencing a positive feedback loop or just system divergence in general. Further evidence can be seen with the PD controller with the largest sampling period, where we see that the control signal oscillates and appears to diverge by the end of the plot. Perhaps this is evidence that the virtual plant that was developed is not the most accurate representation of the physical system, where even our tuned controllers are unable to create a stable output in the physical setup.

Midterm Project

Problem 5Exp: $T_r = 0.05\text{sec}$, $PO_{max} = 15\%$

Controllers

T_s	0.001 sec	0.01 sec	0.1 sec
PID controller	$\frac{27z^2 - 53.43z + 26.44}{z^2 - z}$	$\frac{1.8z^2 - 3.411z + 1.611}{z^2 - z}$	$\frac{0.045z^2 - 0.0756z + 0.03087}{z^2 - z}$

Table 1: Tuned PID controllers for each sampling time.

Virtual plant and Experimental Results

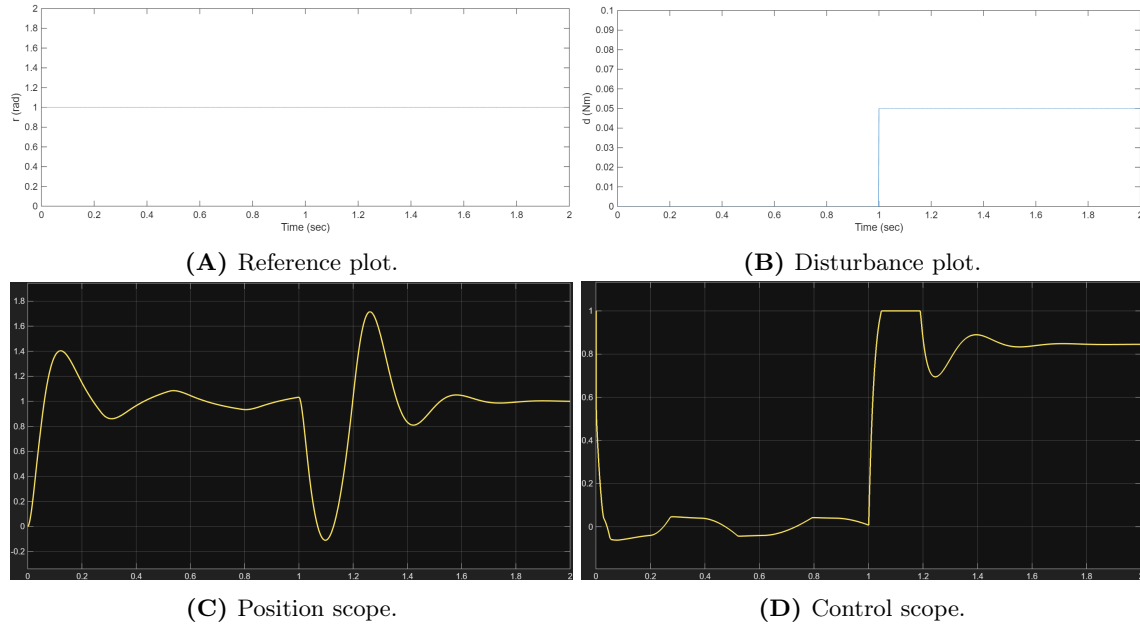
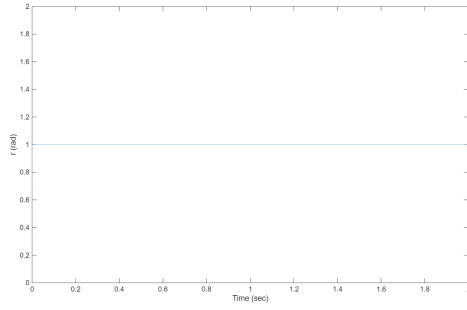
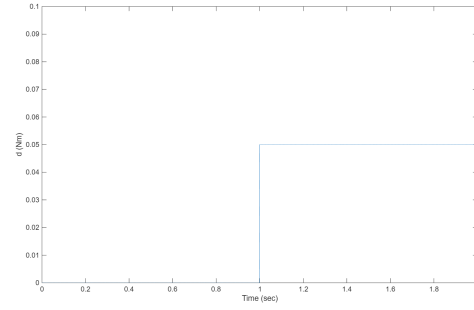


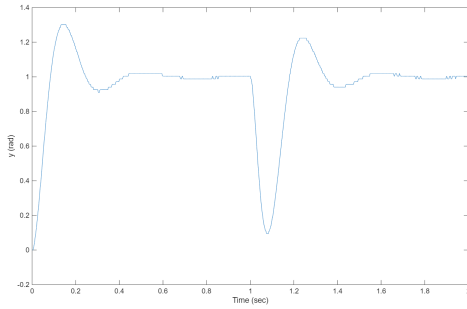
Figure 1: Virtual plant simulation for $T_s = 0.001\text{sec}$.



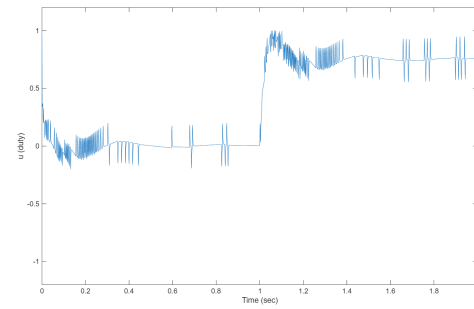
(A) Reference data.



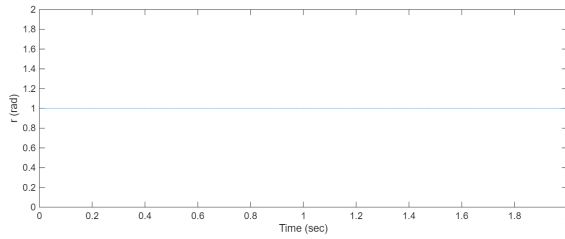
(B) Disturbance data.



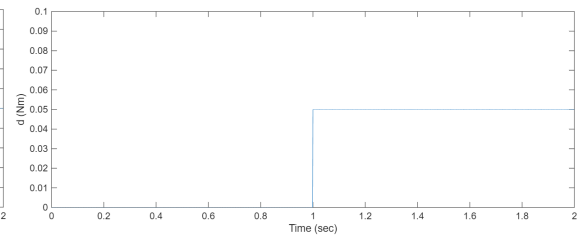
(C) Position data.



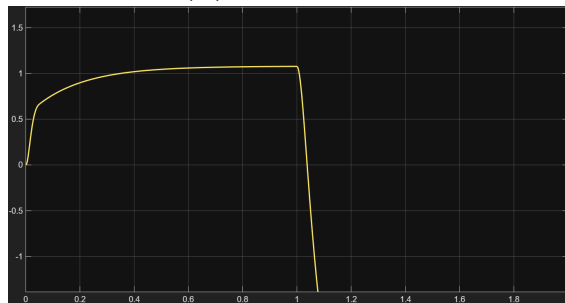
(D) Control data.

Figure 2: Experimental results for $T_s = 0.001\text{sec}$.

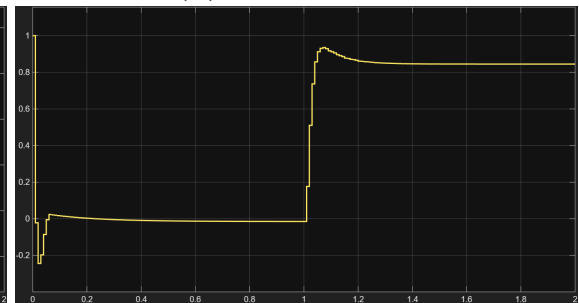
(A) Reference plot.



(B) Disturbance plot.

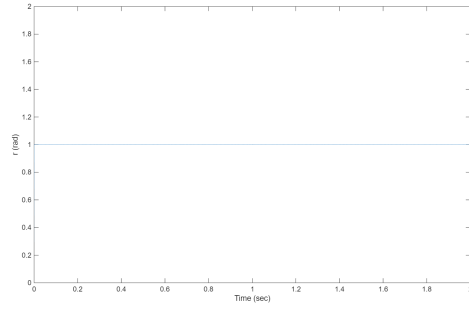


(C) Position scope.

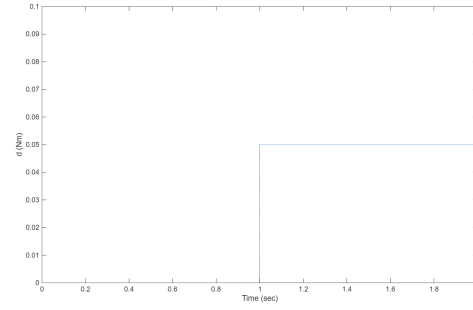


(D) Control scope.

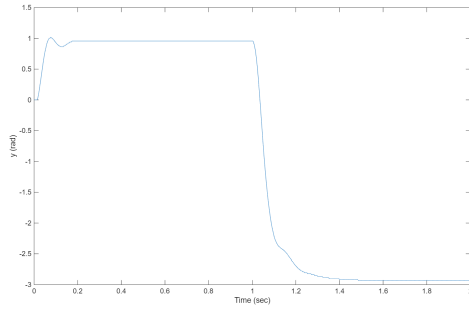
Figure 3: Virtual plant simulation for $T_s = 0.01\text{sec}$.



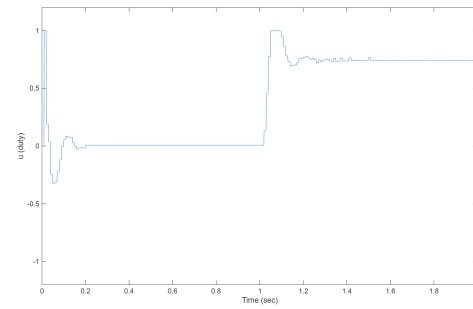
(A) Reference data.



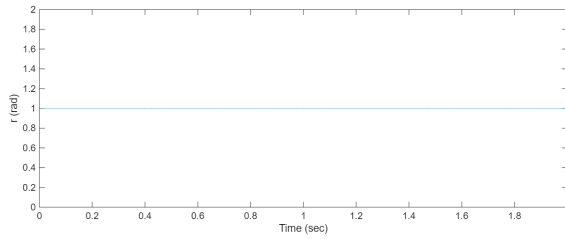
(B) Disturbance data.



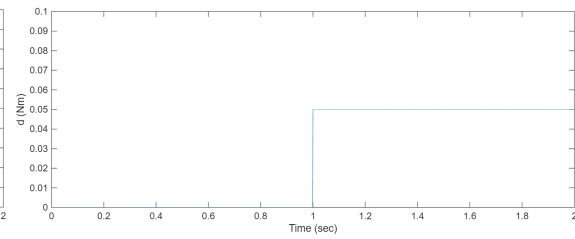
(C) Position data.



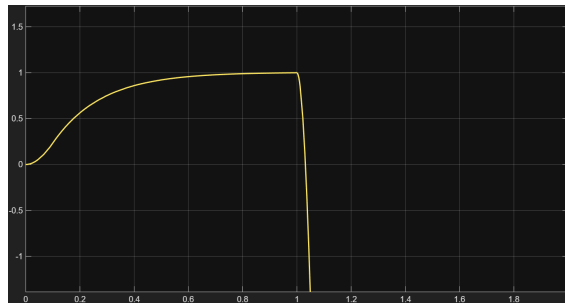
(D) Control data.

Figure 4: Experimental results for $T_s = 0.01\text{sec}$.

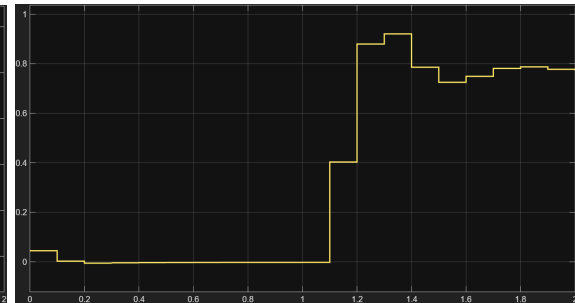
(A) Reference plot.



(B) Disturbance plot.



(C) Position scope.



(D) Control scope.

Figure 5: Virtual plant simulation for $T_s = 0.1\text{sec}$.

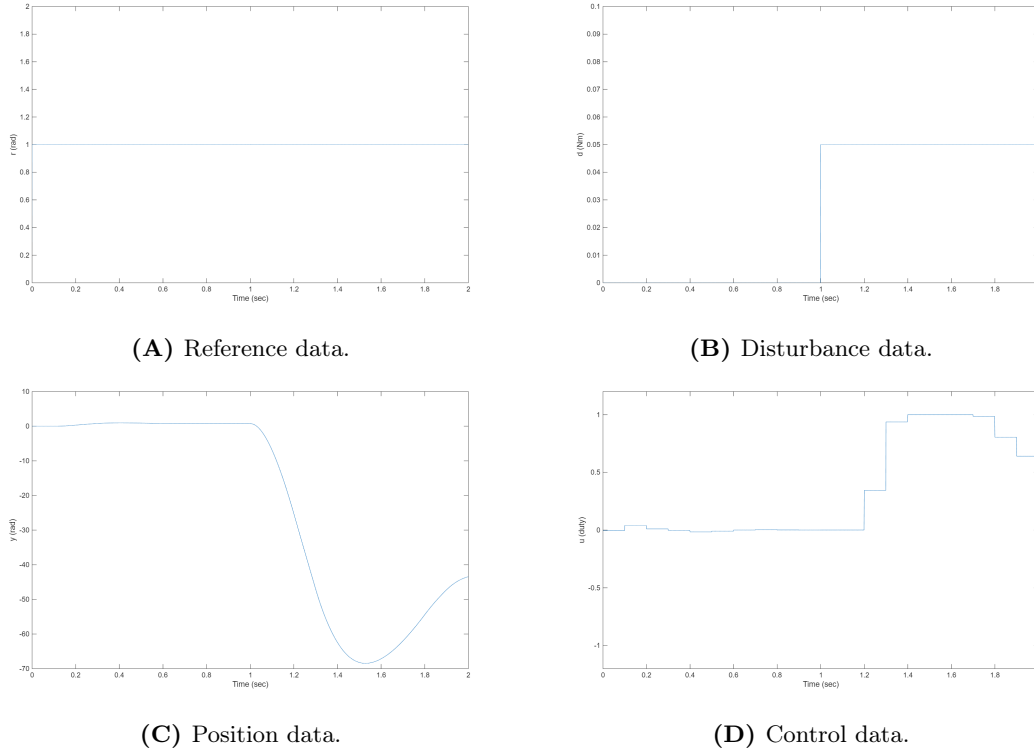


Figure 6: Experimental results for $T_s = 0.1\text{sec}$.

Discussion

This looks pretty good to me. It appears that the virtual plant and the physical system are able to match quite closely, indicating that the plant model and the plant parameters that we identified are quite accurate.

We can see that there is zero error in response to a step reference input. However, the large steady state gain in response to a step disturbance input is also present. The control signal is largely unsaturated, indicating that this error is not due to a discrepancy in the virtual plant model. We can see that the larger sampling period requires a more a transfer function that is more sensitive to input disturbances, as indicated by the Simulink simulations and the physical experiments.

Assignment 6

Problem 11

Laptop 3				
R	κ	τ	K_m	b
10.9	451	0.195	0.0539	-0.000149

Sampling time $T_s = 0.001$ sec

Plant model and State feedback properties

The plant model with identified parameters:

$$\frac{2312.8}{s(s + 11.85)}$$

The plant model in discrete-time with zero order hold and sampling:

$$\frac{0.0011519(z + 0.9961)}{(z - 1)(z - 0.9882)}$$

The state feedback properties are:

- Natural frequency, $\omega_n = 36$ rad/sec
- Damping ratio, $\zeta = 0.5169$
- Continuous poles: $s = -18.61 + 30.82i, -18.61 - 30.82i$
- Discrete poles: $z = 0.9811 + 0.0302i, 0.9811 - 0.0302i$

The state feedback gain, K , and reference gain, N , are:

$$K = \begin{bmatrix} 0.5533 & 0.0110 \end{bmatrix}$$

$$N = 0.5533$$

State observer properties

For observer dynamics of $\omega_n = 100$ Hz and $\zeta = 1$:

- Natural frequency, $\omega_n = 628.31$ rad/sec
- Damping ratio, $\zeta = 1$
- Continuous poles: $s = -628.31, -628.31$
- Discrete poles: $z = 0.5335, 0.5335$

The observer gain L is:

$$\begin{bmatrix} 0.9212 \\ 208.0082 \end{bmatrix}$$

Closed loop transfer functions

$$T_{yr} = \frac{0.00063728(z + 0.9961)}{(z^2 - 1.962z + 0.9635)}$$

$$T_{ur} = \frac{0.55326(z - 1)(z - 0.9882)}{(z^2 - 1.962z + 0.9635)}$$

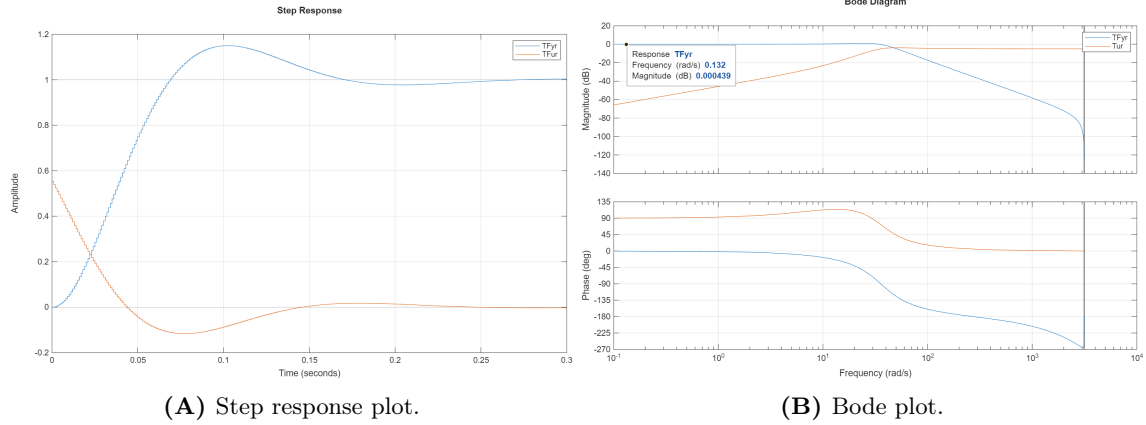


Figure 1: Step and frequency response of control and output to reference.

$$T_{yd} = \frac{0.0011519(z + 0.9961)(z^2 - 1.041z + 0.2806)}{(z - 0.5335)^2(z^2 - 1.041z + 0.9635)}$$

$$T_{ud} = \frac{-0.0032329(z + 0.9961)(z - 0.9571)}{(z - 0.5335)^2(z^2 - 1.962z + 0.9635)}$$

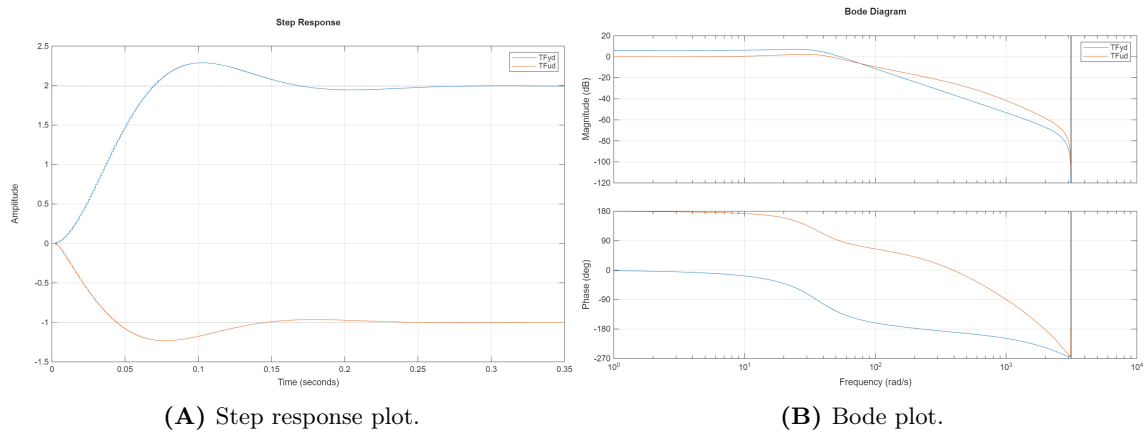


Figure 2: Step and frequency response of control and output to disturbance.

Loop transfer functions

$$L(z) = \frac{0.0032329(z + 0.9961)(z - 0.9571)}{(z - 1)(z - 0.9882)(z^2 - 1.041z + 0.2806)}$$

$$S = \frac{(z - 1)(z - 0.9882)(z^2 - 1.041z + 0.2806)}{(z - 0.5335)^2(z^2 - 1.962z + 0.9635)}$$

- $G_m = 19.3997$
- $P_m = 49.5646$
- $\omega_{cp} = 394.0835$
- $\omega_{eg} = 38.3523$
- $S_{peak} = 1.2882$
- $VGM = 4.4692$
- $\omega_{peak} = 52.1$

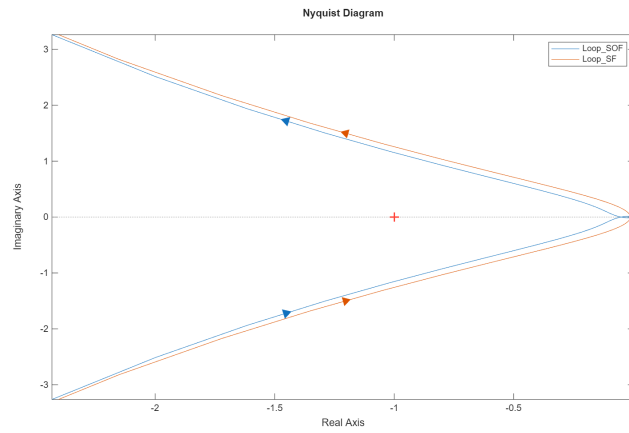


Figure 3: Nyquist plot of $L(z)$

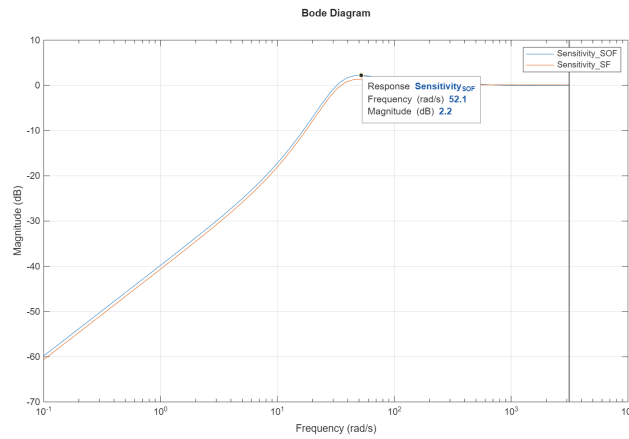


Figure 4: Magnitude plot of S

Simulation

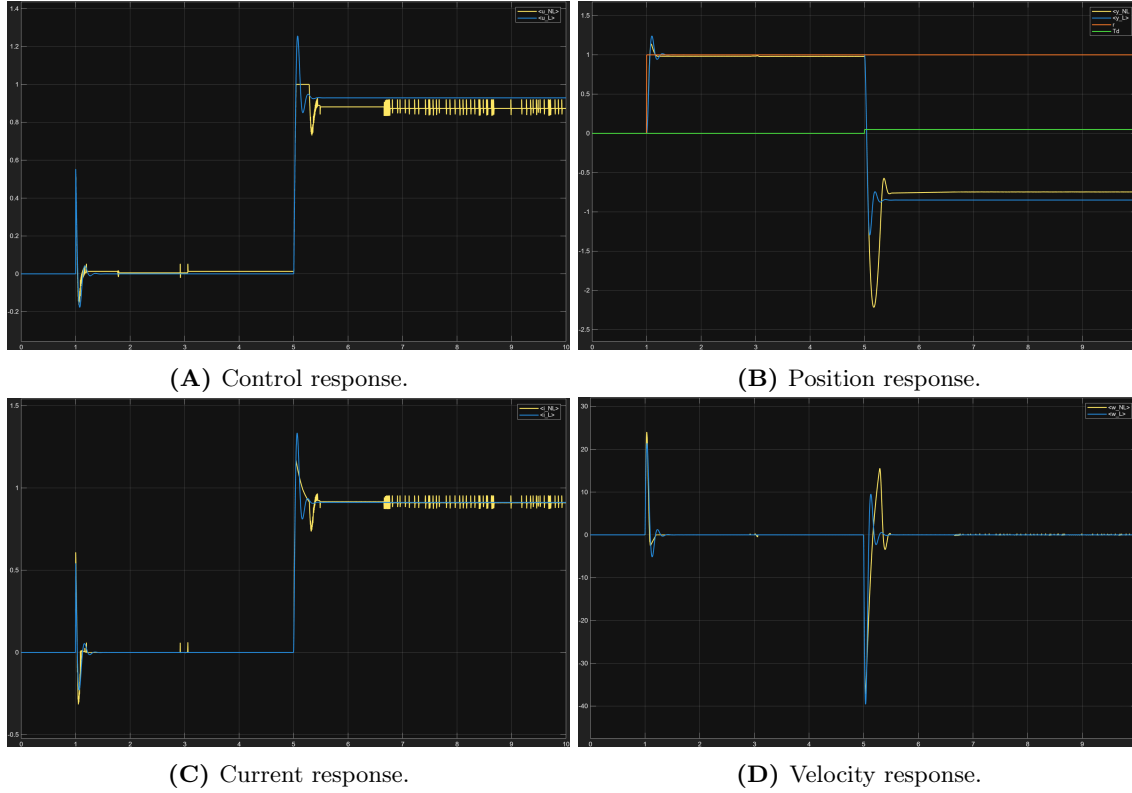


Figure 5: Simulation for $T_s = 0.001\text{sec}$.

Sampling time $T_s = 0.01$ sec

Plant model and State feedback properties

The plant model with identified parameters:

$$\frac{2312.8}{s(s + 11.85)}$$

The plant model in discrete-time with zero order hold and sampling:

$$\frac{0.1112(z + 0.9613)}{(z - 1)(z - 0.8882)}$$

The state feedback properties are:

- Natural frequency, $\omega_n = 36$ rad/sec
- Damping ratio, $\zeta = 0.5169$
- Continuous poles: $s = -18.61 + 30.82i, -18.61 - 30.82i$
- Discrete poles: $z = 0.7911 + 0.2518i, 0.7911 - 0.2518i$

The state feedback gain, K , and reference gain, N , are:

$$K = \begin{bmatrix} 0.4908 & 0.0115 \end{bmatrix}$$

$$N = 0.4908$$

State observer properties

For observer dynamics of $\omega_n = 100$ Hz and $\zeta = 1$:

- Natural frequency, $\omega_n = 628.31$ rad/sec
- Damping ratio, $\zeta = 1$
- Continuous poles: $s = -628.31, -628.31$
- Discrete poles: $z = 0.0019, 0.0019$

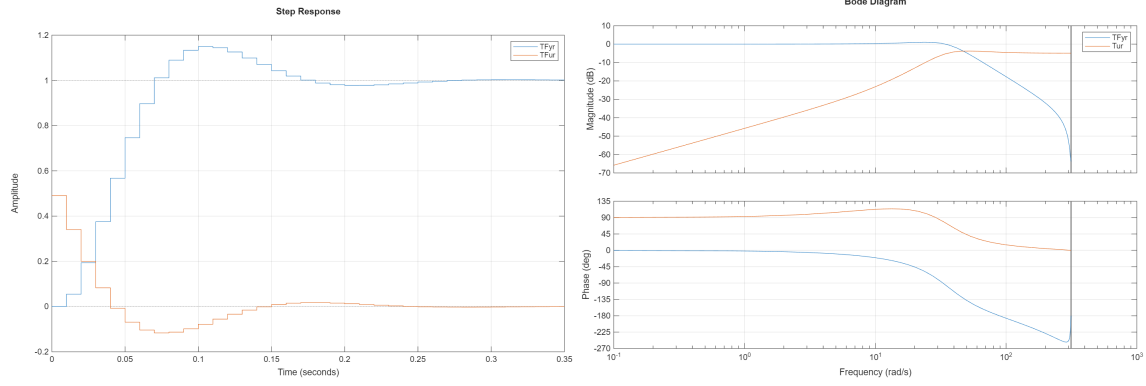
The observer gain L is:

$$\begin{bmatrix} 1.8845 \\ 83.3141 \end{bmatrix}$$

Closed loop transfer functions

$$T_{yr} = \frac{0.0054584(z + 0.9613)}{(z^2 - 1.582 + 0.6892)}$$

$$T_{ur} = \frac{0.49084(z - 1)(z - 0.8882)}{(z^2 - 1.582 + 0.6892)}$$



(A) Step response plot.

(B) Bode plot.

Figure 6: Step and frequency response of control and output to reference.

$$T_{yd} = \frac{0.11121(z + 0.009613)(z^2 + 0.3023z + 0.1681)}{(z - 0.001867)^2(z^2 - 1.582 + 0.6892)}$$

$$T_{ud} = \frac{-0.2097(z + 0.9613)(z - 0.7407)}{(z - 0.001867)^2(z^2 - 1.582 + 0.6892)}$$

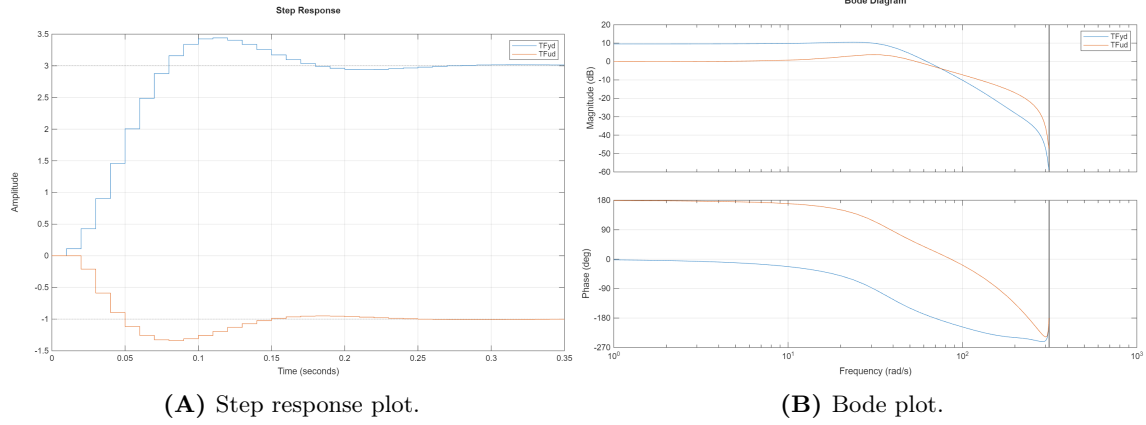


Figure 7: Step and frequency response of control and output to disturbance.

Loop transfer functions

$$L(z) = \frac{0.20979613(z - 0.7407)}{(z - 1)(z - 0.8882)(z^2 + 0.3023z + 0.1681)}$$

$$S = \frac{(z - 1)(z - 0.8882)(z^2 + 0.3023z + 0.1681)}{(z^2 - 0.003735z + 3.487 \times 10^{-6})(z^2 - 1.582z + 0.6892)}$$

- $G_m = 2.8685$
- $P_m = 38.1155$
- $\omega_{cp} = 85.9448$
- $\omega_{eg} = 33.1191$
- $S_{peak} = 1.8387$
- $VGM = 2.1924$
- $\omega_{peak} = 47.3$

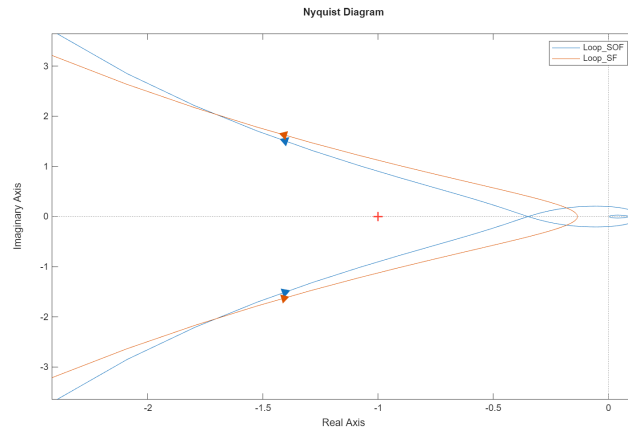
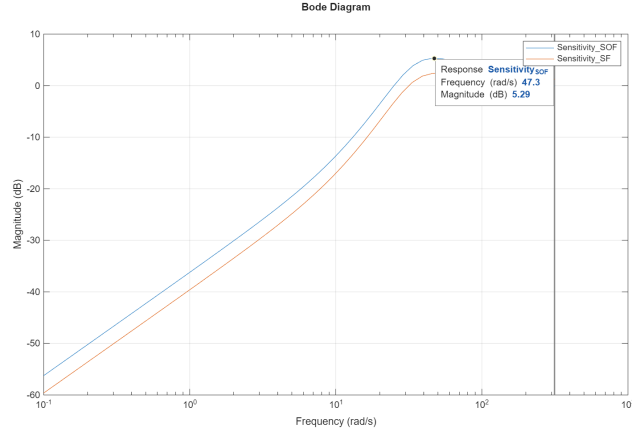
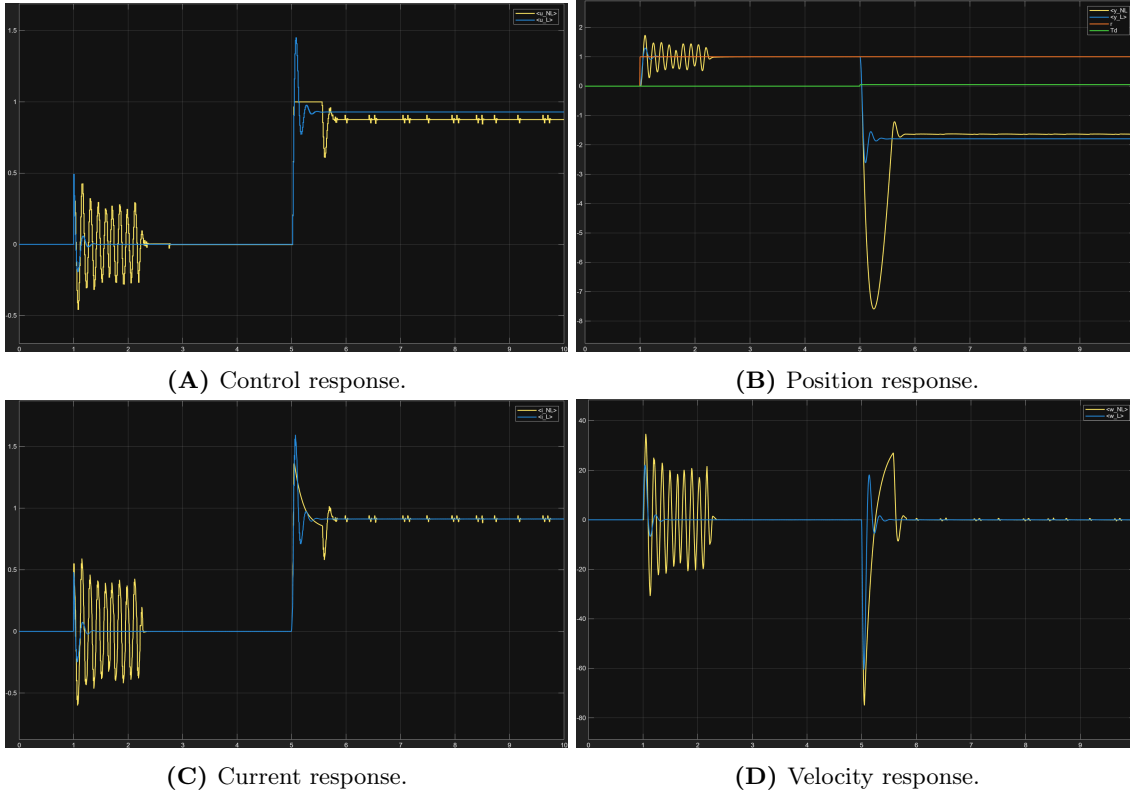


Figure 8: Nyquist plot of $L(z)$

Figure 9: Magnitude plot of S

Simulation

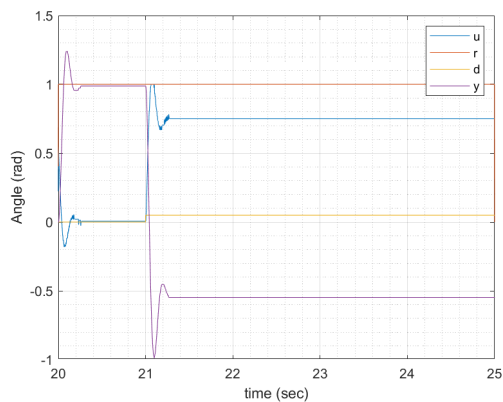
Figure 10: Simulation for $T_s = 0.01\text{sec}$.

Discussion

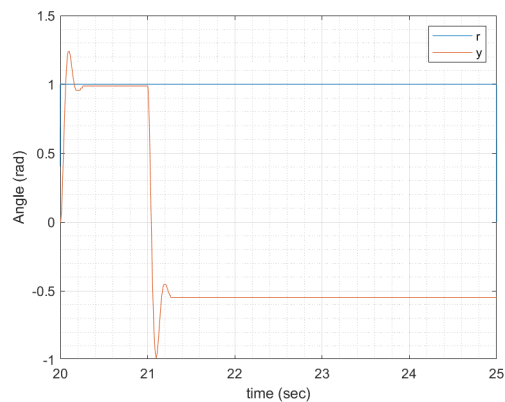
The simulated results seem pretty good. There is very good reference tracking for the position is both the linear plant and the virtual plant. However, we do observe that there ne poor disturbance rejection, with a strong steady state error to a relatively small input disturbance. Moreover, following what we have seen throughout the course, have a shorter sampling period provides cleaner, faster, and more stable responses.

Problem 11Exp

Sampling time $T_s = 0.001$ sec

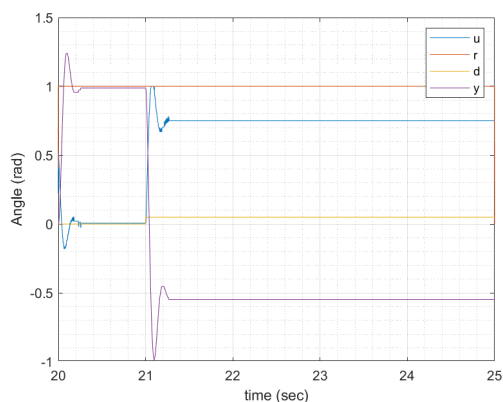


(A) Control scope.

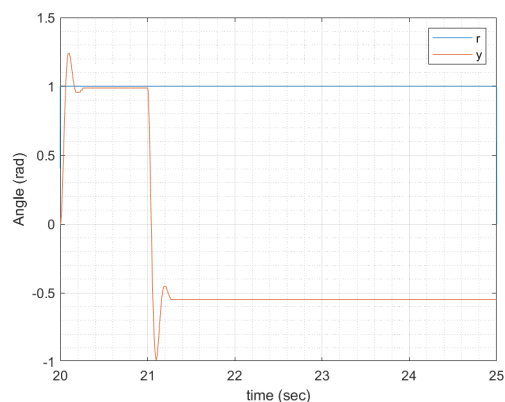


(B) Position scope.

Figure 11: Initial design trial.

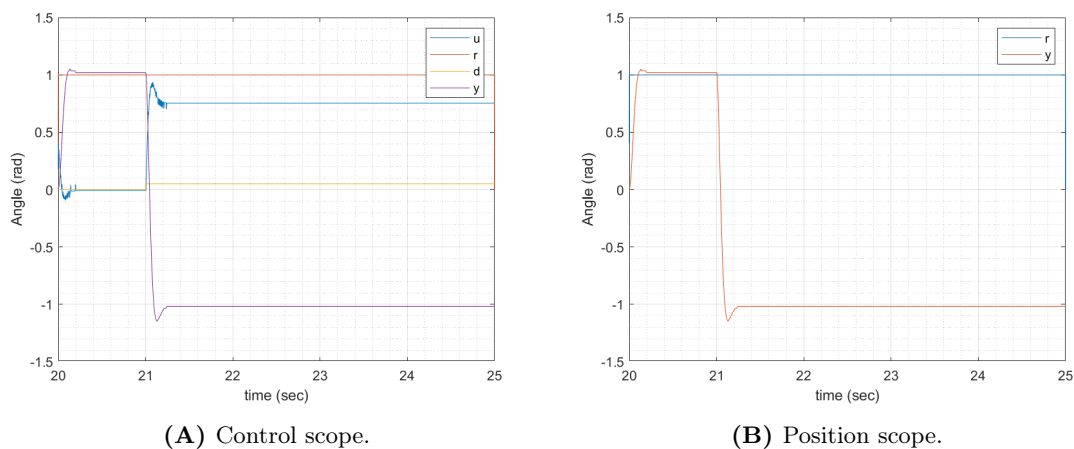


(A) Control scope.



(B) Position scope.

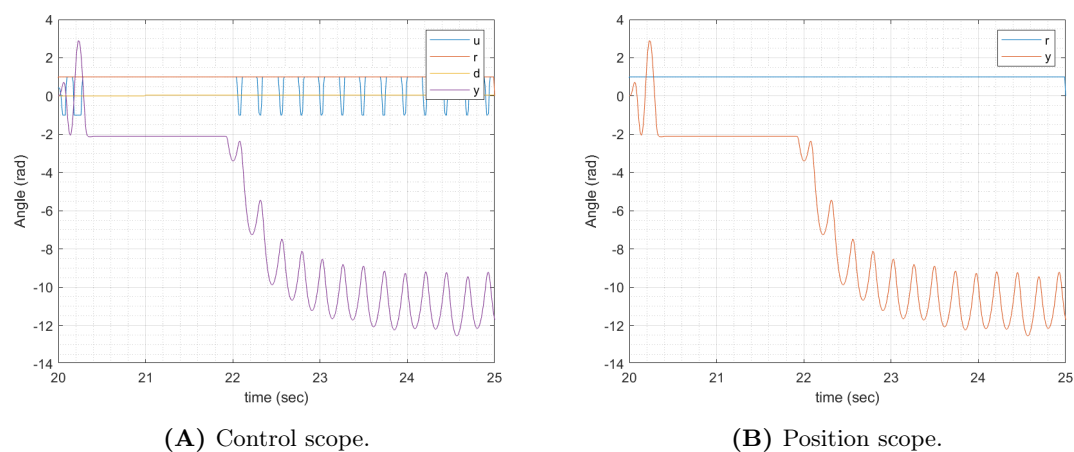
Figure 12: First iteration trial.

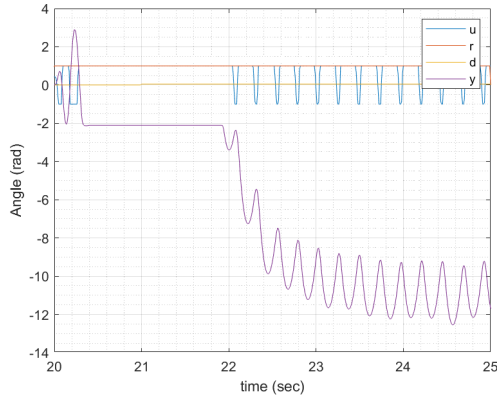
**Figure 13:** Second iteration trial.

$T_s = 0.001$	State feedback pole locations	State observer pole locations	
Initial design	$\begin{bmatrix} 0.5533 & 0.0110 \end{bmatrix}$	0.9212	208.0082
Iteration 1	$\begin{bmatrix} 0.5022 & 0.05 \end{bmatrix}$	0.9916	243.4276
Iteration 2	$\begin{bmatrix} 0.4465 & 0.0169 \end{bmatrix}$	0.9916	243.4276

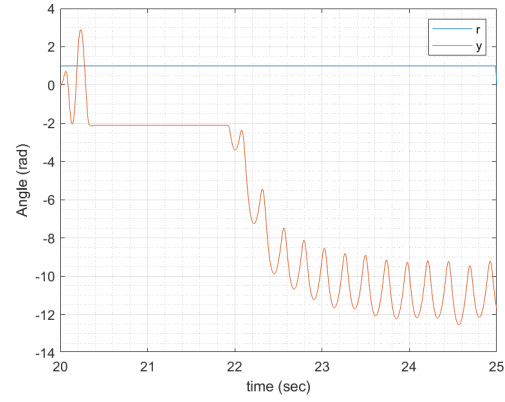
Table 1: State observer feedback pole locations

Sampling time $T_s = 0.01$ sec

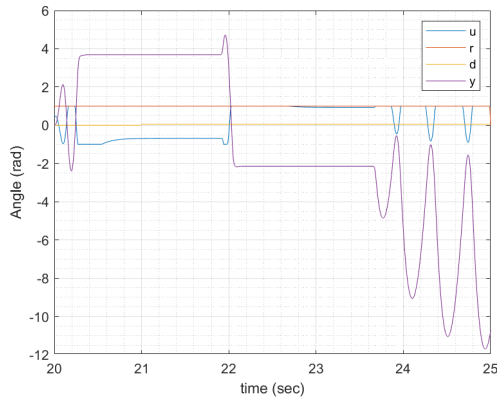
**Figure 14:** Initial design trial.



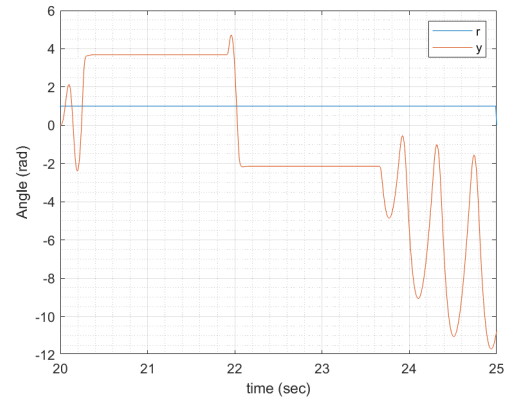
(A) Control scope.



(B) Position scope.

Figure 15: First iteration trial.

(A) Control scope.



(B) Position scope.

Figure 16: Second iteration trial.

$T_s = 0.01$	State feedback pole locations	State observer pole locations	
Initial design	$\begin{bmatrix} 0.4465 & 0.0169 \end{bmatrix}$	0.9916	243.4276
Iteration 1	$\begin{bmatrix} 0.4908 & 0.0115 \end{bmatrix}$	1.8845	83.3141
Iteration 2	$\begin{bmatrix} 0.4 & 0.0075 \end{bmatrix}$	1.6	70

Table 2: State observer feedback pole locations

Discussion

The SOFC system performed much better for the smaller sampling time. It generally produced a faster decaying system in all trials with a relatively small overshoot and quick settling time. In the larger sampling time experiments, it proved to be harder to tune, with it being more sensitive to gain changes and a harsher disturbance response.

In any case, altering the state feedback gain K was shown to alter the steady-state gain of a step reference response, and could be used to match the desired reference. Moreover, altering the observer gains seemed to affect the system less, though there are some visible effects in the transitive responses of the system.

Homework 7

Preliminaries

$$\begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \dot{F}_k \end{bmatrix} = \begin{bmatrix} -1 & 1 & -1 \\ 1 & -1 & 1 \\ 100 & -100 & 0 \end{bmatrix} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ F_k \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} d$$

$$y = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ F_k \end{bmatrix}$$

Part (a)

$$W_c = \begin{bmatrix} B & AB & A^2B \end{bmatrix} = \begin{bmatrix} 1 & -1 & -98 \\ 0 & 1 & 98 \\ 0 & 100 & -200 \end{bmatrix}$$

$$\det(W_c) = -10000$$

The system is **controllable** in continuous-time.

$$A = e^{FT} = \begin{bmatrix} 0.9852 & 0.0148 & -0.0099 \\ 0.0148 & 0.9852 & 0.0099 \\ 0.9868 & -0.9868 & 0.9901 \end{bmatrix}$$

$$F = M^{-1}\Lambda M$$

$$\Lambda = MFM^{-1}$$

Use MATLAB `eig` function to determine M and Λ :

$$M = \begin{bmatrix} -0.7071 & 0.005 - 0.0702i & 0.005 + 0.0702i \\ -0.7071 & -0.005 + 0.0702i & -0.005 - 0.0702i \\ 0 & -0.995 & -0.995 \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 + 14.1067i & 0 \\ 0 & 0 & -1 - 14.1067i \end{bmatrix}$$

$$\begin{aligned}
B &= \int_0^T e^{F\tau} d\tau \cdot G \\
&= \int_0^T M e^{\Lambda\tau} M^{-1} d\tau \cdot G \\
&= M \int_0^T e^{\Lambda\tau} d\tau M^{-1} G \\
&= M \begin{bmatrix} T & 0 & 0 \\ 0 & \frac{1}{\lambda_1}(e^{\lambda_1 T} - 1) & 0 \\ 0 & 0 & \frac{1}{\lambda_2}(e^{\lambda_2 T} - 1) \end{bmatrix} M^{-1} G \\
&= \begin{bmatrix} 0.0099 \\ 0.0001 \\ 0.005 \end{bmatrix} \\
B_d &= M \int_0^T e^{\Lambda\tau} d\tau M^{-1} G_d \\
&= \begin{bmatrix} 0.0001 \\ 0.0099 \\ -0.005 \end{bmatrix}
\end{aligned}$$

$$W_c = \begin{bmatrix} 0.0099 & 0.0097 & 0.0095 \\ 0.0001 & 0.0003 & 0.0005 \\ 0.005 & 0.0146 & 0.0239 \end{bmatrix}$$

$$\det(W_C) = -9.6404 \times 10^{-9}$$

The discrete-time system is **barely controllable**.

Assigning poles to $z = 0.92, z = 0.85, z = 0.8$, we get the following feedback gain:

$$K = \begin{bmatrix} 37.7043 & -13.4231 & 3.3819 \end{bmatrix}$$

Part (c)

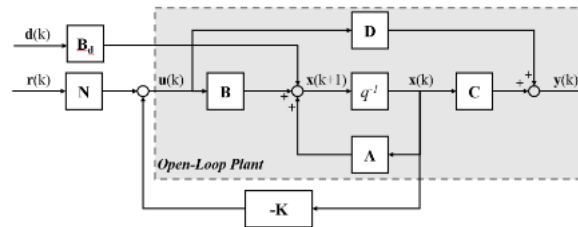


Figure 1: Block diagram for state feedback control system.

$$\frac{y(z)}{r(z)} = \frac{0.000066z^2 - 0.0002745z + 0.000358}{z^3 - 7.71z^2 + 26.41z - 52.74} N$$

For unity D.C. gain:

$$N = -0.00002208$$

$$\frac{y(z)}{d(z)} = \frac{0.009934z^2 - 0.06684z + 0.1967}{z^3 - 7.71z^2 + 26.41z - 52.74}$$

Part (d)

$$L(z) = \frac{0.3904z^2 - 9.989z + 122.9}{z^3 - 26.64z^2 + 342z - 2815}$$

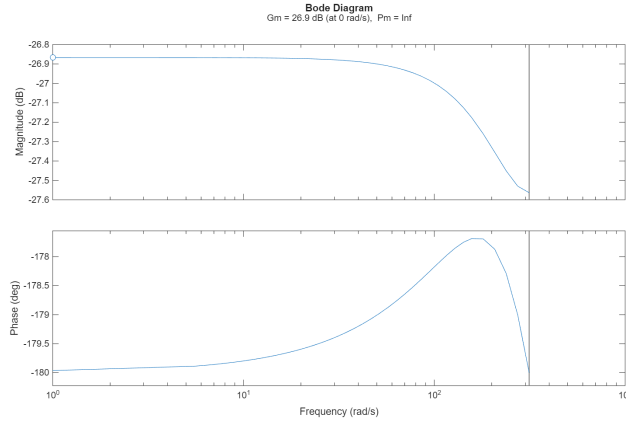


Figure 2: Bode plot of Loop gain transfer function.

The loop gain has a gain margin of 26.9dB at $\omega = 0 \frac{\text{rad}}{\text{s}}$ and infinite phase margin.

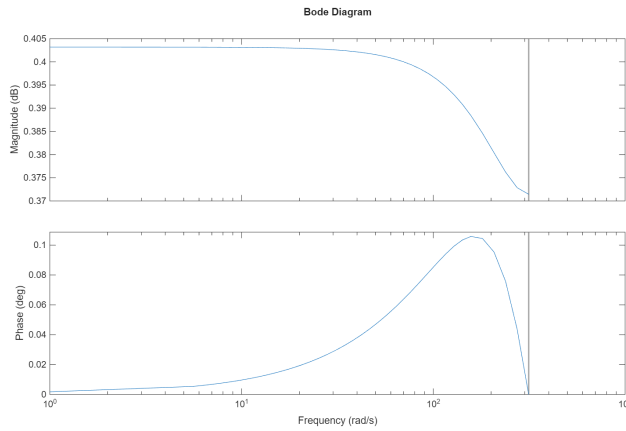


Figure 3: Bode plot of $S = \frac{1}{1+L(z)}$

The vector gain margin is 22.0465.

Part (e)

$$W_o = \begin{bmatrix} 0 & 1 & 0 \\ 0.0148 & 0.9852 & 0.0099 \\ 0.039 & 0.961 & 0.0193 \end{bmatrix}$$

$$\det(W_o) + 0.000097534$$

The discrete system is **barely observable**.

Using a Luenberger observer with poles at $z = 0.75$, $z = 0.7$, $z = 0.65$, we obtain the following observer gain matrix:

$$L = \begin{bmatrix} 1.7953 \\ 0.8604 \\ 20.2503 \end{bmatrix}$$

Part (f)

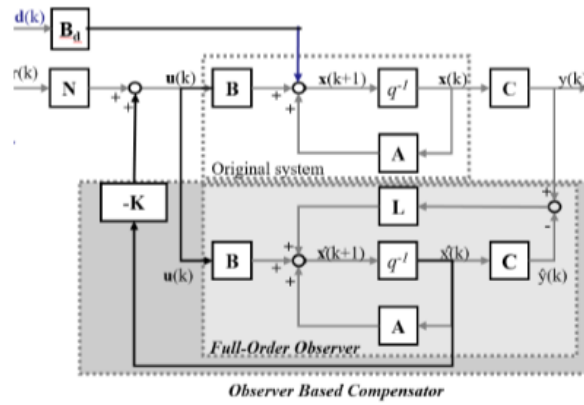


Figure 4: Block diagram of Luenberger observer feedback control system.

We have closed loop poles at (0.92, 0.85, 0.8, 0.75, 0.7, 0.65), as designed.

$$\begin{bmatrix} x(k+1) \\ \tilde{x}(k+1) \end{bmatrix} = \begin{bmatrix} 0.6106 & 0.1482 & -0.0435 & 0.3745 & -0.1333 & 0.0336 \\ 0.0123 & 0.9861 & 0.0096 & 0.0025 & -0.0009 & 0.0002 \\ 0.7998 & -0.9202 & 0.9733 & 0.187 & -0.0666 & 0.0168 \\ 0 & 0 & 0 & 0.9852 & -1.7805 & -0.0099 \\ 0 & 0 & 0 & 0.0148 & 0.1247 & 0.0099 \\ 0 & 0 & 0 & 0.9868 & -21.237 & 0.9901 \end{bmatrix} \begin{bmatrix} x(k) \\ \tilde{x}(k) \end{bmatrix} + \begin{bmatrix} -2193.3 \\ -14.6 \\ -1094.8 \\ 0 \\ 0 \\ 0 \end{bmatrix} r(k) + \begin{bmatrix} 0.0001 \\ 0.0099 \\ -0.005 \\ 0.0001 \\ 0.0099 \\ -0.005 \end{bmatrix} d(k)$$

$$y(k) = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x(k) \\ \tilde{x}(k) \end{bmatrix}$$

$$W_c = \begin{bmatrix} -2193.3 & -1293.9 & -675.8 & -265 & 5.1 & 146.6 \\ -14.6 & -52 & -94.3 & -137.2 & -177.8 & -214.5 \\ -1094.8 & -2806.4 & -3718.5 & -4073.1 & -4050.1 & -3782.5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{rank}(W_c) = 3$$

The closed loop system is **not controllable**.

$$W_o = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0.0123 & 0.9861 & 0.0096 & 0.0025 & -0.0009 & 0.0002 \\ 0.0274 & 0.9653 & 0.0184 & 0.0115 & -0.0125 & 0.001 \\ 0.0433 & 0.939 & 0.026 & 0.0283 & -0.0488 & 0.0022 \\ 0.0588 & 0.9084 & 0.0325 & 0.0527 & -0.1111 & 0.0035 \\ 0.0731 & 0.8746 & 0.0378 & 0.0841 & -0.1929 & 0.0046 \end{bmatrix}$$

$$\text{rank}(W_o) = 6$$

The closed loop system is **observable**.

Part (g)

$$\frac{y(z)}{r(z)} = \frac{-14.61z^2 + 60.6z - 79.04}{z^3 - 7.71z^2 + 26.41z - 52.74}$$

$$\frac{y(z)}{d(z)} = \frac{0.009934z^5 - 0.206z^4 + 2.053z^3 - 13.09z^2 + 59.95z - 210}{z^6 - 21.72z^5 + 227z^4 - 1521z^3 + 7330z^2 + 27070z + 79670}$$

Part (h)

$$C(z) = \frac{124.6z^2 - 252.8z + 128.8}{z^3 - 1.71z^2 + 1.053z - 0.2221}$$

$$L(z) = \frac{0.008244z^4 - 0.008556z^3 - 0.01214z^2 + 0.01675z - 0.004232}{z^6 - 4.67z^5 + 9.054z^4 - 9.346z^3 + 5.428z^2 - 1.685z + 0.2177}$$

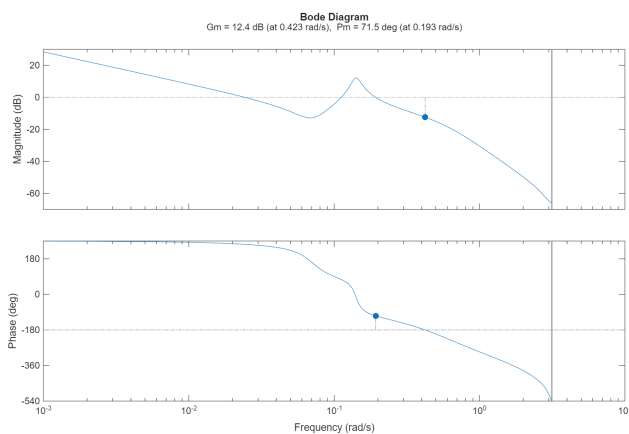


Figure 5: Bode plot of Loop gain transfer function.

The Loop gain transfer function has a gain margin of 12.4dB at $\omega = 0.423 \frac{\text{rad}}{\text{s}}$ and a phase margin of 71.5° at $\omega = 0.193 \frac{\text{rad}}{\text{s}}$.

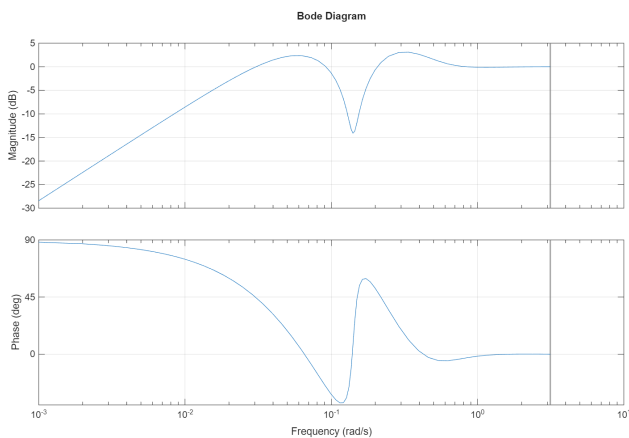


Figure 6: Bode plot of $S = \frac{1}{1+L(z)}$

The closed loop system has a vector gain margin of 3.292.

The margins of the state observer feedback control system are worse than those of the state feedback control system.

Homework 8

Problem 41

NP	SNR_n_y	Hdu(z)	Hdn(z)	IIR order	IIR relative H2 norm error (%)	FIR order	FIR relative H2 norm error (%)
100	0.01	1	1	5,5	0.75	50	5.31
1000	0.01	1	1	5,5	0.27	50	0.53
10000	0.01	1	1	5,5	0.38	50	0.17
100	0.1	1	1	5,5	7.98	50	598.68
1000	0.1	1	1	5,5	9.13	50	5.72
10000	0.1	1	1	5,5	7.43	50	1.61
1000	0.1	Pole=0.8	1	5,5	8.06	50	5.37
1000	0.1	1	Pole=0.8	5,5	8.24	50	6.62
1000	0.1	Pole=0.8	Pole=0.8	5,5	7.78	50	5.53
1000	0.1	1	1	2,2	69.12	20	4.75
1000	0.1	1	1	10,10	3.01	100	9.37

Table 1: System identification parameters and associated error norms of associated IIR and FIR filters.

Reducing the sound-to-noise ratio generally reduced the the error of both the IIR and FIR models. Moreover, increasing the number of data points decreased the error as well. This makes intuitive sense, as both of these actions allow the algorithm to better isolate and study the input/output relation.

Changing the location of the poles and zeros of the plant did not pose any significant changes to the error the algorithm, signifying that the algorithm sufficiently identified the parameters of the plant regardless of the nature of the plant itself.

Changing the order of the filter only had a significant effect on the IIR model, with the error reducing as the order increased. On the other hand, it appeared that the opposite happened with the FIR filter, though the effect is less severe.

Problem 42

Using a 3,3 IIR filter model, we obtain the following transfer function:

$$\frac{Y(z)}{U(z)} = \frac{0.1756z^{-2} + 0.6139z^{-1} + 0.003371}{0.0004176z^{-3} + 0.6139z^{-2} - 1.447z^{-1} + 1}$$

In a zero-pole-gain form:

$$\frac{Y(z)}{U(z)} = 0.0033713 \frac{(1 + 77.43z^{-1})(1 + 0.6728z^{-1})}{(1 + 0.0006792z^{-1})(1 - 1.448z^{-1} + 0.6149z^{-2})}$$

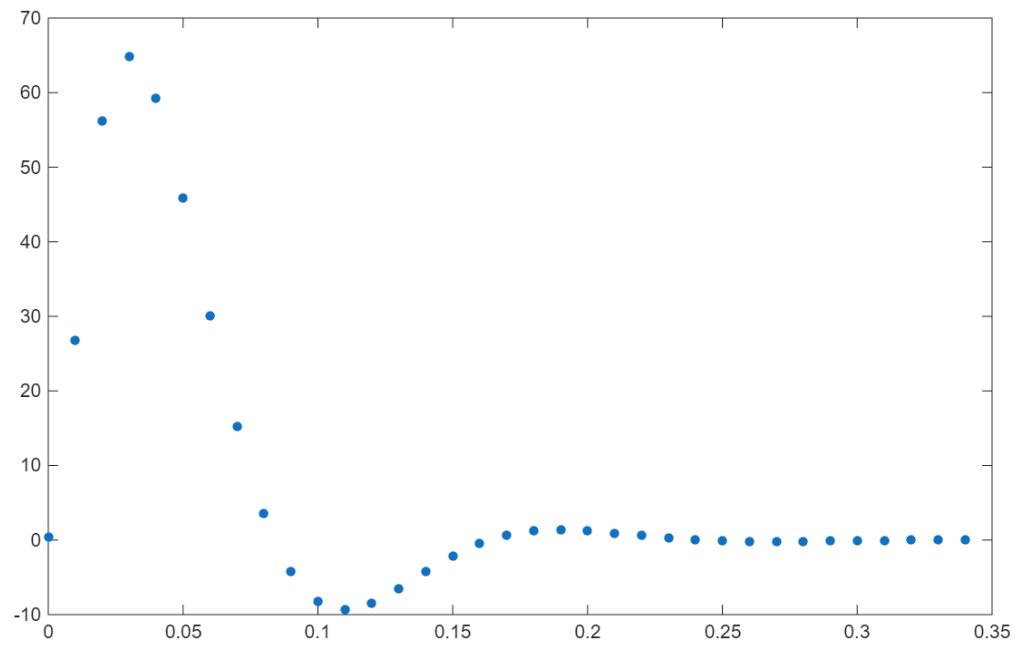


Figure 1: Impulse response of IIR filter model.

Homework 10

Problem 43

Np	SNR_n_y	Hdu(z)	Hdn(z)	Covariance	FIR Order	RLS relative H2 norm error (%)	LS relative H2 norm error (%)
1000	0.01	1	1	10000	50	0.52	0.53
1000	0.1	1	1	10000	50	5.17	5.17

Table 1: RLS and LS comparison against different sound-to-noise ratios.

For the following tests of convergence of the FIR filter, my computing power was limited to $1e4$ steps at a time due to limited memory capacity. To test the final two cases, the computed FIR coefficients and covariance matrix at the final time step is used as the initial parameters of the following RLS computation. This is done 5 and 10 times for the $5e4$ and $1e5$ trials, respectively.

Np	RLS relative H2 norm error (%)
100	25.76
200	15.88
300	11.82
400	9.49
500	7.64
600	6.89
1000	5.17
2000	3.41
5000	2.21
1e4	1.65
5e4	1.65
1e5	1.65

Table 2: Relative H2 norm error at different iteration steps of the recursive least squares algorithm.

The RLS algorithm seems to converge by $1e4$ steps. Successive trials at higher iteration numbers do not significantly alter the identified parameters of the filter, implying that our FIR filter has converged to its most accurate possible configuration.

Problem 44

Np	Plant Ghat	Reference Mref: z^{-d}	FeedFwd Filter Order	Relative tracking error (%)			Remarks on Bode plot of GF
				White noise	Square wave	Sine wave	
1000	Gd	d=0	100	145.97	2.33	2.54e-5	Sine has full match at 1rad/sec Square has good gain match at harmonics, no phase match White noise bad everywhere
1000	Gd	1	100	0.00364	1.63	2.54e-5	White noise has closer gain match, full phase match Square has better gain and phase match with some oscillation Sine looks the same as previous trial
1000	Gd	2	100	0.00385	1.72	2.54e-5	White noise has better gain match (not perfect) and retains phase match Square wave has good matching up to 100rad/sec Sine has better matching beyond 1rad/sec up to 50rad/sec
1000	Gd	1	10	17.54	2.58	1.74e-3	White noise has low frequency oscillation, but is accurate on average Square has very good match up to 200rad/sec, very small oscillations Sine has good match until 10rad/sec
1000	Gd	1	2	91.21	22.46	0.0537	White noise is very bad, looks like sine with no DC matching Square looks like a sine Bode plot Sine still good until 10rad/sec, but diverges more
10000	Gd	1	100	5.65e-4	0.35	2.04e-6	White noise is very good, some oscillation but maintains gain Square is really good, but has bigger oscillations Sine wave looks same as before
1000	Gd2	0	100	397.83	3.51	2.54e-5	White noise is very bad Square still matches at harmonics, but bigger oscillations Sine is same as before
1000	Gd2	1	100	209.84	4.03	2.55e-5	White noise is pretty bad Square has stronger oscillations Sine is same
1000	Gd2	2	100	104.23	3.35	2.54e-5	White noise is better, gain is much closer Square is great until 50rad/sec Sine is a great match until 30rad/sec
1000	Gd2	5	100	13.27	3.36	2.54e-5	White noise has fair matching, within small range of model Square has great gain until 170rad/sec, a number of cycles of phase delay Sine has good gain match until 60rad/sec, full cycle delay
1000	Gd2	10	100	0.4	3.32	2.54e-5	White noise is very good Square is good until 100rad/sec, great DC response Sine is great until 60rad/sec, gain shrinks afterwards
1000	Gd2	1	200	616.24	3.23	2.92e-5	White noise is like previous d=1 case Square behaves like order 100 Sine behaves like other d=1 cases
1000	Gd2	2	200	306.07	2.84	2.92e-5	White noise is okay, DC error before it gets noisy Square is good, very sharp oscillations after 100rad/sec Sine is just like before
1000	Gd2	5	200	38.96	2.85	2.92e-5	White noise is better than before, but the gain magnitude gets pretty noisy Square is good until 170rad/sec, some oscillation after Sine is good until 70rad/sec, gain does not go above unitary
1000	Gd2	10	200	1.15	2.84	2.92e-5	White noise is as good as the previous d=5 case Square is actually worse than before, oscillates by 110rad/sec Sine is as good as before, no improvement
1000	Gd2	1	10	89.49	13.05	2.42e-5	White noise bad, but consistent across frequencies Square has perfect gain match until 200rad/sec Sine diverges at around 15rad/sec, earlier than before
1000	Gd2	2	10	44.06	6.00	0.00187	White noise some DC error, but much more predictable gain magnitude Square is a great match until 70rad/sec, smoother gain curve Sine is a little better, diverges at 20rad/sec
1000	Gd2	5	10	25.44	4.13	0.00113	White noise has mild gain that oscillates slowly Square is great until 190rad/sec Sine is good until 60rad/sec
1000	Gd2	10	10	48.27	6.83	0.0212	White noise is worse than before, maybe overparameterized Square is decent until 35rad/sec Sine matches until 27rad/sec
10000	Gd2	5	10	28.34	2.89	9.89e-5	White noise has better gain matching through 200rad/sec Square is really good, matches until 200rad/sec Sine is good until 50rad/sec, not very remarkable
10000	Gd2	5	100	5.61	1.56	2.04e-6	White noise improved, smaller bounds for oscillation Square is a bit more unstable from 100rad/sec Sine is same as previous case
10000	Gd2	5	200	6.04	1.09	2.06e-6	White noise appears as good as previous trial Square has better matching at lower frequencies, gets erratic at higher Sine sees no noticeable improvements

Table 3: Relative tracking error of adaptive feedforward filter against various RLS configurations.

Generally, the sine wave had great tracking at the wave's frequency (1rad/sec) but eventually had non-unitary gain magnitude as the frequency increased. The square wave had good tracking at the harmonics of its natural frequency (1rad/sec) but experienced oscillations beyond that. The white noise had generally consistent gain across all frequencies and demonstrated a degree of robustness across a broad spectrum of

frequencies, but had non-unitary gain for the most part. Phase tracking was decent for low frequencies, but had error in the multiples of 2π radians.

In general, tracking error improved as the delay increased, though cases where $d = 10$ hardly improved cases where $d = 5$. Increasing the FIR order increased the performance of the white noise and square wave inputs, showing no improvements for the sine wave as it does not need a robust filter to obtain the desired behavior. The causal plant produced better results than the non-causal plant, which is expected. Increasing N_p created better parameter, significantly reducing the error for the white noise and square wave.

Problem 45

Np	Plant Ghat	FeedFwd Filter Order	Relative tracking error (%)			Remarks on Bode plot of GF
			White noise	Square wave	Sine wave	
1000	Gd	2	99.96	62.22	0.0536	White noise is very bad, does not track anywhere Square decent tracking until 10Hz Sine has good tracking until 10Hz
1000	Gd	3	100.08	43.49	0.0112	White noise has no better performance Square has better tracking across more frequencies, up to 100Hz for gain Sine tracks at same frequencies, gain has smaller hump
1000	Gd	5	99.91	45.71	0.0300	White noise has better tracking by 10dB, but still bad Square has same gain tracking, no improvement on phase tracking Sine tracks at same frequencies, gain has smaller hump
1000	Gd	8	103.27	131.74	0.00428	White noise performance in gain looks worse Square has good gain tracking through 230Hz, though phase tracking is poor Sine shows same behavior as before
1000	Gd	10	101.42	132.33	0.00219	White noise tracking looks the same as the previous case Square has good tracking gain until 200Hz, poor phase tracking Sine performs the same as previous trial
1000	Gd	50	108.40	22.05	2.39e-5	White noise has better DC tracking but more unstable tracking from 40Hz Square now resembles AFFF, with gain oscillating through reference model Sine tracks the same frequencies, but has bigger bulge
1000	Gd	100	243.16	2.33	2.54e-5	White noise appears to have same Bode plot as before Square now behaves as theorized, matching reference at harmonic frequencies Sine tracks the same way as previous trial
1000	Gd	200	296.70	1.51	2.91e-5	White noise has more erratic behavior from 200Hz Square has more intersections with the reference model Sine tracks the same as the previous trial
1000	Gd2	2	99.88	67.67	0.0546	White noise looks like sine wave behavior, but with very poor DC gain (-20dB) Square looks like sine wave with small DC gain discrepancy Sine looks the same as original plant Gd
1000	Gd2	3	100.21	48.25	0.00813	White noise has very minor improvements, with -17dB DC gain Square tracks longer, all the way until 20rad/sec when phase stops tracking Sine looks the same as previous with smaller bulge
1000	Gd2	5	99.78	67.33	0.0387	White noise has minor improvements, with -13dB DC gain Square tracks gain until 100rad/sec and phase until 30rad/sec Sine looks the same as before
1000	Gd2	8	100.32	110.08	0.0645	White noise has slight worse tracking, with -15dB DC gain Square tracks gain until 200rad/sec and phase until 30rad/sec Sine looks the same as before
1000	Gd2	10	100.16	109.52	0.00296	White noise has minor improvements, -14dB DC gain Square has same performance as before Sine looks the same as before
1000	Gd2	50	320.31	77.46	2.39e-5	White noise gain becomes erratic from 8rad/sec Square reverts back to tracking at harmonics only Sine looks the same as before
1000	Gd2	100	312.08	3.51	2.55e-5	White noise is still erratic from 8rad/sec, but has better DC gain tracking Square tracks harmonics, but has smaller amplitude oscillations Sine looks the same as before
1000	Gd2	200	627.56	2.85	2.92e-5	White noise looks the same as the FIR order 50 trial Square looks the same as before Sine looks the same as before

Table 4: Relative tracking error of adaptive disturbance rejection control against various RLS configurations.

Overall, there was worse relative tracking error than the adaptive feedforward filter trials. This may attributed to the increased power of the disturbance input, which greatly impacted the error between the reference and the output. The FIR order seemed to only impact the relative error of the periodic inputs, in which the relative error generally decreased with increasing order. There was exception, though, with the order 8 and order 10 filters, which greatly boosted the errors. Perhaps this is due to too much preference being given to weaker parameters of the system, causing a greater variance of the output. Both plants were tracked

to about the same degree of success, with a slight improvement seen in Gd2. This may be because the frequency response of Gd2 has a contour that is better matched by the FIR filter, as opposed to Gd, which has a constant gain and phase response through its whole spectrum.

Final Assignment

Problem 12

Laptop 5				
R	κ	τ	K_m	b
7.2	449	0.154	0.0457	-0.000138

Sampling time $T_s = 0.001$ sec

Plant model

The plant model with identified parameters:

$$\frac{2915.6}{s(s + 12.44)}$$

The plant model in discrete-time with zero order hold and sampling:

$$\frac{0.0014518(z + 0.9959)}{(z - 1)(z - 0.9876)}$$

State feedback and integral properties

The state feedback properties are:

- Natural frequency, $\omega_n = 36$ rad/sec
- Damping ratio, $\zeta = 0.5169$
- Continuous poles: $s = -18.6095 + 30.8170i, -18.6095 - 30.8170i$
- Discrete poles: $z = 0.9811 + 0.0302i, 0.9811 - 0.0302i$
- Integral integral pole: $z = 0.8834$

The state feedback gains, K and K_{int} , and reference gain, N , are:

$$K = \begin{bmatrix} 1.9347 & 0.0481 \end{bmatrix}$$

$$K_{int} = 0.0512$$

$$N = 1.9347$$

State observer properties

For observer dynamics of $\omega_n = 100$ Hz and $\zeta = 1$:

- Natural frequency, $\omega_n = 628.31$ rad/sec
- Damping ratio, $\zeta = 1$
- Continuous poles: $s = -628.31, -628.31$
- Discrete poles: $z = 0.5335, 0.5335$

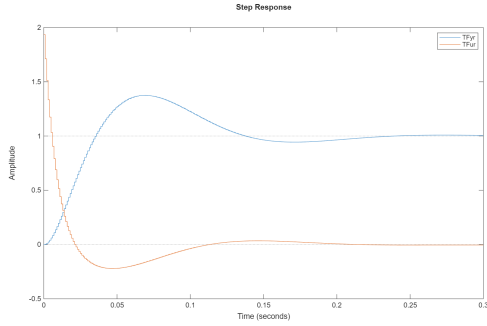
The observer gain L is:

$$\begin{bmatrix} 0.9207 \\ 207.5370 \end{bmatrix}$$

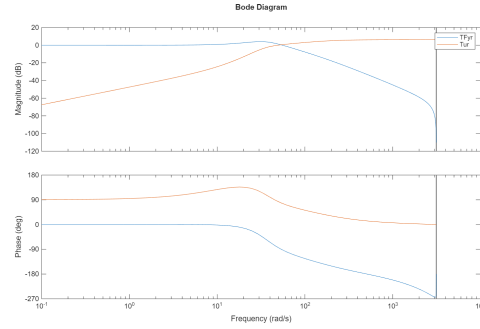
Closed loop transfer functions

$$T_{yr} = \frac{0.0028087(z + 0.9959)(z - 0.9735)}{(z - 0.8834)(z^2 - 1.962z + 0.9635)}$$

$$T_{ur} = \frac{1.9347(z - 1)(z - 0.9876)(z - 0.9735)}{(z - 0.8834)(z^2 - 1.962z + 0.9635)}$$



(A) Step response plot.

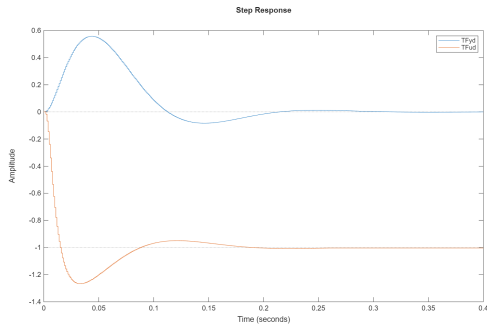


(B) Bode plot.

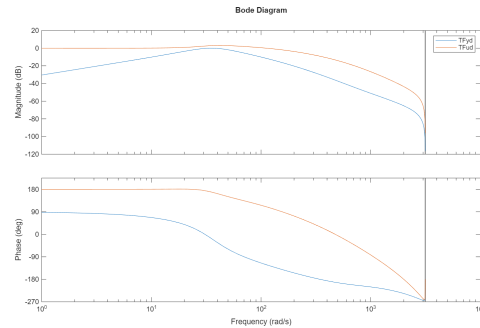
Figure 1: Step and frequency response of control and output to reference.

$$T_{yd} = \frac{0.0014518(z + 0.9959)(z - 1)(z^2 - 0.9249z + 0.2619)}{(z - 0.8834)(z - 0.5335)^2(z^2 - 1.962z + 0.9635)}$$

$$T_{ud} = \frac{-0.017138(z + 0.9959)(z^2 - 1.96z + 0.9612)}{(z - 0.8834)(z - 0.5335)^2(z^2 - 1.962z + 0.9635)}$$



(A) Step response plot.



(B) Bode plot.

Figure 2: Step and frequency response of control and output to disturbance.

Loop transfer functions

$$L(z) = \frac{0.017138(z + 0.9959)(z^2 - 1.96z + 0.9612)}{(z - 1)^2(z - 0.9876)(z^2 - 0.9249z + 0.2619)}$$

$$S = \frac{(z - 0.9876)(z - 1)^2(z^2 - 0.9249z + 0.2619)}{(z - 0.5335)^2(z - 0.8834)(z^2 - 1.962z + 0.9635)}$$

- $G_m = 5.6059$
- $P_m = 55.1651$
- $\omega_{cp} = 474.3849$
- $\omega_{cg} = 97.3741$
- $S_{peak} = 1.3578$
- $VGM = 3.7949$
- $\omega_{peak} = 224 \text{ rad/sec}$

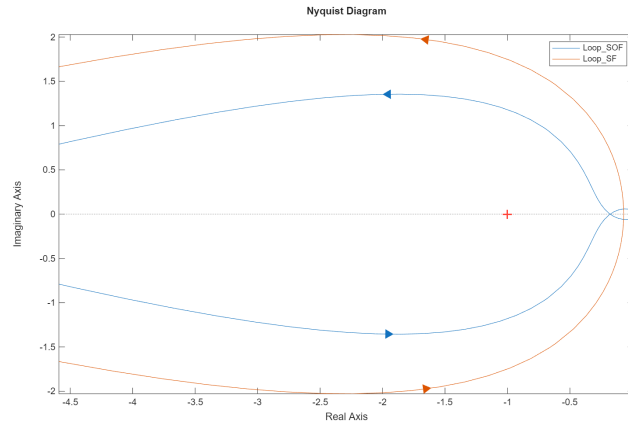


Figure 3: Nyquist plot of $L(z)$

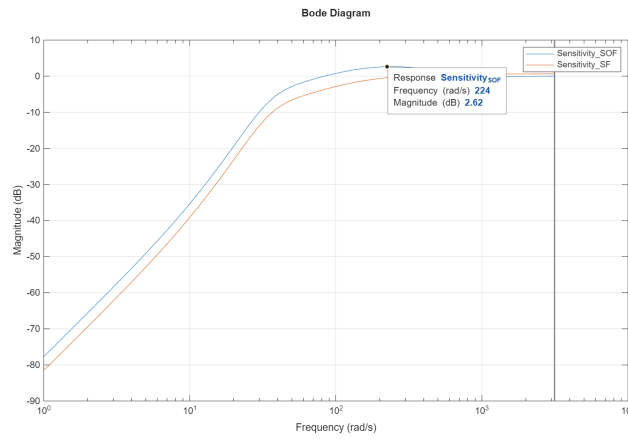


Figure 4: Magnitude plot of S

Simulation

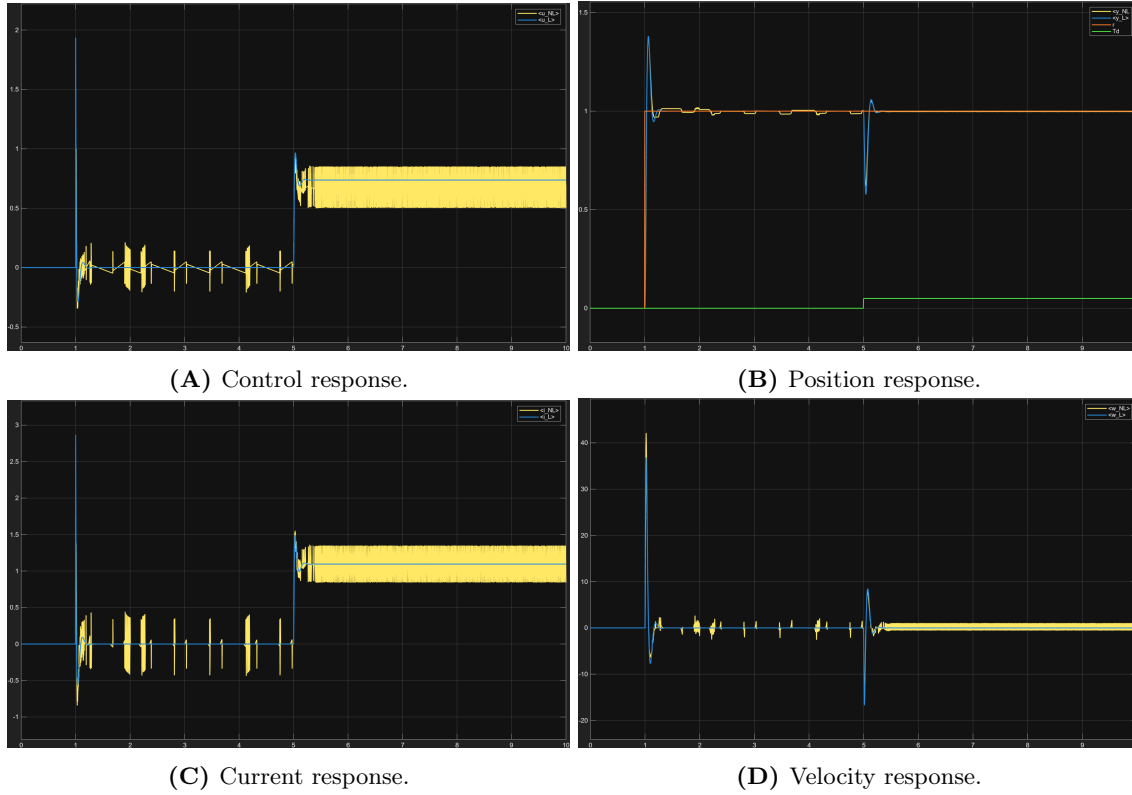


Figure 5: Simulation for $T_s = 0.001\text{sec}$.

Sampling time $T_s = 0.01 \text{ sec}$

Plant model

The plant model with identified parameters:

$$\frac{2915.6}{s(s + 12.44)}$$

The plant model in discrete-time with zero order hold and sampling:

$$\frac{0.13992(z + 0.9594)}{(z - 1)(z - 0.883)}$$

State feedback and integral properties

The state feedback properties are:

- Natural frequency, $\omega_n = 36 \text{ rad/sec}$
- Damping ratio, $\zeta = 0.5169$
- Continuous poles: $s = -18.6095 + 30.8170i, -18.6095 - 30.8170i$
- Discrete poles: $z = 0.7911 + 0.2518i, 0.7911 - 0.2518i$
- Integral discrete pole: $z = 0.7472$

The state feedback gains, K and K_{int} , and reference gain, N , are:

$$K = \begin{bmatrix} 0.7254 & 0.0165 \end{bmatrix}$$

$$K_{int} = 0.0987$$

$$N = 0.7254$$

State observer properties

For observer dynamics of $\omega_n = 100$ Hz and $\zeta = 1$:

- Natural frequency, $\omega_n = 628.31$ rad/sec
- Damping ratio, $\zeta = 1$
- Continuous poles: $s = -628.31, -628.31$
- Discrete poles: $z = 0.0019, 0.0019$

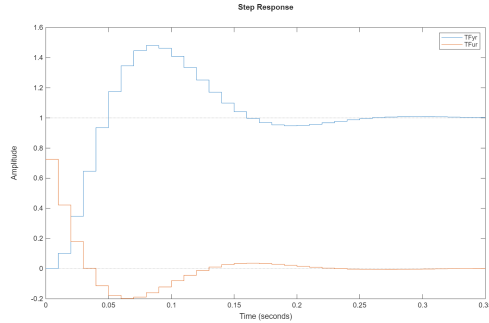
The observer gain L is:

$$\begin{bmatrix} 1.8793 \\ 82.5740 \end{bmatrix}$$

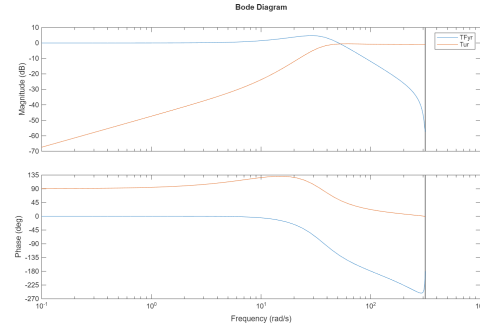
Closed loop transfer functions

$$T_{yr} = \frac{0.1015(z + 0.9594)(z - 0.8639)}{(z - 0.7472)(z^2 - 1.582z + 0.6892)}$$

$$T_{ur} = \frac{0.72543(z - 1)(z - 0.883)(z - 0.8639)}{(z - 0.7472)(z^2 - 1.582z + 0.6892)}$$



(A) Step response plot.

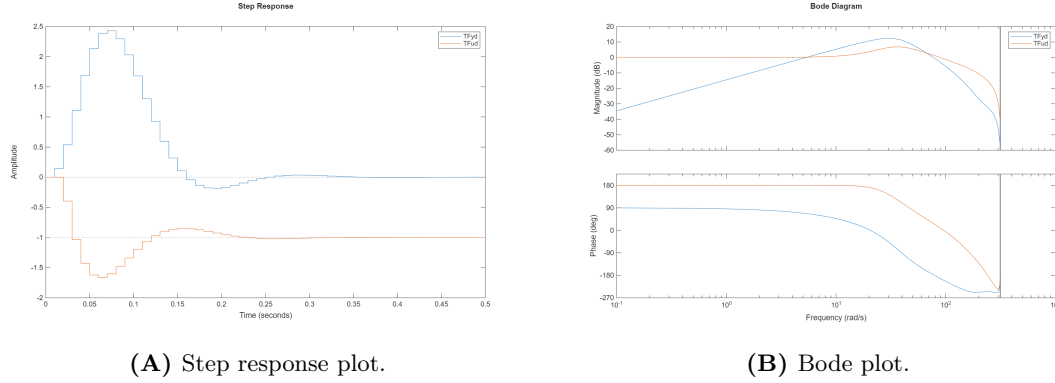


(B) Bode plot.

Figure 6: Step and frequency response of control and output to reference.

$$T_{yd} = \frac{0.13992(z - 1)(z + 0.9594)(z^2 + 0.5499z + 0.3044)}{(z - 0.7472)(z - 0.001867)^2(z^2 - 1.582z + 0.6892)}$$

$$T_{ud} = \frac{-0.39513(z + 0.9594)(z^2 - 1.674z + 0.7091)}{(z - 0.7472)(z - 0.001867)^2(z^2 - 1.582z + 0.6892)}$$



(A) Step response plot.

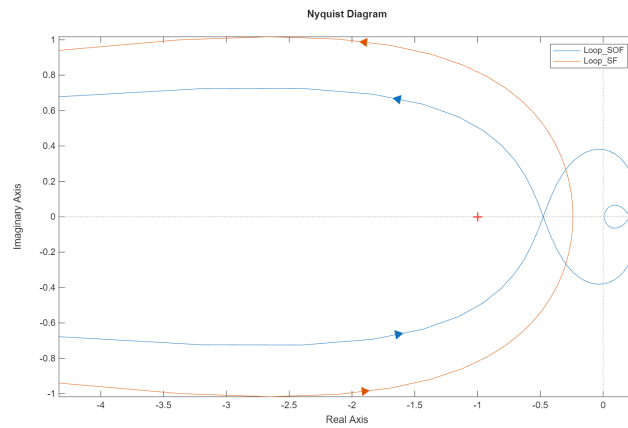
(B) Bode plot.

Figure 7: Step and frequency response of control and output to disturbance.**Loop transfer functions**

$$L(z) = \frac{0.39513(z + 0.9594)(z^2 - 1.674z + 0.7091)}{(z - 1)^2(z - 0.883)(z^2 + 0.5499z + 0.3044)}$$

$$S = \frac{(z - 0.883)(z - 1)^2(z^2 + 0.5499z + 0.3044)}{(z - 0.001867)(z - 0.001867)(z - 0.7472)(z^2 - 1.582z + 0.6892)}$$

- $G_m = 2.0967$
- $P_m = 26.9930$
- $\omega_{cp} = 95.6878$
- $\omega_{cg} = 41.7121$
- $S_{peak} = 2.2922$
- $VGM = 1.7739$
- $\omega_{peak} = 50.7 \text{ rad/sec}$

**Figure 8:** Nyquist plot of $L(z)$

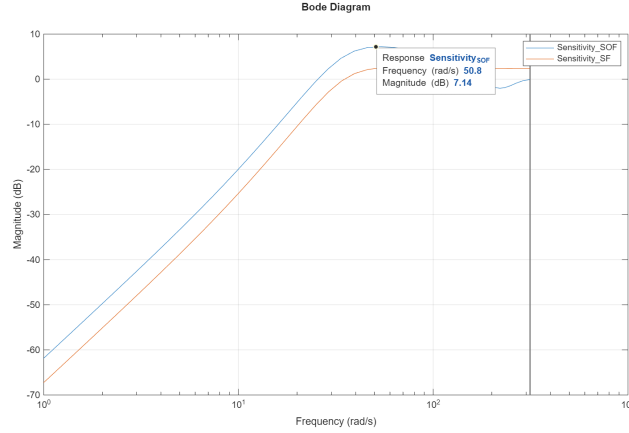


Figure 9: Magnitude plot of S

Simulation

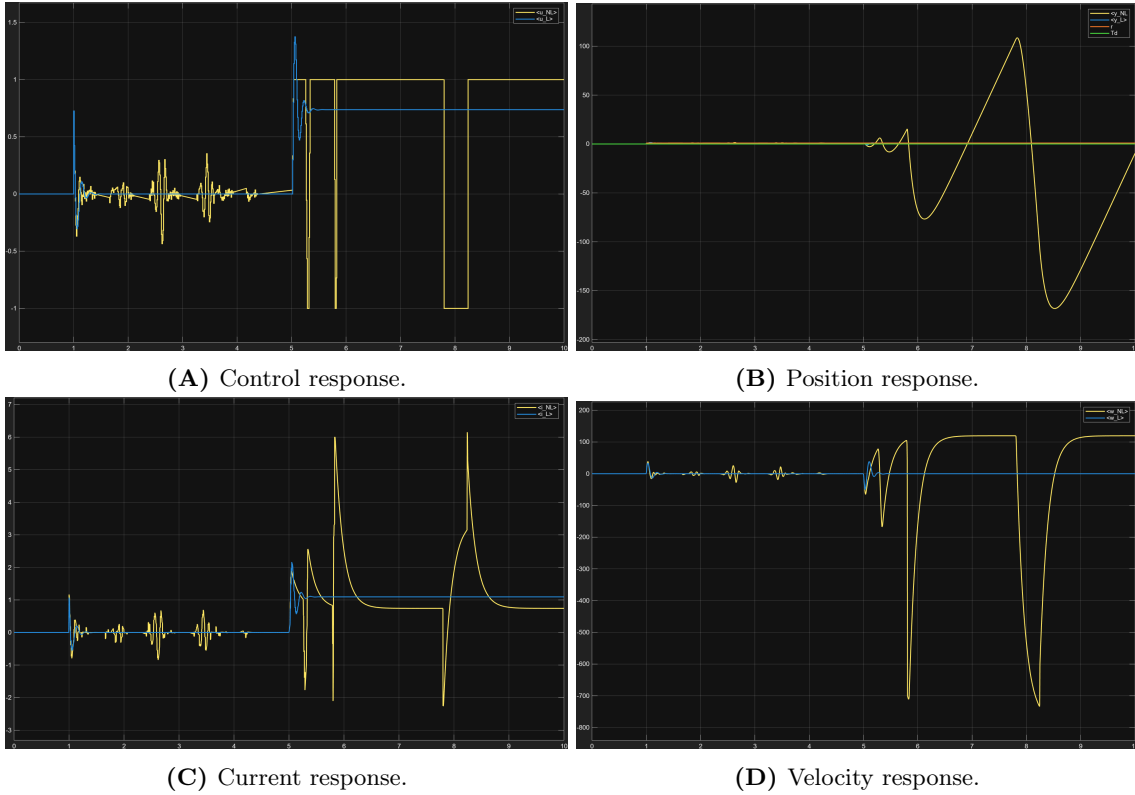


Figure 10: Simulation for $T_s = 0.01\text{sec}$.

Discussion

The higher sampling frequency results look very good. There is excellent reference tracking in both the reduced order and full order plant simulations. Moreover, the simulations demonstrate very good disturbance rejection properties, due to the integrator removing the steady-state error present in Problem 11. Moreover, the control signal does not experience any saturation, even in the virtual plant simulation, signifying that this control scheme will likely perform well in the physical plant setup.

However, the lower sampling frequency simulations look very bad for the virtual plant. The reference tracking is initially good, with mild overshoot. However, the disturbance produced an unstable response, exploding the output of the system. There is clear control saturation, demonstrating that the system's response is very difficult to control when a disturbance is injected.

These results correlate with what we have experienced and theorized throughout this course, with a smaller sampling period producing results that are closer to continuous time behavior and are better represented by the reduced order plant analysis.

Problem 12Exp

Sampling time $T_s = 0.001$ sec

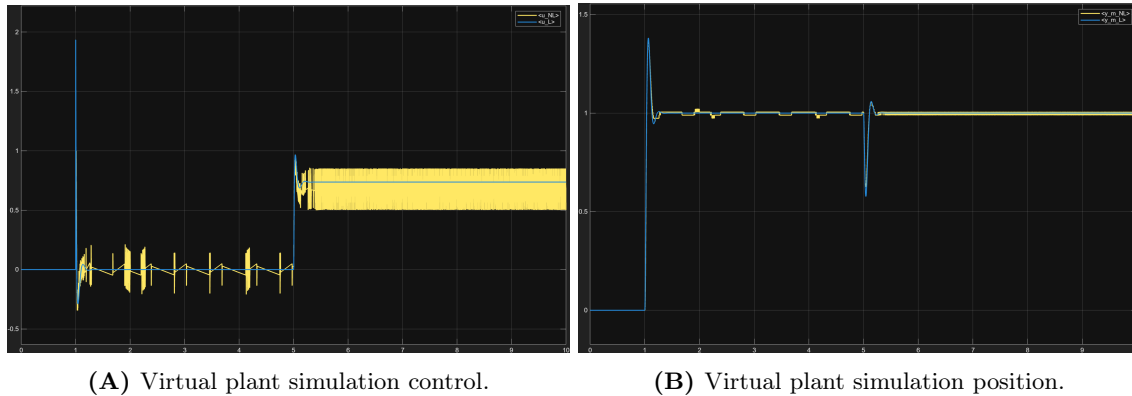


Figure 11: Initial design trial.

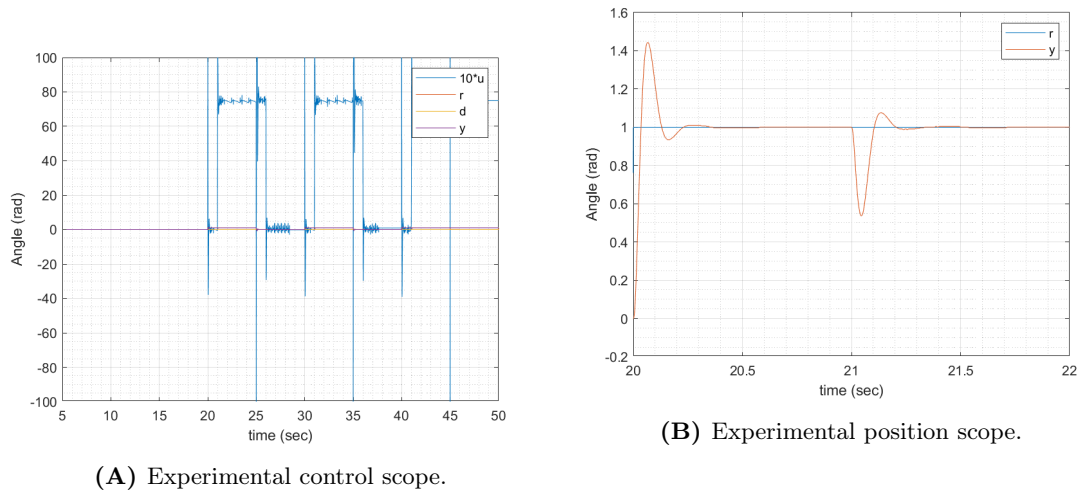
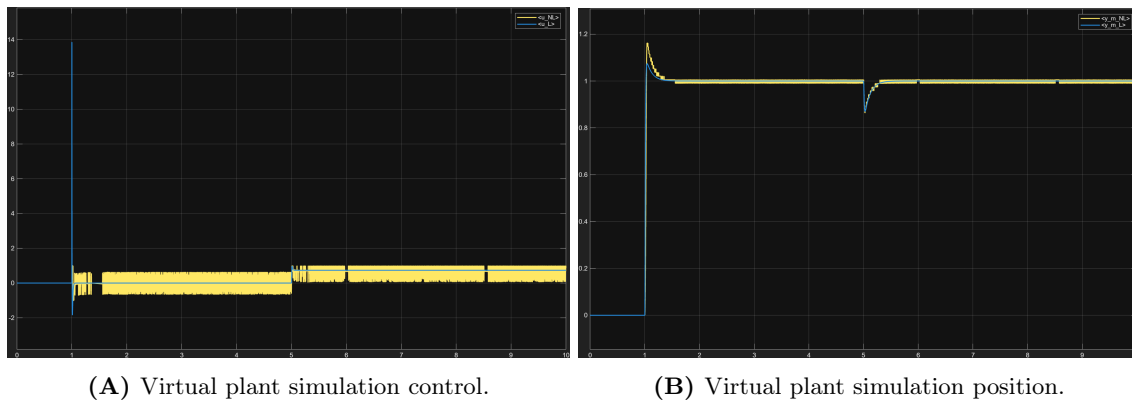
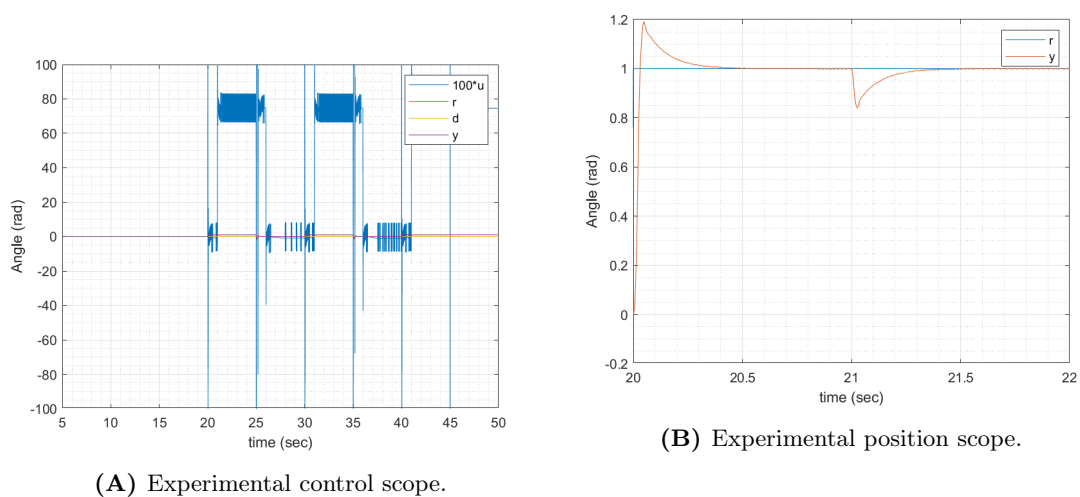


Figure 12: Initial design gains.



(A) Virtual plant simulation control.

(B) Virtual plant simulation position.

Figure 13: Initial design trial.

(A) Experimental control scope.

(B) Experimental position scope.

Figure 14: First iteration gains.

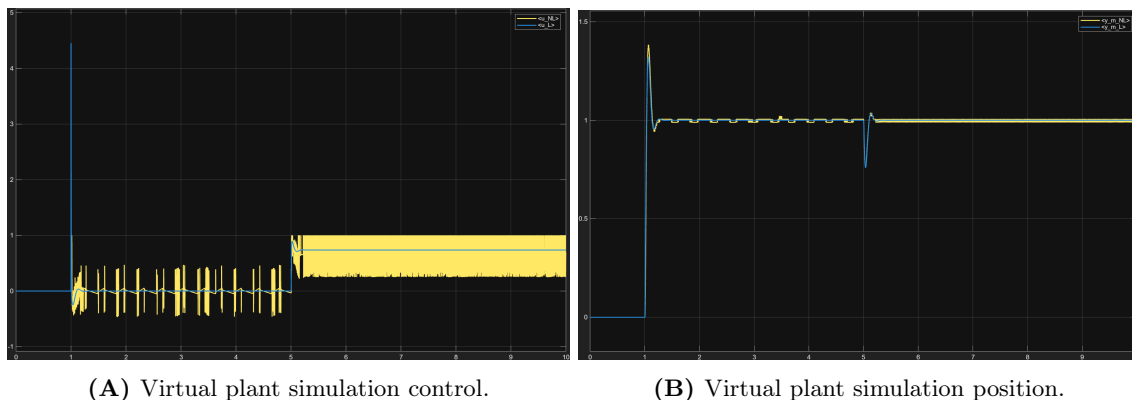


Figure 15: Initial design trial.

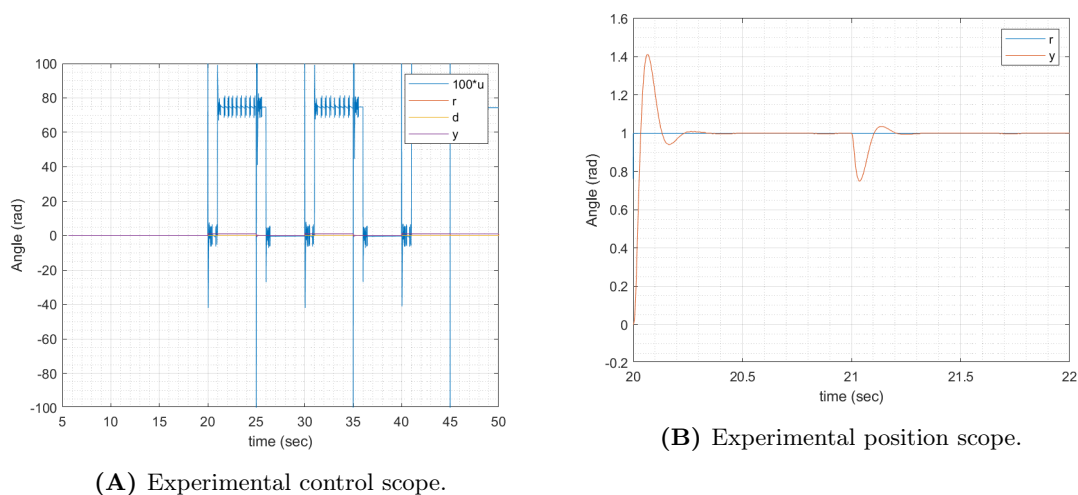


Figure 16: Second iteration gains.

$T_s = 0.001$	State feedback pole locations	State observer pole locations	Integrator pole location
Initial design	$0.9811 + 0.0302i, 0.9811 - 0.0302i$	0.5335, 0.5335	0.8834
Iteration 1	0.99, 0.67	0.5335, 0.5335	0.8910
Iteration 2	$0.9811 + 0.0302i, 0.9811 - 0.0302i$	0.5335, 0.5335	0.6871

Table 1: State observer feedback with integrator pole locations

Sampling time $T_s = 0.01$ sec

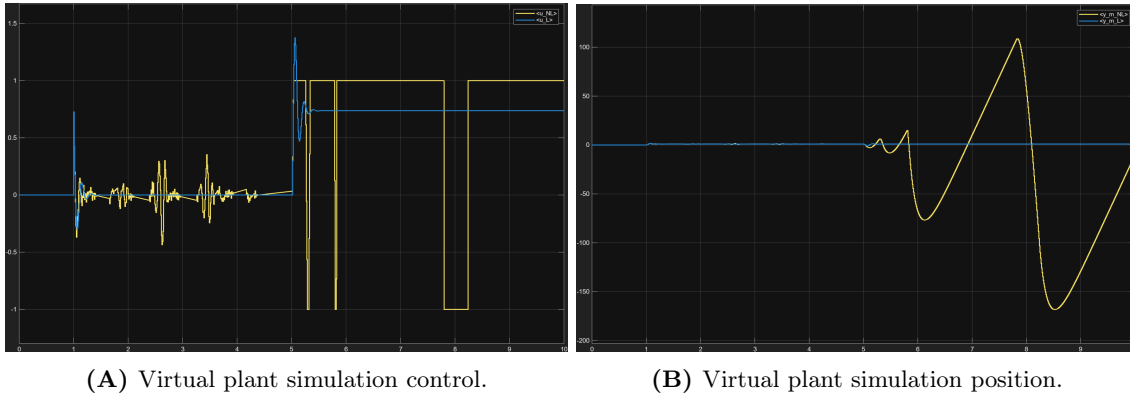


Figure 17: Initial design trial.

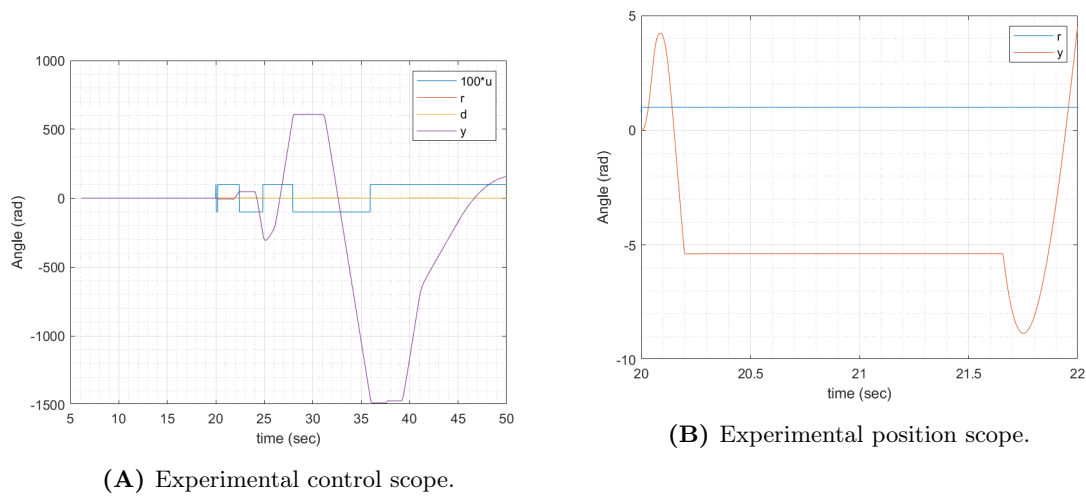
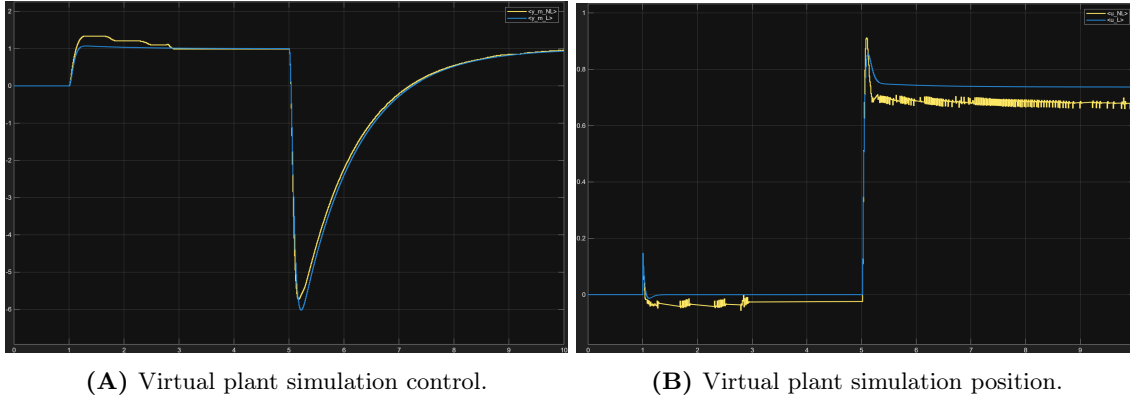
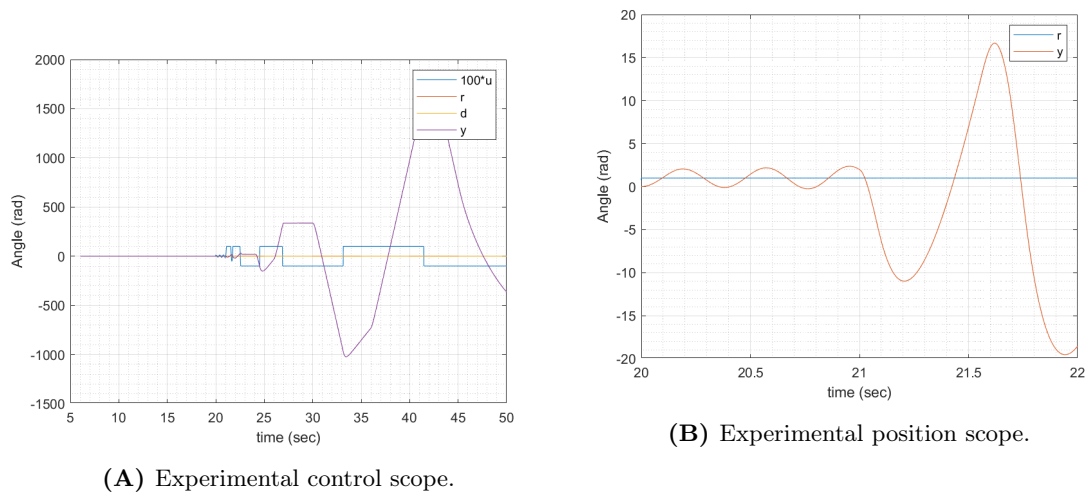
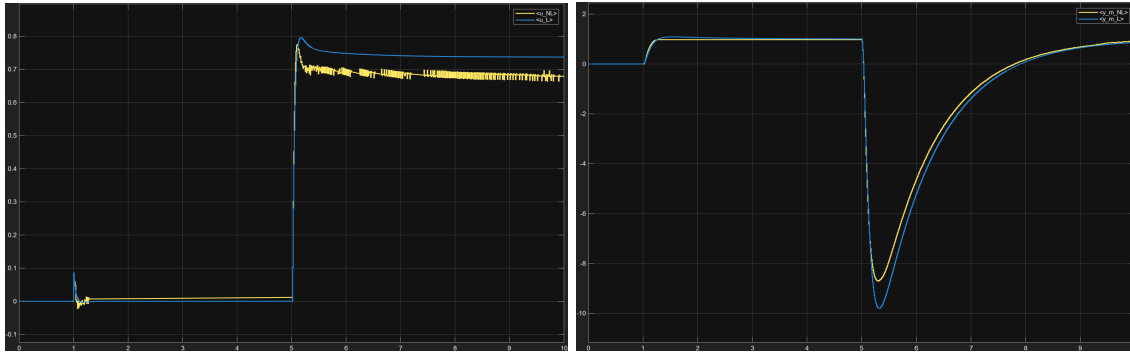


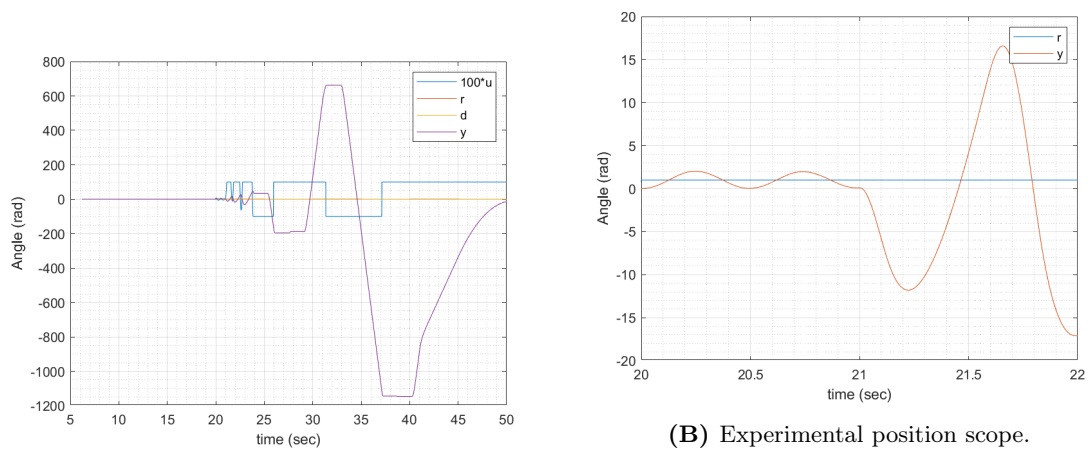
Figure 18: Initial design gains.

**Figure 19:** Initial design trial.**Figure 20:** First iteration gains.



(A) Virtual plant simulation control.

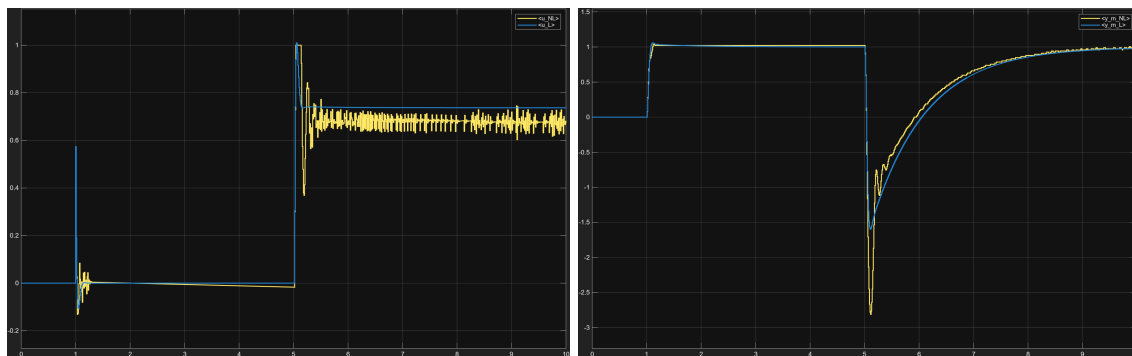
(B) Virtual plant simulation position.

Figure 21: Initial design trial.

(A) Experimental control scope.

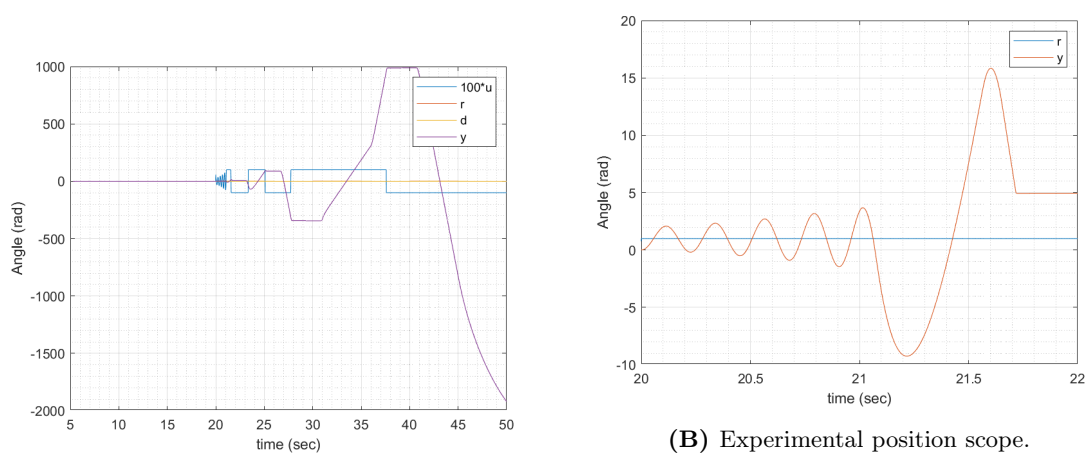
(B) Experimental position scope.

Figure 22: Second iteration gains.



(A) Virtual plant simulation control.

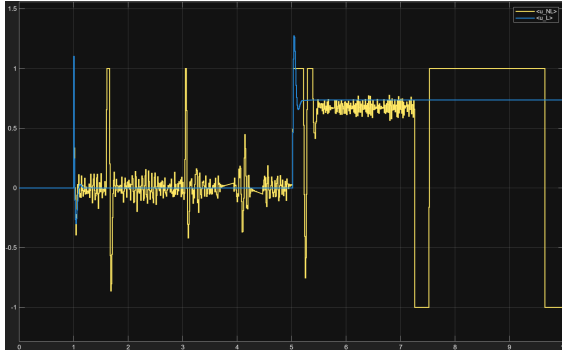
(B) Virtual plant simulation position.

Figure 23: Initial design trial.

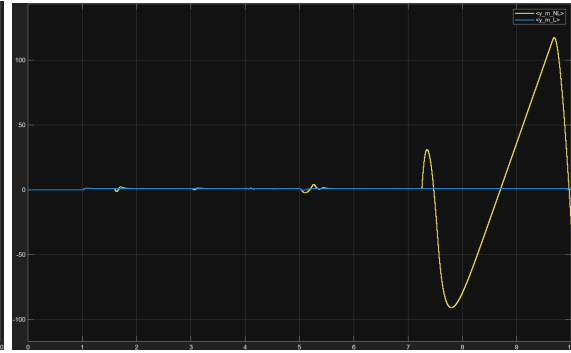
(A) Experimental control scope.

(B) Experimental position scope.

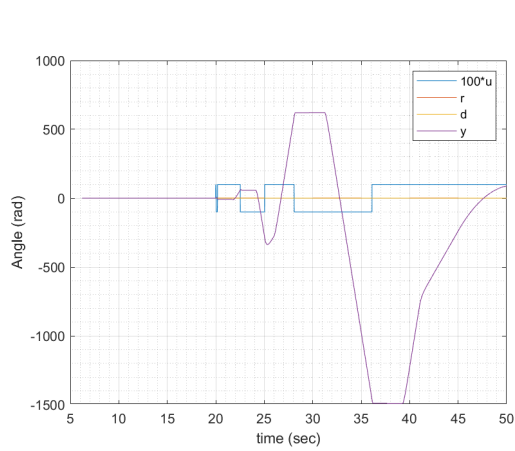
Figure 24: Third iteration gains.



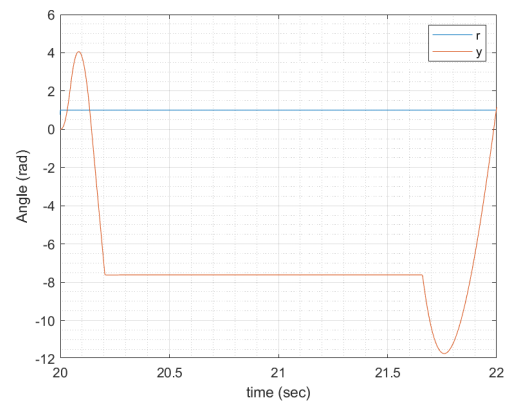
(A) Virtual plant simulation control.



(B) Virtual plant simulation position.

Figure 25: Initial design trial.

(A) Experimental control scope.



(B) Experimental position scope.

Figure 26: Fourth iteration gains.

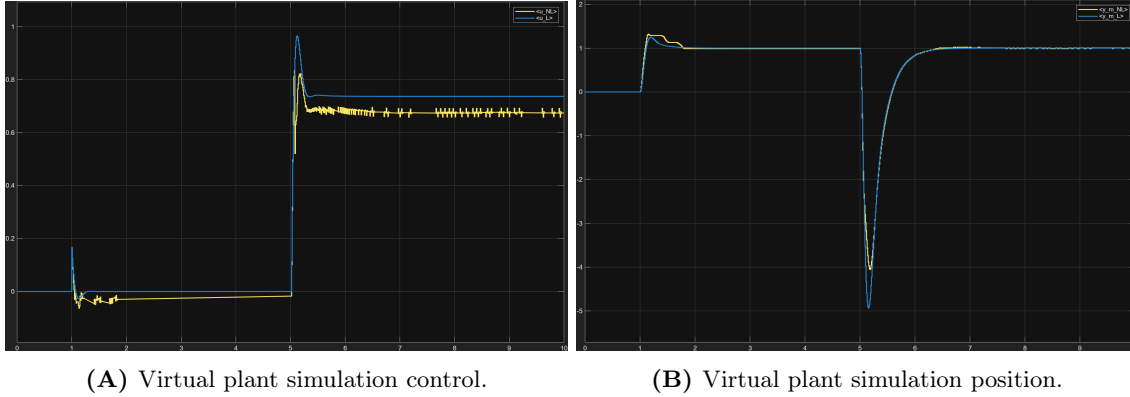


Figure 27: Initial design trial.

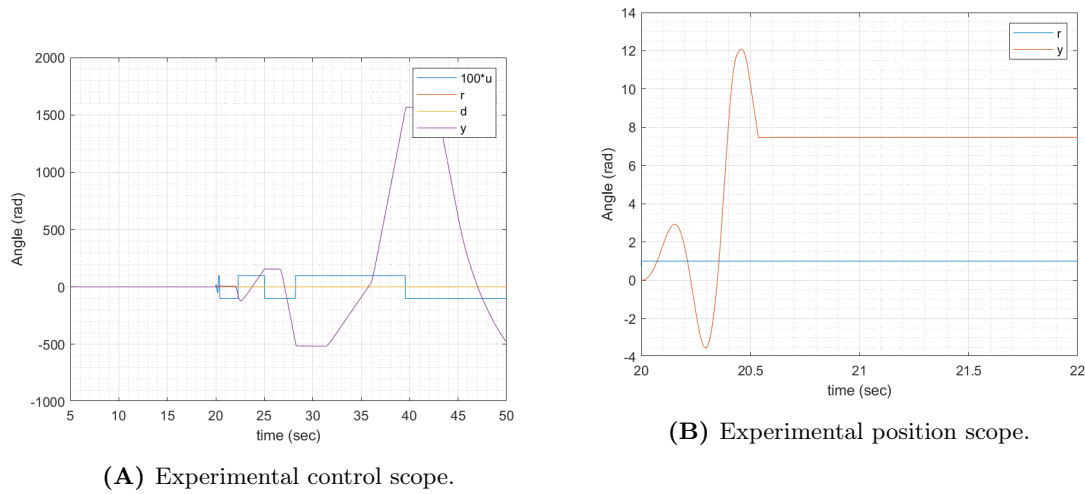


Figure 28: Fifth iteration gains.

$T_s = 0.001$	State feedback pole locations	State observer pole locations	Integrator pole location
Initial design	$0.7911 + 0.2518i, 0.7911 - 0.2518i$	0.0019, 0.0019	0.7472
Iteration 1	0.99, 0.67	0.0019, 0.0019	0.8910
Iteration 2	$0.9811 + 0.0302i, 0.9811 - 0.0302i$	0.0019, 0.0019	0.9405
Iteration 3	0.99, 0.67	0.0019, 0.0019	0.5445
Iteration 4	0.8, 0.56	0.0019, 0.0019	0.6400
Iteration 5	0.95, 0.8	0.0019, 0.0019	0.8550

Table 2: State observer feedback with integrator pole locations

Discussion

The system performed much better for the small sampling time. For the pole locations tested, an asymptotically stable output was produced. Moreover, the behavior matched very closely to what was predicted by the virtual plant. We see that our initial design was successful and further iterations were able to change the transient responses to the system inputs. Both proper reference tracking and disturbance rejection was achieved, proving to be better than SOFC alone.

The low sampling frequency experiments were unsuccessful in creating a stable system. Though many iterations were conducted, the output always diverged in response to the disturbance. What is frustrating is that the simulated virtual plant was able to produce stable behavior towards a disturbance and has successful disturbance rejection. However, using the same poles and gain values were unable to create the same properties. Perhaps this is indicative of an incomplete model of our plant, where there may be properties that are sensitive to the sampling frequency deployed in the experiment.

Regardless, with a satisfactorily high sampling frequency, SOFC with integral action verifies the theoretical properties of the system topology and can be sufficiently simulated without a physical system on-hand.

MS Problem

1 RLS System Identification

For this problem, I have chosen to use the indirect digital PD control system from Problem 4 as the target system for identification. The RLS algorithm will generate parameters for an FIR model. White noise reference and noise inputs were used for RLS learning. However, the power for the signal for the virtual plant system was heavily reduced to avoid duty cycle saturation. The following system and algorithm parameters were selected:

Closed Loop System Parameters	
Sampling Time, T_s	0.01 sec
Natural Frequency, ω_n	36 rad/sec
Damping Ratio, ζ	0.5

Table 1: Target system parameters.

RLS Algorithm Parameters	
Sampling Time, T_s	0.01 sec
N_p	50000
nb_FIR	50
SNR_u_y	0.1
CovPar	10000
ForgetFactor	1
Power_u_linear	1
Power_u_virtual	5e-6

Table 2: RLS Algorithm parameters.

1.1 Transfer function

The identified model transfer function for the linear system is:

$$\begin{aligned}
& 0.00054z^{-1} + 0.3311z^{-2} + 0.3555z^{-3} + 0.265z^{-4} + 0.2211z^{-5} + 0.114z^{-6} + 0.06539z^{-7} - 0.01482z^{-8} - 0.03912z^{-9} \\
& -0.07835z^{-10} - 0.07151z^{-11} - 0.08026z^{-12} - 0.05716z^{-13} - 0.05038z^{-14} - 0.02328z^{-15} - 0.01524z^{-16} \\
& +0.005478z^{-17} + 0.008051z^{-18} + 0.019z^{-19} + 0.01521z^{-20} + 0.01861z^{-21} + 0.0128z^{-22} + 0.01154z^{-23} \\
& +0.005265z^{-24} + 0.00378z^{-25} - 0.001155z^{-26} - 0.001306z^{-27} - 0.005101z^{-28} - 0.003993z^{-29} - 0.004118z^{-30} \\
& -0.001837z^{-31} - 0.002814z^{-32} - 0.001413z^{-33} - 0.0007549z^{-34} + 0.0008145z^{-35} + 0.0005718z^{-36} + 0.0007826z^{-37} \\
& +0.0001853z^{-38} + 0.0003741z^{-39} + 0.0001325z^{-40} + 0.001328z^{-41} + 0.0009961z^{-42} - 0.0002991z^{-43} - 0.0004328z^{-44} \\
& +6.116e-05z^{-45} - 0.001413z^{-46} + 0.0001155z^{-47} + 3.24e-05z^{-48} - 0.0004509z^{-49} + 0.0003047z^{-50}
\end{aligned}$$

The identified model transfer function for the virtual system is:

$$\begin{aligned}
& 0.004302z^{-1} + 0.4214z^{-2} + 0.4065z^{-3} + 0.25z^{-4} + 0.184z^{-5} + 0.05001z^{-6} + 0.01013z^{-7} - 0.06588z^{-8} \\
& - 0.06337z^{-9} - 0.08858z^{-10} - 0.06017z^{-11} - 0.06145z^{-12} - 0.02875z^{-13} - 0.02455z^{-14} + 0.0007195z^{-15} \\
& + 0.001588z^{-16} + 0.01509z^{-17} + 0.01062z^{-18} + 0.01816z^{-19} + 0.009157z^{-20} + 0.0082z^{-21} + 0.0007995z^{-22} \\
& + 0.003012z^{-23} - 0.00173z^{-24} - 0.0008886z^{-25} - 0.00507z^{-26} - 0.004555z^{-27} - 0.007309z^{-28} - 0.004372z^{-29} \\
& - 0.003388z^{-30} + 0.001289z^{-31} + 0.003556z^{-32} + 0.004228z^{-33} + 0.002233z^{-34} + 0.002028z^{-35} + 0.001036z^{-36} \\
& + 0.0004282z^{-37} - 0.001096z^{-38} - 0.001368z^{-39} - 0.001417z^{-40} - 0.0001021z^{-41} + 0.0004201z^{-42} + 0.001394z^{-43} \\
& + 0.000996z^{-44} + 0.0006675z^{-45} + 0.00121z^{-46} + 0.002226z^{-47} + 0.001696z^{-48} + 0.0007691z^{-49} - 0.0006041z^{-50}
\end{aligned}$$

1.2 Model performance

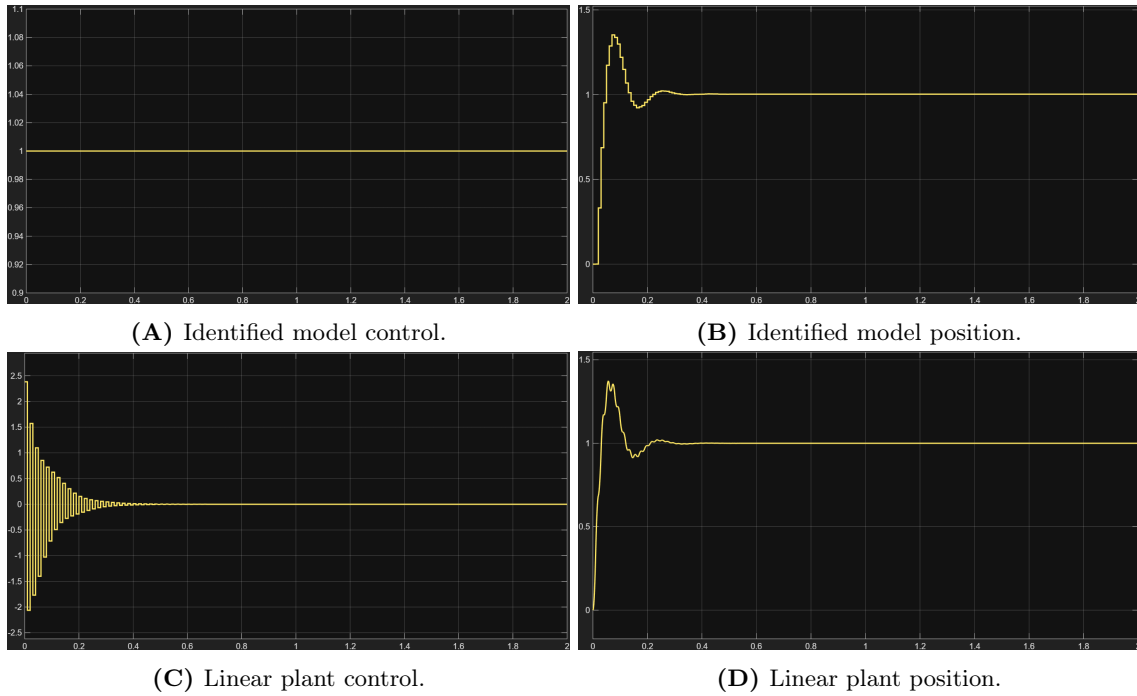


Figure 1: Identified FIR model of linear plant and original linear plant performance. Step size = 1.

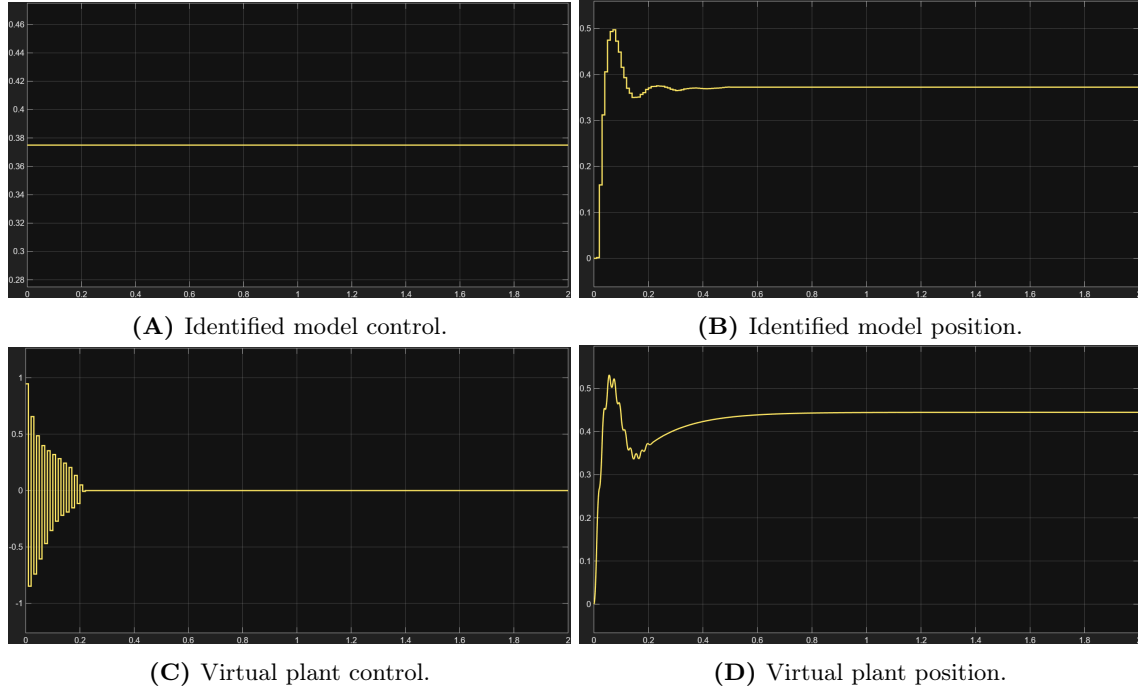


Figure 2: Identified FIR model of virtual plant and original virtual plant performance. Step size = 0.375.

1.3 Discussion

The RLS algorithm was able to create a remarkable model of both closed loop systems. The linear plant model behaved extremely accurately, as sampling the original linear plant would produce a result that is almost exactly replicated by the model. This makes sense, as the FIR model creates a linear system to model a stable linear system. Thus, with enough parameters, RLS algorithm can produce a satisfactory model of the linear system.

The virtual plant has some odd behavior. The model accurately replicates the behavior of the virtual plant system up until 0.2 seconds. However, the model predicts that there will be perfect tracking of the reference/control signal, whereas the actual virtual plant has tracking error. This can be attributed to the dead-zone of the virtual plant simulation — because the reference input had to be reduced to avoid saturation, the dead zone creates a noticeable error that cannot be corrected by feedback. The RLS model predicts no error because it is based on a finite impulse response, and will have perfect tracking to a step input.

2 RLS Feedforward Control

For the adaptive feedforward filters, 100 parameters, an initial covariance of 10,000, and an infinite horizon were used to follow the target model. In the case of a sine wave, the reference input was to be followed exactly. For the square wave input, a low-pass model was used, passing through this reference model:

$$M_{ref} = \left(\frac{1 + z^{-1}}{2} \right)^{10}$$

2.1 Linear plant reference tracking

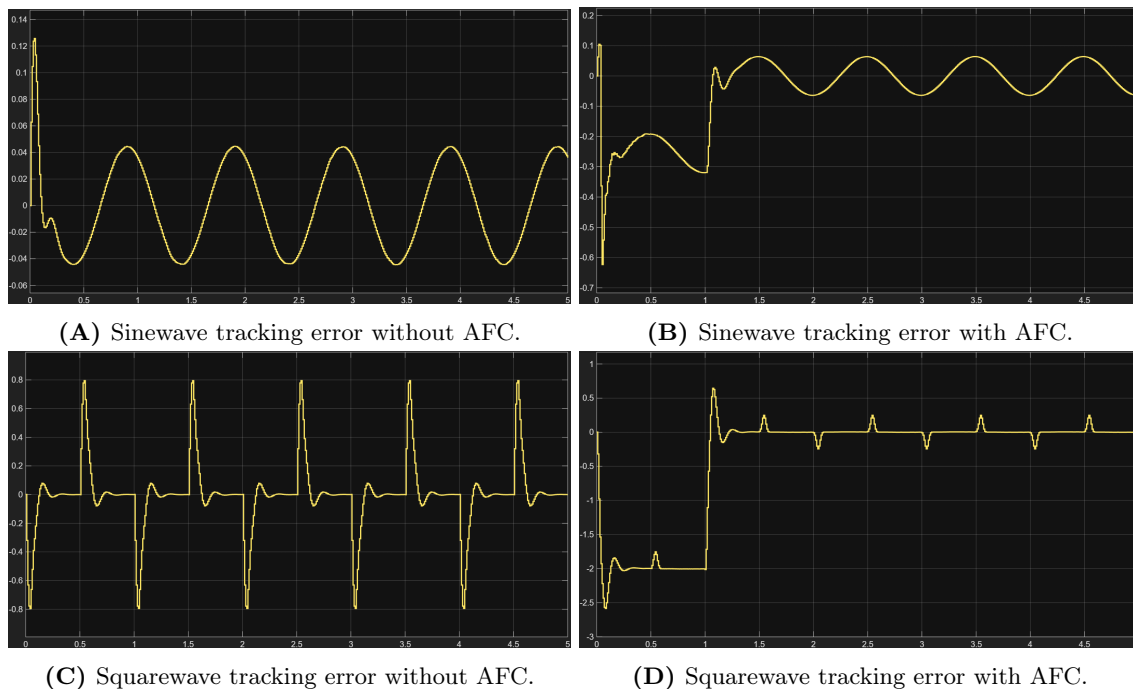


Figure 3: Reference tracking error for linear plant for sine- and square-wave input with amplitude of 1. Tracking error without AFC included.

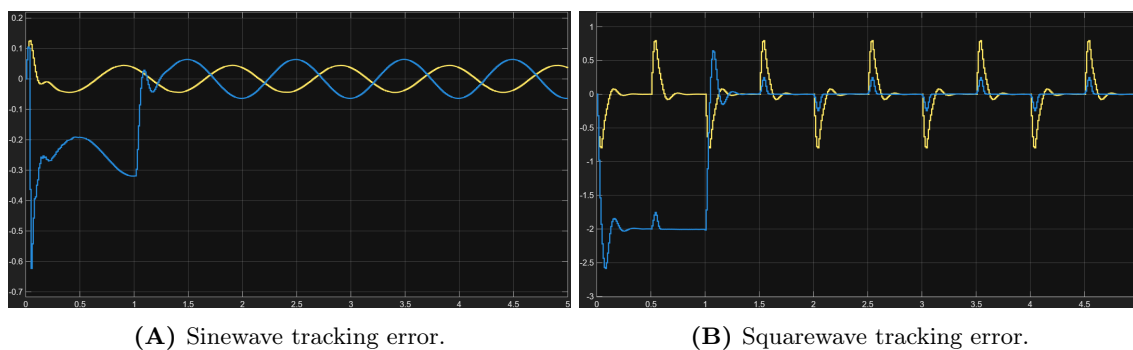


Figure 4: Reference tracking error for the linear plant with AFC (blue) and without AFC (yellow).

Relative tracking error (%)		
Input trajectory	No AFC	With AFC
Sinewave	4.46	6.43
Squarewave	33.45	8.57

Table 3: Relative H2 tracking error of the last 200 sample points for the linear plant system.

2.2 Virtual plant reference tracking

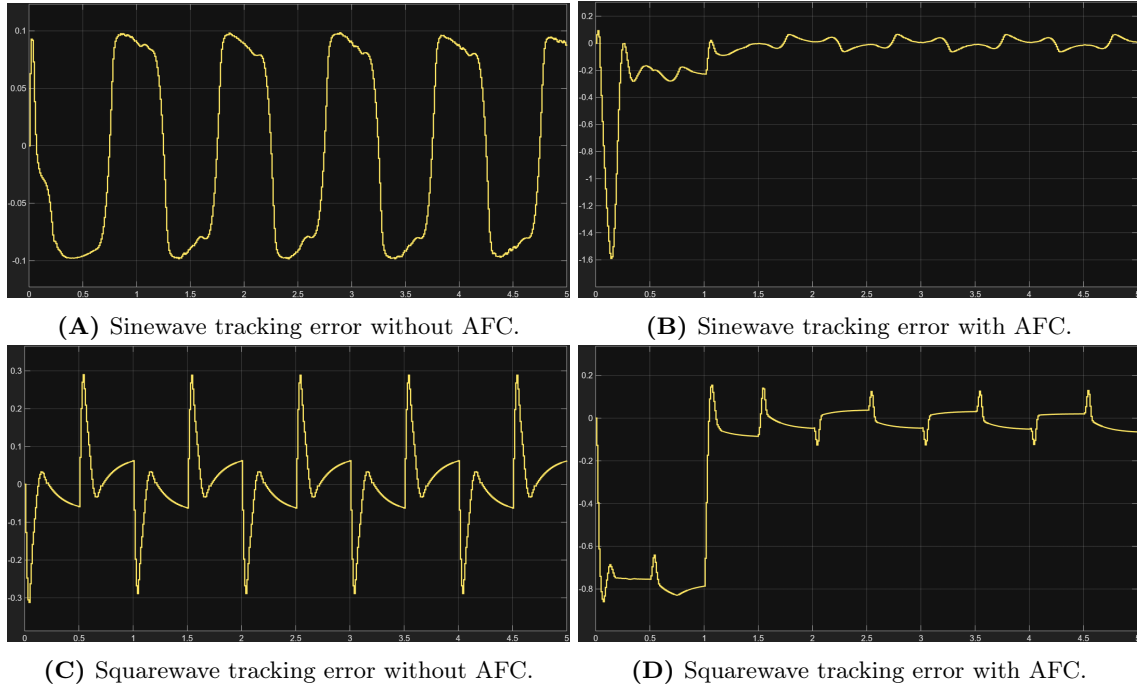


Figure 5: Reference tracking error for virtual plant for sine- and square-wave input with amplitude of 1. Tracking error without AFC included.

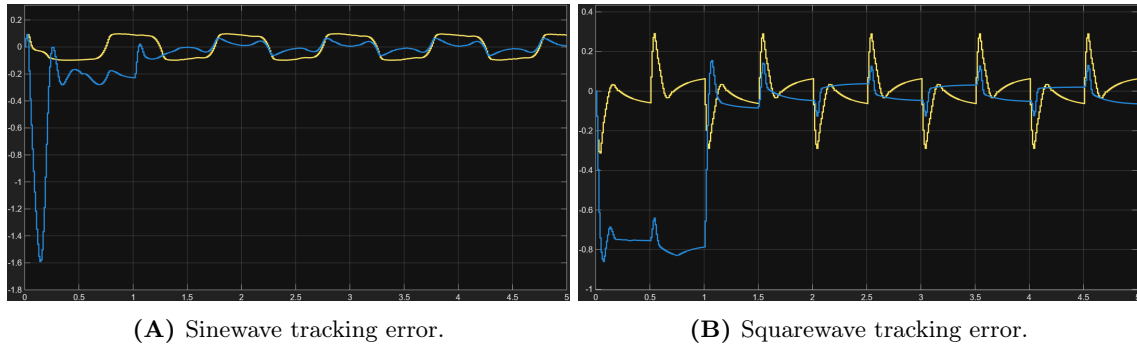


Figure 6: Reference tracking error for the virtual plant with AFC (blue) and without AFC (yellow).

Relative tracking error (%)		
Input trajectory	No AFC	With AFC
Sinewave	11.56	4.51
Squarewave	35.10	17.48

Table 4: Relative H2 tracking error of the last 200 sample points for the virtual plant system.

2.3 Discussion

For most cases, the implementation of the adaptive filter produced better reference tracking improved the tracking behavior of the system in question. In all cases of the virtual plant, the AFC reduced the tracking error of the closed loop system, reducing the relative error by at least 50% in both the sine- and square-wave trajectories. The linear plant also experienced a reduction in tracking error for the squarewave but not in the sinewave case. This may be an indication that the adaptive feedforward control scheme creates better adaptations to impulsive and sudden reference inputs, whereas predictable and smooth inputs are already better tracked by the linear plant.

3 RLS Disturbance Rejection Control

For the ADRC scheme, 100 filter parameters are used with an initial covariance of 10,000 and an infinite horizon were used.

3.1 Linear plant disturbance rejection

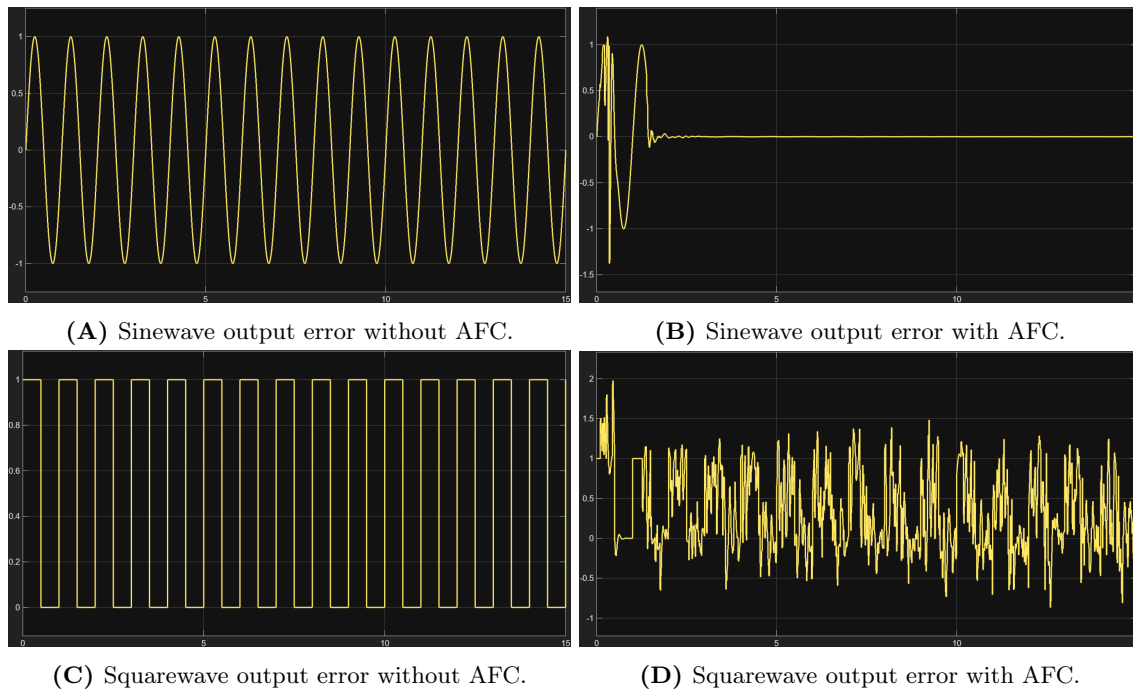


Figure 7: Output error for linear plant for sine- and square-wave input with amplitude of 1. Output error without AFC included.

Relative output error (%)		
Disturbance trajectory	No AFC	With AFC
Sinewave	100	0.0349
Squarewave	100	71.53

Table 5: Relative H2 output error of the last 200 sample points for linear plant system.

3.2 Virtual plant disturbance rejection

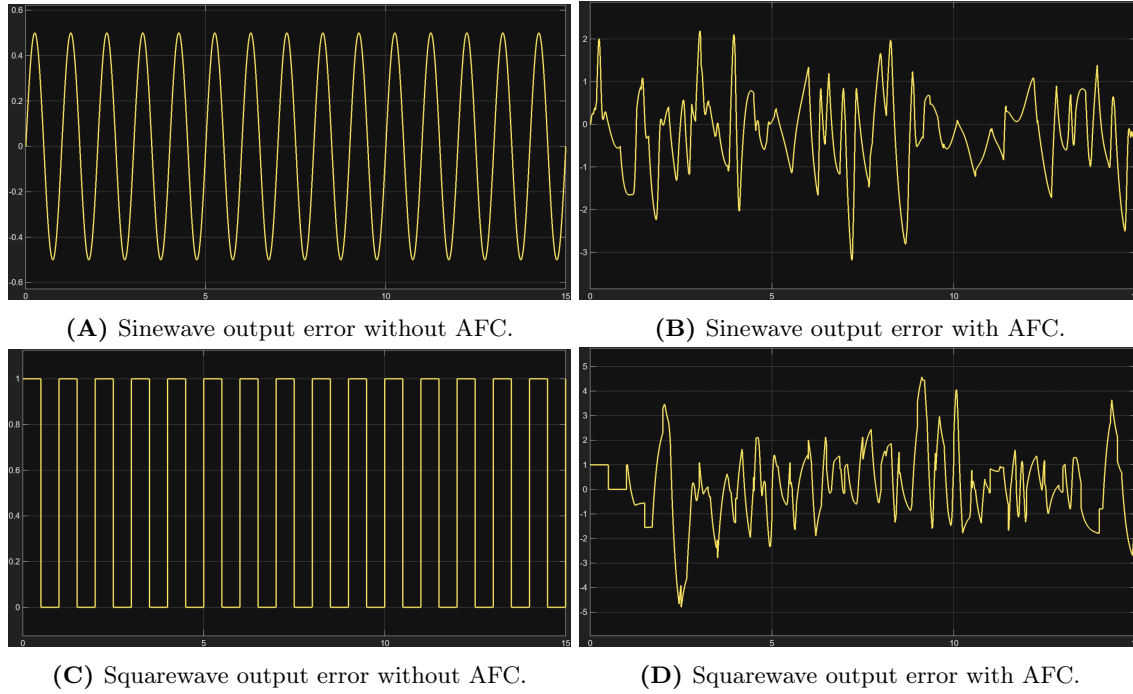


Figure 8: Output error for virtual plant for sine- and square-wave input with amplitude of 1. Output error without AFC included.

Relative output error (%)		
Input trajectory	No AFC	With AFC
Sinewave	100	254
Squarewave	100	227

Table 6: Relative H2 output error of the last 200 sample points for virtual plant system.

3.3 Discussion

The adaptive disturbance rejection control appeared to only provide positive results with the linear plant system, with significant results coming from only the sinewave disturbance input. It seems that the sine trajectory is sufficiently parameterized through our RLS filter. However, the square wave appears to be too impulsive and erratic to be completely rejected, even with 100 filter parameters.

The virtual plant simulations were unable to be controlled through the adaptive filter. Output errors were increased, on average, through the RLS filter, producing vbery erratic and unpredictable output signals. This can be attributed to the inherent differences between the true plant and the identified FIR model, which expects a stable result with no steady state error to a step input. However, this is not the case for the actual plant, so the model can fail to replicate the true behavior of the plant. As such, the differences that are calculated cannot sufficiently reject the disturbance input once passed through the RLS filter.